# Total asphericity and exact abelian colimits

TOTEX2: proof audit and research record

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**Abstract**

Total asphericity does not imply exact abelian colimits. We specify a category of finite linearly ordered 3-uniform hypergraphs with positive predimension $2|V| - 3|E|$ and strict closed embeddings. An alternating-amalgam argument classifies the open stabilizers of its homogeneous limit. Explicit homotopy comparisons then prove that every component of every covariant category of elements is weakly contractible, without restricting the action. Nevertheless, the three point arrows into one hyperedge have no affine equalizer over any nonzero coefficient ring; a column cokernel of projective dimension one has first derived colimit equal to that ring. We also prove that, for every admissible repaired Adámek–Koubek–Reiterman construction, the first binary second homology class survives every stage. All three inclusions from filtered to exact to total to quasi-filtered categories are therefore strict. We prove or correct all nine algebraic and homotopical leads, state the consequent manuscript corrections, and record the exact scope of the literature audit. No priority is claimed.

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# A The main implication and coefficients

## A.1 A counterexample to the main implication

We fix a universe and take all categories to be small in it. We use covariant functors $H : A \to \mathrm{Set}$, with arrows in $\mathrm{El}_A H$ directed by $(a,h) \xrightarrow{f} (b, H(f)h)$. We put $BA = |NA|$. A nonempty connected category $A$ is total if each component of every such category of elements is weakly contractible. Exactness means exactness of $\mathrm{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$. The module $P_p = \mathbb{Z}[A(p,-)]$ is covariant; $\Delta\mathbb{Z}$ is the constant right $\mathbb{Z}A$-module. We write $T_i(M) = \mathrm{Tor}_i^{\mathbb{Z}A}(\Delta\mathbb{Z}, M) = L_i \mathrm{colim}_A M$.

**Theorem A.1** (Refuted: totality does not imply exactness, Part A.1)**.** *Let $\mathcal{C}_\delta$ have as objects all pairs $([n], E)$, where $n \geq 1$, $[n] = \{1 < \cdots < n\}$, $E \subseteq \binom{[n]}{3}$, and*

$$\delta_X(U) = 2|U| - 3|\{e \in E : e \subseteq U\}| > 0 \quad (\varnothing \neq U \subseteq [n]).$$

*An arrow $f : X \to Y$ is a strictly increasing injection which is an induced hypergraph embedding and satisfies*

$$\delta_Y(U) > \delta_Y(fX) \quad (fX \subsetneq U \subseteq Y).$$

*Identities and composition are those of functions. This is a small, nonempty connected total category. Let $V$ be the one-point object, let $W$ be the ordered three-point object with its unique triple as an edge, and let $v_i : V \to W$ select its $i$th vertex. The three arrows $v_1, v_2, v_3$ have no integral affine post-equalizer. Consequently $\mathrm{colim}_{\mathcal{C}_\delta} : \mathrm{Ab}^{\mathcal{C}_\delta} \to \mathrm{Ab}$ is not exact.*

*Proof.* The closure calculus in Lemma A.21 proves that the specified arrows compose and that the category has amalgamation and common successors. Theorem A.34 proves totality for every covariant set-valued action, including infinite actions and actions with noninjective transitions. Its proof uses the finite alternating-amalgam theorem to classify all open stabilizers of a rich homogeneous limit, then gives explicit homotopy comparisons for arbitrary actions. Proposition A.24 proves that the displayed triple has no affine equalizer and exhibits a diagram $Q$ with $T_1(Q) \cong \mathbb{Z}$. These results verify every hypothesis and the asserted failure for this explicitly specified category. □

For two ordinary parallel arrows the affine-equalizer assertion still follows from the audited permutation-module calculation in Lemma B.2. Thus three is the smallest possible size of an obstructing ordinary family. The row and column presentations of Proposition B.5 now give different answers in the very same total category; their distinction is not merely a missing formal step in a proof.

## A.2 An arithmetic reduction

**Proposition A.2** (Proved: fieldwise detection of integral exactness)**.** *For every small category $A$, including the empty category and categories with several components, the following conditions are equivalent:*

1. $\operatorname{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ *is exact;*
2. $\operatorname{colim}_A : (\mathbb{Q}\text{-Mod})^A \to \mathbb{Q}\text{-Mod}$ *is exact and, for every prime $p$, the functor* $\operatorname{colim}_A : (\mathbb{F}_p\text{-Mod})^A \to \mathbb{F}_p\text{-Mod}$ *is exact.*

*No totality hypothesis is needed. Consequently, Parts A.1 and A.2 taken together ask equivalent universal questions. A negative answer to A.1 necessarily gives a failure over $\mathbb{Q}$ or over at least one prime field, although the obstructing diagram need not remain the same after reduction.*

*Proof.* The ordinary abelian-group colimit of a diagram of $R$-modules, for $R = \mathbb{Q}$ or $\mathbb{F}_p$, carries the unique compatible $R$-module structure, and its underlying abelian group is the ordinary abelian-group colimit. This follows from the direct-sum-and-relations construction, since the subgroup of relations is an $R$-submodule. Forgetting scalars is exact, so (1) implies (2).

Assume (2). We use the normalized bar complex

$$C_n(A; M) = \bigoplus_{a_0 \to \cdots \to a_n}^{\text{nondegenerate}} M(a_0),$$

with its usual face differential; its homology is $T_n(M)$. If $M$ is a $\mathbb{Q}$- or $\mathbb{F}_p$-module diagram, this is, after restriction of scalars, exactly the bar complex computing derived colimits over that field. Thus such diagrams have $T_i(M) = 0$ for $i > 0$. This comparison is an equality of the specified complexes, not an assumption that arbitrary base change is flat.

We recall the three closure facts used here. A direct sum or filtered colimit of acyclic diagrams is acyclic because the bar complex commutes with these operations and both operations are exact in Ab. If $0 \to U \to V \to W \to 0$ is objectwise exact, the bar complexes form a short exact sequence; their long exact homology sequence shows closure under extensions and under kernels of epimorphisms between acyclic diagrams.

Let $M : A \to \mathrm{Ab}$ be arbitrary. For each prime $p$, its $p$-primary torsion subfunctor is the union of $M[p^n]$, where $M[p^n](a) = \{m : p^n m = 0\}$. The quotient $M[p^n]/M[p^{n-1}]$ is an $\mathbb{F}_p$-module diagram. Induction on $n$, then filtered colimits, prove acyclicity of the $p$-primary subfunctor. The full torsion subfunctor $M_{\mathrm{tor}}$ is their direct sum and is therefore acyclic. Put $M_{\mathrm{tf}} = M/M_{\mathrm{tor}}$. It is objectwise torsion free, so the natural map

$$M_{\mathrm{tf}} \longrightarrow \mathbb{Q} \otimes_{\mathbb{Z}} M_{\mathrm{tf}}$$

is a monomorphism in $\mathrm{Ab}^A$. Its cokernel is objectwise torsion: each element is a finite rational linear combination, and a common denominator kills its class. Both the middle diagram and this cokernel are acyclic. Closure under kernels of epimorphisms makes $M_{\mathrm{tf}}$ acyclic. Finally the sequence $0 \to M_{\mathrm{tor}} \to M \to M_{\mathrm{tf}} \to 0$ makes $M$ acyclic. Therefore $T_1(M) = 0$ for every $M$, which is equivalent to exactness of the right-exact colimit functor. □

**Corollary A.3** (Refuted: both coefficient implications, Part A.2)**.** *The same category $\mathcal{C}_\delta$ has nonexact colimits over every nonzero unital coefficient ring $R$. In particular the implication from totality to exactness is false over $\mathbb{Q}$ and over $\mathbb{F}_p$ for each prime $p$, with one category and one triple working simultaneously for all these coefficients. Explicitly, put*

$$D_R : R[\mathcal{C}_\delta(W, -)] \longrightarrow R[\mathcal{C}_\delta(V, -)]^2, \qquad g \longmapsto (gv_2 - gv_1, gv_3 - gv_1).$$

*Then $D_R$ is injective, $Q_R = \operatorname{coker} D_R$ has projective dimension exactly one, and $T_1(Q_R) \cong R$.*

*Proof.* Proposition A.24 proves injectivity over every $R$ by a private vertex in each finite nonempty support. The resulting two-term projective resolution has colimit differential zero, so $T_1(Q_R) = R \neq 0$. Its projective dimension is at most one by that resolution and is not zero by this nonvanishing. The totality proof concerns the underlying category and is independent of $R$. □

## A.3 Proof and construction routes investigated

The accompanying report records the precise scope of each attempted route. The model-category observation that totality makes ordinary simplicial-set colimits homotopy invariant does not by itself identify the underlying set colimit with the abelian-group colimit. A free abelian cotriple resolution has permutation modules as its terms, but some face maps send a basis vector to a linear combination; it is not the linearization of a simplicial set diagram. We therefore do not infer acyclicity of its colimit complex from totality. The row/column formulas below identify the nonzero simultaneous-intersection obstruction in the separator, and explain why these proof-side routes cannot imply the proposed universal vanishing.

**Proposition A.4** (Refuted: universal membership of the column module in the proposed resolving class)**.** *For a small category $A$, let $\mathcal{R}_A$ be the smallest class of covariant integral modules containing every $\mathbb{Z}[H]$, for $H : A \to \mathrm{Set}$, and closed under extensions, kernels of epimorphisms between members, arbitrary direct sums, direct summands, filtered colimits, and $F \mapsto F \otimes_{\mathbb{Z}} \mathbb{Z}[H]$. Every member of $\mathcal{R}_A$ is objectwise torsion-free. There is a finite filtered category $A$ and three parallel arrows $a_0, a_1, a_2 : p \to q$ for which*

$$Q_a = \operatorname{coker}\big(P_q \longrightarrow P_p^2,\ t \longmapsto (t(a_1 - a_0), t(a_2 - a_0))\big) \notin \mathcal{R}_A.$$

*Thus the first proposed target of approach P1 cannot work uniformly, even for filtered categories. This does not refute P1's alternative of finding a different module with the same $T_1$, and it is not a separator between totality and exactness.*

*Proof.* Permutation modules are objectwise free. Subgroups, extensions, direct sums and direct summands of torsion-free abelian groups are torsion-free. A filtered colimit is torsion-free because an equation $n[x] = 0$, $n \neq 0$, holds at some later stage, where it implies that the representative itself is zero. Tensoring with a free abelian group is a direct sum of copies. All operations are evaluated objectwise, so the class of objectwise torsion-free modules contains $\mathcal{R}_A$.

Here is the promised category completely. Its objects are $p, q, x, t$. Besides identities, its arrows are

$$a_0, a_1, a_2 : p \to q, \qquad g_0, g_1, g_2, g_3 : q \to x, \qquad c_0, c_1, c_2 : p \to x,$$

and one arrow $u_y : y \to t$ for each $y = p, q, x$. There are no other arrows. Identities compose as identities, every composition ending at $t$ is the indicated unique arrow, and the other compositions are specified by

| | $a_0$ | $a_1$ | $a_2$ |
|---|---|---|---|
| $g_0$ | $c_0$ | $c_1$ | $c_2$ |
| $g_1$ | $c_0$ | $c_2$ | $c_1$ |
| $g_2$ | $c_1$ | $c_0$ | $c_2$ |
| $g_3$ | $c_1$ | $c_2$ | $c_0$ |

Associativity holds: a composable triple of nonidentity arrows must follow $p \to q \to x \to t$, and both composites are $u_p$. All other associativity equations contain an identity. The object $t$ is terminal, so the category is nonempty, connected and filtered. It is total directly: for any

$H : A \to \mathrm{Set}$, the object $(a, h)$ of its category of elements maps to $(t, H(u_a)h)$. Its connected component is exactly the fibre of this value in $H(t)$ and has that terminal object; the assertion is also valid if $H(t)$ is empty.

At $x$, $P_q(x) = \mathbb{Z}^4$ on the $g_j$ and $P_p(x) = \mathbb{Z}^3$ on the $c_i$. The column map has its image in $\ker(\mathrm{aug} : \mathbb{Z}^3 \to \mathbb{Z})^2$, a rank-four direct summand of $\mathbb{Z}^6$. With bases $c_1 - c_0, c_2 - c_0$ in both augmentation kernels, its matrix is

$$D = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

Its determinant is 3, and its upper-left $3 \times 3$ minor has determinant $-1$. Hence its Smith invariant factors are $1, 1, 1, 3$, and

$$Q_a(x) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/3.$$

More explicitly, $v = (1, 0, 1, 0)^{\mathrm{t}}$ represents a nonzero element of order 3 in the reduced cokernel: one has $D(1, 1, -1, -1)^{\mathrm{t}} = 3v$, while the unique rational preimage of $v$ is $(1, 1, -1, -1)^{\mathrm{t}}/3$, which is not integral. Thus $Q_a$ cannot belong to $\mathcal{R}_A$. The program `code/algebra_column_torsion.py` reconstructs the category, checks all 52 composable pairs and 121 composable triples, verifies the identities and terminal object, and recomputes the Smith form and this order-three certificate. Its output is `code/algebra_column_torsion.out`. The original colimit is evaluation at $t$, so $T_1(Q_a) = 0$ in this example; no nonexactness is being inferred from its objectwise torsion. □

**Proposition A.5** (Refuted: rational direct membership in P1)**.** *There is a small filtered category $A$ and a triple $a_0, a_1, a_2 : p \to q$ such that its rational column module*

$$Q_a = \mathrm{coker}\big(\mathbb{Q}[A(q, -)] \longrightarrow \mathbb{Q}[A(p, -)]^2, \quad t \longmapsto (t(a_1 - a_0), t(a_2 - a_0))\big)$$

*does not belong to the smallest class of rational diagrams containing all permutation modules and closed under extensions, kernels of epimorphisms, arbitrary direct sums, direct summands, filtered colimits, and tensoring with permutation modules. This strengthens the integral torsion obstruction: all values of this example are rational vector spaces. It does not refute the alternative in P1 of using another module with the same $T_1$. Moreover, the same category and triple give an integral column module with torsion-free values which lies outside the original integral resolving class.*

*Proof.* First fix an automorphism $\sigma : v \to v$ in a category. For a rational diagram $M$, put $T_M = M(\sigma)$ and consider the condition

$$\ker(T_M^{2n} - 2^{2n} \,\mathrm{id}_{M(v)}) = 0 \qquad \text{for every integer } n \geq 1. \tag{1}$$

Every permutation module satisfies this condition. A finite-support eigenvector of a permutation has support a finite union of finite orbits. On an orbit of length $r$ its eigenvalue must have $r$th power one; the eigenvalue $2^{2n}$ cannot occur.

Condition (1) is preserved by subobjects, extensions, arbitrary sums and summands. It is preserved by filtered colimits because evaluation at $v$ and the displayed operator commute with them, and filtered colimits of rational vector spaces are exact. In particular it is preserved by kernels of epimorphisms.

We verify the additional tensor operation explicitly. Let $P$ be the permutation of a set $S$ and give $M(v) \otimes \mathbb{Q}[S]$ the automorphism $T_M \otimes P$. Fix $n \geq 1$ and a finite-support vector $z$ in the kernel of $(T_M \otimes P)^{2n} - 2^{2n}$. Decompose its support into $P^{2n}$-orbits. On an infinite orbit,

the eigenvector equation and invertibility of $T_M$ propagate any nonzero coefficient indefinitely, contradicting finite support. On an orbit of length $r$, iterating the coefficient equation gives

$$(T_M^{2nr} - 2^{2nr})m = 0.$$

The old condition forces $m = 0$. Thus $z = 0$, which proves tensor closure. The use of all even powers is essential; injectivity of $T_M + 2$ alone would not suffice for this operation.

Now define $A$. Its objects are $p, q, v, t$, with $t$ terminal. The only endomorphisms of $p, q, t$ are identities, and $\mathrm{End}(v) = \{\sigma^n : n \in \mathbb{Z}\}$ is infinite cyclic. The other nonterminal hom-sets are

$$A(p,q) = \{a_0, a_1, a_2\}, \quad A(q,v) = \{h_{j,n} : j = 1, 2,\ n \in \mathbb{Z}\}, \quad A(p,v) = \{b_n : n \in \mathbb{Z}\}.$$

There are no further arrows except the unique arrows to $t$. Composition with $\sigma^m$ adds $m$ to the subscript, and

$$h_{j,n}a_i = b_{n+e_{j,i}}, \qquad (e_{1,0}, e_{1,1}, e_{1,2}) = (0,1,2), \quad (e_{2,0}, e_{2,1}, e_{2,2}) = (0,2,1).$$

All composites with terminal arrows are the unique terminal arrow. These formulas define an associative category: all nonidentity nonterminal associativity checks are associativity of integer addition or the equality $m + (n + e_{j,i}) = (m + n) + e_{j,i}$; terminal composites are unique. The object $t$ makes $A$ filtered.

Set $R = \mathbb{Q}[z, z^{-1}]$, using $z$ to represent the action of $\sigma$. At $v$, the domain of the column presentation is $R^2$ and each of its two target factors is $R$. The presentation matrix is

$$D = \begin{pmatrix} z-1 & z^2-1 \\ z^2-1 & z-1 \end{pmatrix}.$$

The sum and difference bases diagonalize it over $R$, giving entries $(z-1)(z+2)$ and $-z(z-1)$. Since $-z$ is a unit,

$$Q_a(v) \cong R/((z-1)(z+2)) \oplus R/(z-1).$$

For an explicit nonzero $-2$ eigenvector, let $\xi$ be the class of $(z-1, z-1)$. Then

$$(z+2)(z-1, z-1) = D(1,1),$$

so $(T_{Q_a} + 2)\xi = 0$ and consequently $(T_{Q_a}^2 - 4)\xi = 0$. The functional obtained by evaluating at $z = -2$ and summing the two coordinates annihilates both columns of $D$ but takes value $-6$ on $(z-1, z-1)$. Thus $\xi \neq 0$.

The column module fails (1) for $n = 1$, whereas the entire specified resolving class satisfies it. Hence it does not belong to that class. Since $t$ is terminal, $\mathrm{colim}_A Q_a = Q_a(t) = \mathbb{Q}^2$ and $T_i(Q_a) = 0$ for $i > 0$. This example concerns direct membership only, not the main implication.

**Proved integral strengthening.** The same invariant and closure argument work over $\mathbb{Z}$. For the tensor operation on an infinite permutation orbit, choose an extreme nonzero coefficient in its finite support. The coefficient equation one step beyond that support says that $T_M^{2n}$ kills that coefficient, contrary to invertibility of $T_M$. This argument does not divide by 2; the finite-orbit argument is unchanged.

Let $R_{\mathbb{Z}} = \mathbb{Z}[z, z^{-1}]$. The integral column matrix at $v$ is the same $D$, and

$$\det D = -z(z-1)^2(z+2) = -z^4 + 3z^2 - 2z.$$

It is nonzero over the domain $R_{\mathbb{Z}}$, so $0 \to R_{\mathbb{Z}}^2 \xrightarrow{D} R_{\mathbb{Z}}^2 \to Q_a(v) \to 0$ is exact. Its reduction modulo any prime $p$ is also injective: the determinant has leading coefficient $-1$, so it remains

nonzero in the domain $\mathbb{F}_p[z, z^{-1}]$. The two free abelian terms in this presentation therefore give

$$\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/p, Q_a(v)) = \ker(D \bmod p) = 0 \qquad \text{for every prime } p.$$

Equivalently, $Q_a(v)$ has no element killed by a prime and is torsion free. At $p$ and $t$ the column module is $\mathbb{Z}^2$; at $q$ it is $\mathbb{Z}^5$, since the single presentation column has a coordinate equal to 1. Thus every value is torsion free. The same vector $\xi = [(z-1, z-1)]$ is killed by $z+2$ and is nonzero: evaluate at $z=-2$ in $\mathbb{Z}[1/2]$, then sum the coordinates, obtaining $-6$. It violates the integral invariant for $n=1$. Hence the integral column module also fails membership in P1's full resolving class, although all its higher colimits vanish because $t$ is terminal. □

**Exact symbolic certificate.** The program `finite_rational_p1.py` checks every composable triple of the finitely many arrow families using three independent integer exponent variables. Equality of the resulting affine expressions proves the displayed associativity identities for all integer exponents; no finite truncation of the category is substituted. It also verifies the rational change of basis, the two polynomial relations for $\xi$, and the nonzero detecting functional. The full tensor-closure argument is the orbit proof above, not a numerical test.

### A.4 Ramsey classes and integral affine equalizers

The following argument is included with full proof. We do not claim priority for the implication; the literature search did not identify an earlier statement in these terms.

**Theorem A.6** (Proved: a Ramsey criterion for integral exactness)**.** *Let $A$ be a small category with finite hom-sets. Suppose that any two objects have a common successor. Suppose also that $A$ has the following morphism Ramsey property: for any objects $p, q$ and positive integer $c$ there is an object $r$ such that every coloring*

$$\chi : A(p, r) \longrightarrow \{1, \dots, c\}$$

*admits $g : q \to r$ for which $\chi$ is constant on $g \circ A(p, q)$. Then $A$ has exact colimits in abelian groups. More precisely, if $a_0, \dots, a_k : p \to q$ and $k \geq 1$, every target $r$ with the stated Ramsey property for $p, q, c = (3k+1)^k$ admits an integral affine combination $t \in \mathbb{Z}[A(q, r)]$ satisfying $ta_0 = \cdots = ta_k$.*

*Proof.* Fix the family and such a target. Put $X = A(p, r)$, $Y = A(q, r)$, and define the homomorphism between finite-rank free abelian groups

$$D : \mathbb{Z}[Y] \longrightarrow \mathbb{Z}[X]^k, \qquad D[g] = ([ga_i] - [ga_0])_{i=1}^k.$$

Let $\epsilon : \mathbb{Z}[Y] \to \mathbb{Z}$ be augmentation, and write $\epsilon(\ker D) = d\mathbb{Z}$ with $d \geq 0$. We must prove $d = 1$. Suppose otherwise. Choose

$$\theta = \begin{cases} 1/2, & d = 0, \\ \lfloor d/2 \rfloor / d, & d \geq 2. \end{cases}$$

Then $1/3 \leq \theta \leq 1/2$ and $d\theta = 0$ in the circle group $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Consequently

$$\phi : \operatorname{im} D \longrightarrow \mathbb{T}, \qquad \phi(Dt) = \theta\epsilon(t) \pmod{\mathbb{Z}}$$

is well-defined: two preimages differ by an element of $\ker D$, whose augmentation is a multiple of $d$.

This homomorphism extends to a homomorphism $\widetilde{\phi} : \mathbb{Z}[X]^k \to \mathbb{T}$. To see the extension directly, choose a Smith normal form for the subgroup $\operatorname{im} D$ of the finite-rank free group. In a

suitable basis $e_1, \dots, e_m$ it is generated by $s_1 e_1, \dots, s_\ell e_\ell$, where $s_j > 0$. Choose an $s_j$th root in $\mathbb{T}$ of $\phi(s_j e_j)$ as the value on $e_j$, and assign arbitrary values to the remaining basis vectors. Such roots exist because multiplication by every positive integer on $\mathbb{T}$ is surjective.

Write $\lambda_i(x) \in [0, 1)$ for the representative of the value of $\widetilde{\phi}$ on the $x$-basis vector in block $i$. For every $g \in Y$, the construction gives

$$\sum_{i=1}^{k} (\lambda_i(g a_i) - \lambda_i(g a_0)) = \theta \pmod{\mathbb{Z}}. \tag{2}$$

Set $L = 3k + 1$ and color $x \in X$ by the $k$-tuple

$$(\lfloor L\lambda_1(x) \rfloor, \dots, \lfloor L\lambda_k(x) \rfloor) \in \{0, \dots, L-1\}^k.$$

Ramsey supplies $g : q \to r$ on whose $p$-subobjects this color is constant. In particular, $g a_i$ and $g a_0$ have the same color for every $i$. Thus

$$\left| \sum_{i=1}^{k} (\lambda_i(g a_i) - \lambda_i(g a_0)) \right| < \frac{k}{3k+1} < \frac{1}{3}.$$

Every real representative of $\theta$ modulo $\mathbb{Z}$ has absolute value at least $1/3$, contradicting (2). Therefore $d = 1$, which means precisely that some $t \in \ker D$ has augmentation 1. A family of one arrow requires only $1_q$. Together with the common-successor hypothesis, simultaneous integral affine equalization is Isbell–Mitchell's exactness criterion (Isbell and Mitchell, 1976, Theorem 3.1 and Proposition 3.3). □

**Remark A.7** (Proved: scope of the finite-hom-set hypothesis)**.** It is used above only to give an elementary Smith normal form proof of extension to the circle. Divisibility makes $\mathbb{T}$ an injective abelian group, so the same argument works without finite hom-sets if one invokes that standard injectivity theorem. The finite version already covers the ordered-graph application below.

**Corollary A.8** (Proved: ordered finite graphs are exact)**.** *Let $\mathcal{G}$ have one representative of each finite linearly ordered simple graph as objects and increasing induced graph embeddings as arrows, composed as functions. Then $\mathcal{G}$ has exact colimits in abelian groups. In particular it is not a counterexample to totality implying exactness.*

*Proof.* The category is small, has finite hom-sets, and has common successors: take the disjoint union of two graphs and concatenate their orders. Finite simple graphs with an arbitrary linear order have the Ramsey property by (Hubička and Nešetřil, 2019, Corollary 4.3, printed p. 51). The source defines embeddings by preservation and reflection of relations (printed p. 6), so these are induced embeddings in the graph case. Because a finite linear order is rigid, an isomorphic induced copy of $p$ is specified by a unique embedding from $p$; hence its Ramsey property for copies is exactly the morphism Ramsey property used in Theorem A.6. That theorem applies. □

## A.5 Explicit small ordered-graph certificates

Here $q$ is the increasing induced path with vertex set $\{0 < 1 < 2\}$ and edge set $\{01, 12\}$. A triple $(a, b, c)$ denotes the embedding of $q$ with images $a < b < c$. These examples were first found by exact finite computation, then checked by the displayed integer identities. They are independent of the Ramsey theorem.

**Proposition A.9** (Proved: three vertex maps)**.** *For the three embeddings $v_0, v_1, v_2$ of the one-vertex graph into $q$, an integral affine equalizer exists in the ordered graph $r_6$ on $\{0 < \cdots < 5\}$ with edge set*

$$\{01, 12, 13, 15, 23, 24, 34, 35\}.$$

*It is*

$$t = (012) - (015) + (124) - (134) + (235).$$

*Proof.* Each displayed triple has its consecutive pairs in the edge set and its outer pair absent, so it is an increasing induced embedding. The coefficient sum is $1 - 1 + 1 - 1 + 1 = 1$. The three vertex marginals are respectively

$$[0] - [0] + [1] - [1] + [2] = [2], \quad [1] - [1] + [2] - [3] + [3] = [2], \quad [2] - [5] + [4] - [4] + [5] = [2].$$

Thus $tv_0 = tv_1 = tv_2$ over $\mathbb{Z}$, and after change of coefficients over every unital coefficient ring. □

**Proposition A.10** (Proved: two edge maps)**.** *For the embeddings $e_0, e_1$ of the ordered edge into the two edges of $q$, an integral affine equalizer exists in the ordered graph $r_7$ on $\{0 < \cdots < 6\}$ with edge set*

$$\{01, 12, 23, 35, 56, 14, 45\}.$$

*It is*

$$t = (012) + (123) + (235) + (356) - (014) - (145) - (456).$$

*Proof.* Each displayed triple is an increasing induced path by inspection of the specified edge set. The augmentation is $4 - 3 = 1$. In the free abelian group on embeddings of the edge, one has

$$(e_1 - e_0)_*((012) + (123) + (235) + (356)) = [56] - [01],$$
$$(e_1 - e_0)_*((014) + (145) + (456)) = [56] - [01].$$

Indeed the two telescoping sums are respectively

$$([12] - [01]) + ([23] - [12]) + ([35] - [23]) + ([56] - [35]),$$
$$([14] - [01]) + ([45] - [14]) + ([56] - [45]).$$

Subtracting gives $te_0 = te_1$. Precomposition with each of the two vertex maps into the edge also gives $tv_0 = tv_1 = tv_2$. □

**Finite computations and their limits.** The supplied script `finite_ordered_graphs.py` enumerates every graph on the fixed ordered vertex set $\{0, \ldots, n-1\}$ for $1 \le n \le 6$. It uses exact rational and finite-field arithmetic to compare the rank of $D$ with the rank after adjoining augmentation as an extra row. The counts of targets admitting affine equalizers are zero for both families over all three fields when $n \le 5$. For $n = 6$, among all $2^{15} = 32768$ ordered graphs, the counts are

| | $\mathbb{Q}$ | $\mathbb{F}_2$ | $\mathbb{F}_3$ |
|---|---|---|---|
| three vertex maps | 42 | 34 | 42 |
| two edge maps | 8 | 0 | 8 |

The first vertex certificate has ranks $5, 6$ over all three fields; the displayed seven-vertex edge certificate has ranks $6, 7$ over all three. A separate script, `finite_ordered_graphs_verify.py`, rebuilds the graph maps directly, checks all reported coefficients and marginals, and independently computes these ranks using `DomainMatrix` over the three exact coefficient domains. The exhaustive computation shows that the smallest target size for the vertex family

over $\mathbb{Z}$ is 6, and for the edge family over $\mathbb{Z}$ is 7: an integral certificate reduces modulo 2, so the enumerated $\mathbb{F}_2$ obstructions give the latter lower bound. These minimal-size assertions are certified finite computations, not conceptual lower-bound proofs.

A separately recorded sample of 20000 seven-vertex graphs, using seed 20261113, is expressly not an exhaustive seven-vertex search. None of these finite searches by itself proves universal exactness or totality. The universal conclusion in Corollary A.8 instead follows from the circle-group argument and the opened structural Ramsey theorem.

### A.6 A second homogeneous candidate: independent linear and cyclic orders

**Proposition A.11** (Refuted: the linear–cyclic order candidate)**.** *Let $C$ be a skeleton of the category of nonempty finite sets with an independent linear order and cyclic order, whose arrows are injections preserving both orders. Cyclic orders on one or two points are taken to be unique. This small category is neither total nor exact. More precisely, $BC$ is simply connected and $H_2(BC;\mathbb{Z}) \cong \mathbb{Z}$, and there is a five-arrow parallel family with no affine equalizer over any nonzero commutative ring. No assertion that $C$ is quasi-filtered is made.*

*Proof.* Take $p$ to have two points and $q$ to have linear order $1 < 2 < 3 < 4$ and cyclic order $(1, 3, 2, 4)$. Every increasing pair $ij$ defines an arrow $a_{ij} : p \to q$. For any target $r$, choose a cut in its cyclic order, giving a linear reading of that circle. For an arrow $u : p \to r$, define $\chi(u) \in \{0, 1\}$ to be one if the larger of its two points in the given linear order of $r$ occurs first in the circular reading, and zero otherwise. Extend $\chi$ linearly to $R[C(p, r)] \to R$ for any commutative ring $R$.

For every $g : q \to r$, the circular order induced on its image is $(g1, g3, g2, g4)$. Examining the four possible positions of the restricted cut gives

$$\chi(ga_{12}) + \chi(ga_{23}) - \chi(ga_{13}) = 1,$$
$$\chi(ga_{12}) + \chi(ga_{24}) - \chi(ga_{14}) = 0.$$

These identities can also be read from the negative orientation of the triple 123 and the positive orientation of 124. For $t \in R[C(q, r)]$ annihilating the differences of the five arrows

$$a_{12}, a_{23}, a_{13}, a_{24}, a_{14},$$

write $c$ for the common value $\chi(ta_{ij})$. The first identity gives $c = \mathrm{aug}(t)$ and the second gives $c = 0$. Thus every simultaneous annihilator has augmentation zero, at every target. In particular there is no affine equalizer when $1_R \neq 0$. The column exact sequences of Proposition B.5, with $R$ in place of $\mathbb{Z}$, give $T_1(Q_a) \cong R$ directly, without assuming condition I. In the integral case the smaller four-arrow family $a_{12}, a_{23}, a_{34}, a_{14}$ already has an obstruction:

$$\chi(ga_{12}) + \chi(ga_{23}) + \chi(ga_{34}) - \chi(ga_{14}) = 1$$

forces $2c = 1$ for an affine equalizer.

To check totality independently, let $X_n$ be the set of cyclic orders on the linearly ordered set $\{0, \ldots, n\}$, with face maps given by deletion followed by increasing relabeling. This is a semisimplicial set. Its category of simplices is exactly $C$: an increasing injection is an arrow precisely when the cyclic order on its source is the restriction of that on its target. The usual simplex-category comparison gives $BC \simeq |X|$. One may obtain this comparison by realizing the diagram of standard simplices over the simplex category. Its objectwise collapse to points is a homotopy-colimit equivalence, and its augmentation to $|X|$ is the simplex-category subdivision comparison. Equivalently, barycentric subdivision with the face occurrences retained gives the nerve of that category. This is the same semisimplicial comparison used in the finite sphere test, and does not identify the unrelated full cyclic category with $C$.

There is one zero-simplex, one one-simplex, two two-simplices, and six three-simplices. The cellular differentials in these degrees, with the alternating deletion convention, are

$$d_1 = 0, \qquad d_2 = (1,1) : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad d_3 = 0 : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^2.$$

For $d_2$, all three faces of either cyclic triangle are the unique edge, with coefficient sum $1 - 1 + 1 = 1$. For $d_3$, the six cyclic orders fixing the initial point have alternating triangle-face sums zero in each of the two orientation classes. The program `cyclic_linear_cuts.py` prints and checks all six columns, as well as all four cut identities above. It follows that $H_2(|X|;\mathbb{Z}) = \ker d_2 \cong \mathbb{Z}$. Its one-skeleton is a circle, and either two-simplex attaches with degree one on that circle. Hence $\pi_1(|X|) = 0$. Already the constant singleton $C$-set has element category $C$, whose nonzero second homology excludes totality. □

## A.7 A compact inverse-limit reformulation of a stronger Ramsey condition

This subsection characterizes a compact inverse-limit condition which implies exactness. It is strictly stronger than the totality assumption needed in the main question: the counterexample below proves that totality does not imply this condition.

**Definition A.12** (Proved: agreement of the two Ramsey formulations for finite hom-sets)**.** A small category $A$ has the finite-pattern morphism Ramsey property if, for every $p, q \in A$, finite subset $S \subset A(p,q)$, and positive integer $c$, there is $r \in A$ such that every coloring

$$\chi : A(p,r) \longrightarrow \{1, \dots, c\}$$

admits $g : q \to r$ for which $\chi$ is constant on $gS$. When all hom-sets are finite, this is equivalent to the morphism Ramsey property in Theorem A.6: take $S = A(p,q)$.

**Theorem A.13** (Proved: compact inverse limits and finite-pattern Ramsey)**.** *Let $A$ be a small nonempty category with common successors. The following are equivalent.*

1. *Every contravariant diagram $X : A^{\mathrm{op}} \to \mathrm{CompHaus}$ of nonempty compact Hausdorff spaces has a nonempty strict inverse limit.*
2. *$A$ has the finite-pattern morphism Ramsey property.*

*No bound on the hom-sets is assumed.*

*Proof.* Suppose (2) fails for $p, q, S, c$. Necessarily $S$ is nonempty. For each $a$, let $X(a)$ consist of the colorings of $A(p,a)$ that are nonconstant on $gS$ for every $g : q \to a$. This is a closed subset of the compact product $\{1, \dots, c\}^{A(p,a)}$: each forbidden monochromaticity event concerns finitely many coordinates and is clopen. The failure of (2) says that every $X(a)$ is nonempty. Precomposition with $A(p,u)$, for $u : a \to b$, defines a continuous map $X(b) \to X(a)$, since a monochromatic $gS$ after restriction would be a monochromatic $(ug)S$ before restriction. Thus $X$ is a diagram as in (1). A compatible family $(\chi_a)_a$ would satisfy

$$\chi_a(h) = \chi_p(1_p) \qquad (h : p \to a).$$

In particular $\chi_q$ would be constant on $S$, contrary to its defining condition for $g = 1_q$. This proves (1) ⇒ (2).

For the converse, first observe that a Ramsey target has a finite witness set. If $r$ works for $p, q, S, c$, the clopen sets

$$U_g = \{\chi : \chi \text{ is constant on } gS\}, \qquad g : q \to r,$$

cover the compact coloring space. A finite subcover gives a finite $G \subset A(q,r)$ such that some member of $G$ witnesses every coloring.

It follows that finitely many finite-pattern requirements can be satisfied simultaneously. To spell this out, let $S_j \subset A(p_j,q)$ and $c_j$ be given for $1 \le j \le m$. Put $r_0 = q$. At stage $j$ apply (2) to the finite pattern

$$S_j^{(j-1)} = \bigcup_{h_i \in G_i,\, 1 \le i < j} h_{j-1} \cdots h_1 S_j \subset A(p_j, r_{j-1}),$$

where $S_1^{(0)} = S_1$. Obtain a target $r_j$ and a finite witness set $G_j \subset A(r_{j-1}, r_j)$. Given colorings at $r_m$, choose $h_m \in G_m$ first, then $h_{m-1} \in G_{m-1}$ for the appropriate pullback coloring, and continue backwards. The composite $g = h_m \cdots h_1 : q \to r_m$ makes every $gS_j$ monochromatic. Indeed the prefix $h_{j-1} \cdots h_1 S_j$ is contained in $S_j^{(j-1)}$, so the choice at stage $j$ controls every possible subsequent choice of that prefix.

Now let $X$ be as in (1). Its limit is cut out in the compact product $\prod_a X(a)$ by the closed equations

$$x_a = X(f)x_b \qquad (f : a \to b).$$

Fix finitely many such equations, indexed by a finite set $E$ of arrows. For each source $a$ in $E$, fix a neighborhood $U_a$ of the diagonal in $X(a) \times X(a)$. Choose a finite cover of $X(a)$ by sets whose Cartesian squares lie in $U_a$, and assign each point to one containing cover member. Such a cover exists by the compatible uniformity of a compact Hausdorff space. This gives a finite coloring of $X(a)$; continuity of this coloring is not required.

Choose a common successor $q$ of the finitely many objects involved and arrows $j_a : a \to q$. For each source $a$ use the finite pattern

$$S_a = \{j_a\} \cup \{j_b f : (f : a \to b) \in E\}.$$

Apply the simultaneous Ramsey assertion with the cover color counts to obtain $r$. Choose $z \in X(r)$ and color $h : a \to r$ by the cover color of $X(h)z$. The resulting $g : q \to r$ makes the assignments $x_a = X(gj_a)z$ satisfy

$$(x_a, X(f)x_b) \in U_a \qquad ((f : a \to b) \in E).$$

Choose arbitrary values for the other coordinates. Thus every finite set of equations can be satisfied to arbitrary uniform accuracy. Use closed entourages for each equation: their inverse images in the product have the finite intersection property, while the intersection over all such entourages is exactly the equality condition. Compactness therefore supplies a point satisfying every equation. This proves (1). □

**Corollary A.14** (Proved: the compact assertion implies integral exactness)**.** *A nonempty small category with common successors satisfying either condition of Theorem A.13 has exact abelian colimits.*

*Proof.* The proof of Theorem A.6 uses monochromaticity only on the specified finite family $\{a_0, \dots, a_k\}$, so its Ramsey input can be replaced by Definition A.12. For arbitrary hom-sets, replace its finite-rank Smith normal form step by extension of a character into $\mathbb{R}/\mathbb{Z}$. For completeness, that extension needs only divisibility: choose a maximal partial extension on a subgroup by Zorn's lemma. If its domain is not the whole group, adjoin an element $x$. If a least positive $n$ has $nx$ in the old domain, choose an $n$th root of its assigned value; if no such $n$ exists, choose any value for $x$. This extends the character in either case, a contradiction. The circle argument then gives an integral affine equalizer for every finite parallel family.

To see directly that these equalizers suffice, two parallel affine arrows have finite supports with augmentation one. Equalizing all ordinary arrows in the union of those supports by

a further affine arrow makes the two affine composites equal. Common successors and this property make $\mathrm{Aff}_{\mathbb{Z}}(A)$ filtered. The exactness criterion of Isbell and Mitchell (1976, Theorem 3.1 and Proposition 3.3) applies. □

**Proposition A.15** (Proved: uniformly bounded finite fibers)**.** *Let $A$ be nonempty, connected, and quasi-filtered. Every contravariant diagram $X : A^{\mathrm{op}} \to \mathrm{Set}$ of nonempty finite sets with $|X(a)| \le N$ for one finite integer $N$ has a nonempty inverse limit.*

*Proof.* Let $A_X$ have the objects of $A$, with parallel arrows identified precisely when they induce the same function $X(b) \to X(a)$. Functoriality makes this an equivalence relation compatible with composition. Conditions I and IV descend from $A$ by choosing representatives, applying their required square or affine equality in $A$, and then passing to the quotient. Hence $A_X$ is quasi-filtered. For each fixed $a$ one has

$$|A_X(a,b)| \le |X(a)|^{|X(b)|} \le |X(a)|^N \qquad \text{for every } b.$$

By Isbell and Mitchell (1976, Proposition 4.5), every finite parallel family out of $a$ is equalized by an ordinary arrow in $A_X$. Connectedness and I give common successors, so $A_X$ is filtered.

The diagram $X$ factors through $A_X^{\mathrm{op}}$, with the same strict limit. Every finite set of compatibility equations for this diagram has a solution: filteredness gives a cocone on the finite diagram in question, and any point at its vertex supplies that solution. The compatibility conditions are closed in the compact product of the finite discrete values. They have the finite intersection property, so their intersection is nonempty. □

**Example A.16** (Proved: coefficient pinning can destroy a colimit relation)**.** Let $A = \Delta_{\mathrm{inj}}$, which is exact and total, and let $B(n)$ be the free Boolean algebra on the generators $x_0, \dots, x_{n-1}$, also regarded as its associated Boolean ring with addition given by symmetric difference. An increasing injection $f : n \to m$ acts by $x_i \mapsto x_{f(i)}$. These are injective unital Boolean-algebra homomorphisms: the corresponding restriction maps on the sets of Boolean valuations are surjective.

In $B(2)$ the elements $x_0, x_1$ have the same class in $\mathrm{colim}_A^{\mathrm{Set}} UB$, since the two maps $1 \to 2$ carry the sole generator to them. Nevertheless, with $c = x_0$, the pairs $(c, x_0)$ and $(c, x_1)$ lie in different components of $\mathrm{El}_A(UB \times UB)$. Equality of the two coordinates is preserved and reflected by every structure map; it holds for the first pair and fails for the second.

The failure persists for the abelian colimit after pinning $c$. Consider the component $E$ of $\mathrm{El}_A UB$ containing $(2, x_0)$. Its objects are exactly the pairs $(n, x_j)$. Indeed renaming variables sends a Boolean expression to a generator only if that expression was itself a generator, as can be checked on valuations. On $E$, evaluation at the valuation which sends $x_j$ to 1 and all other generators to 0 defines a natural homomorphism of additive groups

$$B(n) \longrightarrow \mathbb{F}_2.$$

For an arrow $f : (n, x_j) \to (m, x_{f(j)})$, the target valuation restricts to the specified source valuation, proving naturality. At $(2, x_0)$ this map takes $x_0$ to 1 and $x_1$ to 0. Consequently $x_0$ and $x_1$ remain distinct in the abelian colimit over $E$.

Thus moving to a category of elements in order to hold a multiplicative coefficient fixed need not preserve even a set-colimit relation available before that move. In a proposed Boolean-algebra argument, one must prove the needed lifted tuple relations; totality alone cannot be cited for their existence. This example does not rule out a different homological argument exploiting the full Boolean identity.

**Corollary A.17** (Refuted: the proposed compact strengthening of totality)**.** *Totality does not imply either equivalent compact-limit or finite-pattern Ramsey condition of Theorem A.13. In particular there is a countable total category with common successors admitting a contravariant diagram of nonempty compact Hausdorff spaces whose inverse limit is empty.*

*Proof.* Take the ordered predimension category $\mathcal{C}_\delta$ of Theorem A.34. It is countable, total, and has common successors, but is not exact. Either condition of Theorem A.13 would make it exact by Proposition A.14, a contradiction. The compact condition therefore fails, and its equivalent Ramsey condition fails as well. Proposition A.15 remains valid: a finite-valued obstruction in this quasi-filtered category must have unbounded fiber cardinalities. □

## A.8 Totality does not give compact convex inverse limits

The following result rules out a possible averaging shortcut in the AKR persistence problem and its proposed strengthening under totality. The same positive-approximation obstruction holds in every repaired AKR category and in the total predimension category $\mathcal{C}_\delta$.

**Proposition A.18** (Proved: a uniform obstruction to positive approximation)**.** *Let $A$ be either a repaired AKR category $K^*$ covered by Proposition C.4, or the ordered predimension category $\mathcal{C}_\delta$. For every object $X$ and every finitely supported family of nonnegative real coefficients $(t_g)_{g:W\to X}$ with $\sum_g t_g = 1$, put*

$$\mu_i(v) = \sum_{g:\, gv_i = v} t_g \quad (v \in A(V,X),\ i = 1,2,3).$$

*Then*

$$\|\mu_2 - \mu_1\|_1 + \|\mu_3 - \mu_1\|_1 \geq 1.$$

*In particular the initial triple has no arbitrarily accurate simultaneous equalizers obtained from probability combinations of ordinary arrows, even as the target varies.*

*Proof.* For either category, every nonempty finite set $S$ of $W$-arrows into one object satisfies

$$\left|\bigcup_{g\in S}\{gv_1, gv_2, gv_3\}\right| > \tfrac{3}{2}|S|.$$

For $K^*$ this is the proved counting invariant from Proposition C.4. For $\mathcal{C}_\delta$, the arrows in $S$ have distinct hyperedge images, since an increasing three-point embedding is determined by its image. If $U$ is their vertex union, positive predimension gives

$$2|U| > 3|E_X(U)| \geq 3|S|.$$

Every point in this union is represented by a composite $gv_i$, even though other raw vertices of $X$ need not be $d$-closed. In both cases distinct underlying points give distinct point maps, since the category is concrete with singleton $V$. For a point map $v$, set

$$w(v) = \max\big(\{t_g : gv_i = v \text{ for some } i\} \cup \{0\}\big).$$

Apply the counting invariant to the finite threshold sets $S_s = \{g : t_g > s\}$. Integrating the elementary identities for the lengths of the intervals $0 < s < t_g$ and $0 < s < w(v)$ gives

$$\sum_v w(v) = \int_0^\infty \left|\bigcup_{g\in S_s} g(W)\right| ds \geq \frac{3}{2}\int_0^\infty |S_s|\, ds = \frac{3}{2}.$$

All sums and integrals reduce to finite step functions on the support of $t$. Nonnegativity gives $w(v) \leq \max_i \mu_i(v)$. Moreover,

$$\max_i \mu_i(v) \leq \mu_1(v) + (\mu_2(v) - \mu_1(v))_+ + (\mu_3(v) - \mu_1(v))_+.$$

Every $\mu_i$ has total mass one. Thus the sums of the positive and negative parts of $\mu_i - \mu_1$ agree, and

$$\frac{3}{2} \le \sum_v \max_i \mu_i(v) \le 1 + \frac{1}{2}(\|\mu_2 - \mu_1\|_1 + \|\mu_3 - \mu_1\|_1).$$

Rearrangement proves the bound. The integrated inequality is in fact strict for each finite probability support; only its uniform lower bound is used. No positive lower bound on $\inf_m a_m - 3/2$ is required. □

**Theorem A.19** (Refuted: totality guarantees compact convex limits)**.** *There is a nonempty connected quasi-filtered small category $A$ and a contravariant diagram of nonempty finite-dimensional compact convex sets, with continuous affine transition maps, whose strict inverse limit is empty. One may take $A = K^*$ as above, or the countable category $A = \mathcal{C}_\delta$, which is total by Theorem A.34. Thus the proposed implication fails even under totality.*

*Proof.* For either choice of $A$, write $E_X = A(W, X)$ and $P_X = A(V, X)$. Both sets are finite because the objects are finite sets and all arrows are injective functions. In the finite-dimensional real vector space $\mathbb{R}[P_X]^2$ put

$$D_X[g] = ([gv_2] - [gv_1],\ [gv_3] - [gv_1]) \qquad (g \in E_X).$$

If $E_X$ is nonempty, Proposition A.18 says that the compact convex hull $C_X = \operatorname{conv}\{D_X[g] : g \in E_X\}$ does not meet the open unit ball for the $\ell^1$ norm. Finite-dimensional convex separation gives a nonzero linear functional $\lambda$ with

$$\inf_{z \in C_X} \lambda(z) \ge \sup_{\|b\|_1 < 1} \lambda(b) = \|\lambda\|_\infty.$$

Here one separates a convex set from a disjoint open convex set; the last equality is the elementary duality of the two specified norms. Dividing by $\|\lambda\|_\infty$ gives coefficient functions $\phi_1, \phi_2 : P_X \to [-1, 1]$ such that

$$\phi_1(gv_2) - \phi_1(gv_1) + \phi_2(gv_3) - \phi_2(gv_1) \ge 1 \qquad (g \in E_X). \tag{3}$$

Define $\mathcal{C}(X)$ to be the set of all pairs $(\phi_1, \phi_2) \in [-1, 1]^{2P_X}$ satisfying (3). It is a closed convex subset of a finite-dimensional cube and is nonempty by the preceding separation. If $E_X$ is empty, the conditions are vacuous, so use the whole cube, which is nonempty also when $P_X$ is empty.

For an arrow $u : X \to Y$, define

$$\mathcal{C}(u) : \mathcal{C}(Y) \longrightarrow \mathcal{C}(X), \qquad (\phi_i)_i \longmapsto \big(v \longmapsto \phi_i(uv)\big)_i.$$

This is continuous and affine. It preserves the inequalities because the inequality for $g : W \to X$ after restriction is the inequality for $ug : W \to Y$ before restriction. Restrictions compose in the required order, so $\mathcal{C}$ is the asserted contravariant diagram.

Suppose a compatible family $(\phi_{i,X})_{X,i}$ existed. For each $v : V \to X$, compatibility along $v$ evaluated at $1_V$ would give

$$\phi_{i,X}(v) = \phi_{i,V}(1_V).$$

Hence each coefficient function is constant on all point maps, with the same constant across objects. At $X = W$, the inequality for $g = 1_W$ would then read $0 \ge 1$, a contradiction. Thus the limit is empty. Quasi-filteredness and connected nonemptiness of $K^*$ are the established properties of the repaired AKR construction; the corresponding properties and totality of $\mathcal{C}_\delta$ are proved in Theorem A.34. □

**Scope for the persistence problem.** The theorem refutes the general compact convex inverse-limit lemma suggested as a way to extend locally bounded cycle functionals, including its proposed totality-specific version. A particular diagram of such functionals could still have a compatible point for additional reasons. Neither quasi-filteredness nor totality alone supplies the required compatible bounded potentials. The construction does not assume the stronger compact Hausdorff condition of Theorem A.13.

## A.9 Uniformly finite coefficient diagrams

**Proposition A.20** (Proved: bounded finite coefficients)**.** *Let $A$ be a nonempty connected small quasi-filtered category, and let $M : A \to \mathrm{Ab}$ satisfy $|M(a)| \le N$ for every object $a$, for one finite integer $N$. Then $T_1^A(M) = 0$. If $A$ is total, then $T_i^A(M) = 0$ for every $i > 0$.*

*Consequently, a prime-field column module $Q_a$ with nonzero first derived colimit over a quasi-filtered category must have unbounded dimensions as the object varies.*

*Proof.* Define $B$ to have the same objects as $A$ and the typed hom-sets

$$B(a,b) = \mathrm{im}\big(A(a,b) \longrightarrow \mathrm{Hom}_{\mathrm{Ab}}(M(a), M(b))\big).$$

Composition is composition of homomorphisms. The quotient functor $u : A \to B$ is the identity on objects and surjective on every hom-set, and $M = u^*\overline{M}$ for a functor $\overline{M} : B \to \mathrm{Ab}$. Conditions I and IV descend along $u$: lift an ordinary span or parallel pair, solve the corresponding problem in $A$, and apply $u$. Furthermore

$$|B(a,b)| \le N^{|M(a)|} \le N^N.$$

Thus Isbell and Mitchell (1976, Proposition 4.5) supplies an ordinary post-equalizer for every finite parallel family in $B$. Connectedness and I give common successors, so $B$ is filtered.

For $b \in B$, the comma category $b \downarrow u$ is the category of elements of the covariant $A$-set $B(b, u(-))$. Its object $(b, \mathrm{id}_b)$ is weakly initial, by hom-surjectivity of $u$. Hence this comma category is nonempty and connected. It is simply connected under quasi-filteredness, by the audited element-category form of I and IV; under totality it is weakly contractible.

For completeness, the homological comparison uses these comma homology groups, not ordinary finality alone. Choose the projective right bar resolution $P_* \to \Delta\mathbb{Z}$ on $A$, with

$$P_n = \bigoplus_{a_0 \to \cdots \to a_n} \mathbb{Z}[A(-, a_0)], \qquad K_* = \mathrm{Lan}_{u^{\mathrm{op}}} P_*.$$

Left Kan extension sends each right representable to the corresponding right representable on $B$. Thus $K_*$ is a nonnegative complex of projective right $\mathbb{Z}B$-modules. At $b$ its degree-$n$ group is

$$K_n(b) = \bigoplus_{a_0 \to \cdots \to a_n} \mathbb{Z}[B(b, u(a_0))],$$

with the nerve differential of $b \downarrow u$. Consequently $H_0K_* \cong \Delta\mathbb{Z}$ naturally and $H_1K_* = 0$; under totality, $H_jK_* = 0$ for every $j > 0$.

Coend adjunction identifies the two complexes

$$K_* \otimes_{\mathbb{Z}B} \overline{M} \cong P_* \otimes_{\mathbb{Z}A} u^*\overline{M}.$$

The first-quadrant hyper-Tor spectral sequence is therefore

$$E^2_{p,q} = \mathrm{Tor}_p^{\mathbb{Z}B}(H_qK_*, \overline{M}) \implies T^A_{p+q}(M).$$

Since $B$ is filtered, $\Delta\mathbb{Z}$ is flat on $B$. Both terms of total degree one vanish: $(p,q) = (1,0)$ by flatness and $(0,1)$ by $H_1K_* = 0$. This proves the first assertion. Under totality the

spectral sequence has only its $q = 0$ row, and flatness proves all positive-degree vanishings. Uniform finite dimension over $\mathbb{F}_p$ implies the required uniform cardinal bound, proving the consequence. □

This does not cover arbitrary finite-valued diagrams with unbounded cardinalities, or arbitrary prime-field modules. In particular, a finite set of coefficient vectors need not generate a uniformly bounded subdiagram. Passing to a category of elements that pins a tuple also need not preserve the relations of an original bar cycle; Proposition A.16 gives the relevant warning about such a change of category.

## A.10 A full ordered predimension category

For a finite simple 3-uniform hypergraph $X$ and a vertex subset $U$, put

$$\delta_X(U) = 2|U| - 3|E_X(U)|,$$

where $E_X(U)$ consists of the hyperedges contained in $U$. Let $\mathcal{C}_\delta$ be a small skeleton of the category of nonempty finite linearly ordered such hypergraphs satisfying $\delta_X(U) > 0$ for every nonempty $U$. Its arrows are increasing induced embeddings with *d-closed* image, where

$$A \leq_d B \quad \Longleftrightarrow \quad \delta_B(U) > \delta_B(A) \quad \text{for every } A \subsetneq U \subseteq B. \tag{DC}$$

Isomorphisms are allowed. Including the empty object instead makes no difference to the assertions below. The strict closure terminology is used for sparse graphs in Evans et al. (2019, Definition 4.4); the weighted hypergraph assertions here will be proved directly.

**Lemma A.21** (Proved: closure calculus and free amalgamation)**.** *The relation $\leq_d$ is preserved by restriction to subsets, is transitive, and is closed under finite intersections inside an ambient object. Free amalgams of d-closed embeddings are positive-predimension hypergraphs, and both factors are d-closed in the amalgam. The orders can be amalgamated. In particular, $\mathcal{C}_\delta$ satisfies condition I and has common successors.*

*Proof.* Predimension is submodular:

$$\delta(U) + \delta(V) \geq \delta(U \cup V) + \delta(U \cap V). \tag{SM}$$

The vertex term is modular, and the edge count is supermodular. If $A \leq_d B$, $X \subseteq B$, and $A \cap X \subsetneq U \subseteq X$, then $U \cap A = A \cap X$ and $U \cup A \supsetneq A$. Hence

$$\delta(U) - \delta(A \cap X) \geq \delta(U \cup A) - \delta(A) > 0.$$

Thus $A \cap X \leq_d X$. For $A \leq_d B \leq_d C$ and $A \subsetneq U \subseteq C$, the case $U \subseteq B$ is immediate. Otherwise restriction gives $U \cap B \leq_d U$ and

$$\delta(U) > \delta(U \cap B) \geq \delta(A).$$

This proves transitivity. Intersections follow by restriction and transitivity.

Let $A \leq_d B, C$ and let $D = B \cup_A C$ be the free hypergraph amalgam. For every $U \subseteq D$,

$$\delta_D(U) = \delta_B(U \cap B) + \delta_C(U \cap C) - \delta_A(U \cap A).$$

If $U$ meets $C \setminus A$, restriction makes the difference of the last two terms positive. The first is nonnegative. If $U$ does not meet $C \setminus A$, it is a subset of $B$. Therefore every nonempty $U$ has positive predimension. If $B \subsetneq U \subseteq D$, then

$$\delta_D(U) - \delta(B) = \delta_C(U \cap C) - \delta(A) > 0.$$

Thus $B \leq_d D$, and symmetrically $C \leq_d D$. Merge the two linear orders separately in each interval determined by their common ordered subset $A$. This extends both orders. Amalgamation gives I; the same construction over the empty structure gives common successors, even if the empty structure is not an object. □

**Lemma A.22** (Proved: the long-necklace predimension estimate)**.** *Let $f, g : S \rightrightarrows T$ be arrows of $\mathcal{C}_\delta$. Choose $n \geq 2$ bounding the vertex lengths of the nonfixed paths of the partial bijection $f(x) \mapsto g(x)$, and put $w = 2n - 2$. For $k \geq n$, let $Z$ be the covered, actual-length repaired point necklace of Lemma C.32, of length $s = 4k + 1$, with copies $\alpha_i : T \to Z$. Give $Z$ precisely the hyperedges that occur in those copies. If*

$$s > w\big(\delta(T) + 2|S|\big), \tag{PD}$$

*then $Z$ is a positive-predimension hypergraph and every $\alpha_i$ is an induced embedding with $d$-closed image.*

*Proof.* We use the exact consecutive overlaps and visibility intervals from Lemma C.32. A fixed point has full-circle visibility. Every other point, including a private point, has a proper cyclic interval of at most $w$ indices. Since $w \leq 2k - 2$, every proper interval is shorter than half the index circle and contains the unique shortest index path between any two of its indices.

First, each copy is induced. If an edge from copy $j$ has all its vertices in copy $i$, the shortest index path from $j$ to $i$ lies in all three vertex visibility sets. At a consecutive overlap the edge is contained in the indicated $S$-image. Because $f$ and $g$ are induced embeddings, it comes from an edge of $S$ and propagates to the next copy. It therefore occurs in copy $i$. This also shows that the visibility of an edge is exactly the intersection of the three vertex visibility sets. That intersection is a connected proper interval, or the full circle when all three vertices are fixed. All edge and point identifications are consequently generated by the neighboring overlaps.

Cut one overlap of the necklace. Its linear amalgam $L$ is obtained by successive free amalgamations over $S$. By Lemma A.21, it is positive-predimension, and every $T$-copy is $d$-closed in it. Restoring the omitted overlap gives a quotient $q : L \to Z$ which identifies at most $|S|$ vertex pairs. If $U \subseteq Z$ and $\widetilde{U} = q^{-1}(U)$, every edge of $Z[U]$ lifts to an edge of $L[\widetilde{U}]$. Hence

$$\delta_Z(U) \geq \delta_L(\widetilde{U}) - 2|S|. \tag{1}$$

Indeed, if $d_v, d_e$ are the numbers of vertex and edge duplications removed by $q$ on these induced substructures, then the exact difference is $2d_v - 3d_e$, with $0 \leq d_v \leq |S|$ and $d_e \geq 0$.

Suppose first that the union of the proper visibility intervals of the vertices in $U$ misses an index $r$. Cut the overlap from $r - 1$ to $r$. No proper vertex fiber of $U$ is cut into two components, and a full-circle fiber becomes a connected path. All such vertices therefore have unique lifts. Edges have unique lifts as well, either by the connected visibility description or by simplicity of the hypergraph once their three vertices lift uniquely. Thus $L[\widetilde{U}] \cong Z[U]$. This proves $\delta_Z(U) > 0$ for nonempty $U$. If $U$ properly contains a copy $\alpha_i(T)$, its lift properly contains that same $T$-copy, so $\delta_Z(U) > \delta(T)$.

It remains to consider the case where those proper visibility intervals cover all $s$ indices. Cut any overlap and number the linear copies $0, \ldots, s-1$. Build $L[\widetilde{U}]$ by successively adjoining its intersection $U_j$ with copy $T_j$. For $j > 0$, the intersection of $T_j$ with the preceding amalgam is exactly its incoming $S$-image $A_j$. Its predimension contribution is therefore

$$\delta(U_j) - \delta(U_j \cap A_j).$$

This is zero if $U_j \subseteq A_j$. Otherwise submodularity and $A_j \leq_d T_j$ give

$$\delta(U_j) - \delta(U_j \cap A_j) \geq \delta(U_j \cup A_j) - \delta(A_j) \geq 1. \tag{2}$$

The initial contribution $\delta(U_0)$ is at least one whenever $U_0$ is nonempty.

Let $J$ be the set of indices at which at least one nonfixed vertex of $\widetilde{U}$ first appears. Each such vertex has a linear visibility interval of at most $w$ indices. Consequently all the nonfixed vertices first appearing at $j \in J$ are visible only among $j, \dots, j+w-1$. Every copy index contains a nonfixed vertex of $U$ by the case assumption, and its lift has some first-appearance index in $J$. The intervals $[j, j+w-1]$, for $j \in J$, therefore cover all $s$ indices. Thus $|J| \geq \lceil s/w \rceil$. Every such stage contributes at least one in (2), or in the initial contribution, and all other stages contribute nonnegatively. We obtain

$$\delta_L(\widetilde{U}) \geq \lceil s/w \rceil, \qquad \delta_Z(U) \geq \lceil s/w \rceil - 2|S| > \delta(T) > 0,$$

using (1) and (PD). This counts first-appearance indices, not individual new vertices; no bound on $|T|$ is needed. The two cases prove positivity for every nonempty subset and strict relative positivity over every copy of $T$. □

**Proposition A.23** (Proved: quasi-filteredness of the full category)**.** *The category $\mathcal{C}_\delta$ is quasi-filtered. The same is true of its full subcategory consisting of objects in which every point and every hyperedge is d-closed.*

*Proof.* For $f, g : S \rightrightarrows T$, choose $k$ satisfying Lemma A.22 and also exceeding the eventual ordering threshold of Proposition C.33. The latter proposition orders the very same point target $Z$ so that every $\alpha_i$ is increasing. Hence every $\alpha_i$ is an arrow in $\mathcal{C}_\delta$.

In the integral linearization, set

$$t = \sum_{i=0}^{2k} \alpha_i - \sum_{i=2k+1}^{4k} \alpha_i.$$

It has augmentation one. The exact necklace equations give

$$tg = \sum_{i=1}^{2k} \alpha_i f - \sum_{i=2k+1}^{4k-1} \alpha_i f = tf - \alpha_0 f + \alpha_{4k} f = tf.$$

This is IV. Condition I and connectedness were proved in Lemma A.21.

For the asserted full subcategory, free amalgams and the necklace targets are covered by old copies. Every new point or edge is $d$-closed in some old copy, hence in the target by transitivity. Both constructions therefore remain in this subcategory. □

**Proposition A.24** (Proved: the initial triple remains unequalized)**.** *Let $V$ be one point, let $W$ be the ordered three-point single-edge hypergraph, and let $v_1, v_2, v_3 : V \to W$ be its three point arrows. In either category of Proposition A.23, this triple has no affine post-equalizer over any nonzero unital ring $R$. For*

$$D : R[A(W,-)] \longrightarrow R[A(V,-)]^2, \qquad D(g) = (gv_2 - gv_1, gv_3 - gv_1), \qquad Q = \operatorname{coker} D,$$

*one has $T_1^A(Q) \cong R$. In particular neither category is exact.*

*Proof.* Each $v_i$ is $d$-closed: its predimension is two and its proper extensions inside $W$ have predimension four or three. Consider a nonzero coefficient vector supported on $r$ distinct arrows $g : W \to X$. Their images are $r$ distinct hyperedges, since an increasing three-point embedding is determined by its image. If their union has $v$ vertices, positivity gives

$$2v > 3|E_X(U)| \geq 3r.$$

The $r$ support edges have $3r$ incidences, so some vertex belongs to exactly one support edge. Its coefficient $c \neq 0$ occurs in $D$ as $(-c,-c)$, $(c,0)$, or $(0,c)$ and cannot cancel. Its corresponding $V$-arrow exists by composition $V \to W \to X$. Thus $D$ is objectwise injective over every $R$.

The two displayed representable modules are projective. Their colimits are $R$ and $R^2$, and the colimit of $D$ is zero because each difference of two composites has augmentation zero. The long exact derived-colimit sequence of this projective presentation gives $T_1(Q) \cong R$. Objectwise injectivity of $D$ also directly precludes any nonzero, hence any augmentation-one, simultaneous equalizer of the three arrows. □

**Scope and the subsequent totality proof.** The preceding arguments establish quasi-filteredness and the all-ring triple obstruction. The separate finite-support and all-action arguments below prove totality of both categories in Theorem A.34, giving the stated negative answer to A.1.

In the unrestricted category, a raw vertex or hyperedge need not be $d$-closed, so representable $V$- and $W$-actions cannot silently be identified with all vertices and hyperedges. The preceding argument uses only actual arrows. Also, the full category has additional old-target arrows and additional amalgams. The selected AKR class does not automatically survive upon passing to it.

## A.11 The binary cycle obstruction vanishes in the full category

We retain either category $A$ of Proposition A.23. Let $G_X$ have vertices $A(V,X)$ and an edge $e : W \to X$ directed from $ev_1$ to $ev_2$. Put

$$K(X) = \ker(\mathbb{Z}[A(W,X)] \xrightarrow{v_2 - v_1} \mathbb{Z}[A(V,X)]).$$

Thus $K(X)$ is the integral cycle group of $G_X$. The augmentation $\epsilon : K(X) \to \mathbb{Z}$ sums its edge coefficients.

**Lemma A.25** (Proved: adjoining a loose path of length three)**.** *Let $B$ be a positive-predimension hypergraph and let $x, c$ be two distinct vertices of $B$. Adjoin five distinct new vertices $r, s, a, b, d$ and precisely the three new hyperedges*

$$\{x, r, a\}, \qquad \{r, s, b\}, \qquad \{s, c, d\}. \tag{LP}$$

*Then $B$ is $d$-closed in the resulting positive-predimension hypergraph. No assumption that $\{x, c\}$ is $d$-closed is needed.*

*Proof.* For a nonempty subset $N$ of the five new vertices, let $e(N)$ be the number of the three displayed edges whose new vertices all lie in $N$. The relative predimension over the whole of $B$ is $2|N| - 3e(N)$. If $e(N) = 0$, this is positive. If $e(N) = 1$, then $|N| \geq 2$, so it is at least one. If $e(N) = 2$, the union of the new vertices required by those edges has size at least four, so it is at least two. If $e(N) = 3$, all five new vertices occur and it is one. Thus every proper extension of $B$ has strictly larger predimension.

For an arbitrary subset $U$ meeting the new vertices, submodularity gives

$$\delta(U) - \delta(U \cap B) \geq \delta(U \cup B) - \delta(B) > 0.$$

Since the old subsets have nonnegative predimension, this proves positivity of every nonempty subset of the extension. □

**Lemma A.26** (Proved: atoms in a loose cycle)**.** *Let $C$ be a loose hypergraph cycle of length $\ell \geq 4$: it has binary vertices $b_0, \ldots, b_{\ell-1}$, distinct private vertices $p_0, \ldots, p_{\ell-1}$, and edges $\{b_i, b_{i+1}, p_i\}$ with cyclic indices. Then $C$ has positive predimension, and every point and every hyperedge is $d$-closed.*

*Proof.* For a chosen subset $B'$ of the binary vertices, including a private vertex lowers predimension only when both its binary endpoints are included; in that case its net contribution is $-1$. Otherwise its contribution is 2. Thus the minimum at a fixed nonempty $B'$ is

$$2|B'| - e(B'),$$

where $e(B')$ counts the edges of the ordinary cycle induced on $B'$. A proper nonempty $B'$ induces a forest, so this minimum is at least $|B'| + 1$; the whole set gives $\ell$. With no binary vertices, every nonempty subset has predimension at least two. This proves positivity.

A proper extension of a hyperedge either adds a private vertex with its binary endpoints unchanged, increasing predimension by two, or has at least three binary vertices. In the latter case the preceding bound is at least four, including the full-cycle case $\ell \geq 4$. Its predimension therefore exceeds the edge's value three. Every nonempty subset of at least two vertices has predimension at least three: with two or more binary vertices use the same bound, and with at most one binary vertex it has no edge. Thus every proper extension of a point exceeds its predimension two as well. □

**Proposition A.27** (Proved: the binary cycle colimit is exactly its augmentation)**.** *For either category $A$ of Proposition A.23, augmentation induces an isomorphism*

$$\operatorname{colim}_A K \xrightarrow{\epsilon} \mathbb{Z}.$$

*Consequently, for $H = A(V, -)/(ev_1 \sim ev_2)$,*

$$H_2(B \operatorname{El}_A H; \mathbb{Z}) = 0.$$

*In particular, the binary class used to prove nontotality for the selected generated AKR category vanishes upon passage to this full predimension category.*

*Proof.* Fix $z = \sum_e c_e e \in K(X)$, with nonempty finite edge support; the zero cycle needs no construction. Let $B$ be the set of its binary endpoints. These are actual $V$-arrows and hence $d$-closed points of $X$. Adjoin a new centre $c$, and for each $x \in B$ a loose length-three path

$$P_x = (x \longrightarrow r_x \longrightarrow s_x \longrightarrow c)$$

using five new vertices $r_x, s_x, a_x, b_x, d_x$ as in (LP). All these new vertices are distinct for distinct $x$. Call the resulting hypergraph $Z$. Starting from the disjoint union of $X$ and $\{c\}$, repeated applications of Lemma A.25 prove that $Z$ is positive-predimension and $X \leq_d Z$.

For a support edge $e$ write its ordered vertices as $u < v < p$. Its binary endpoints are $u, v$. Let

$$F_e = e \cup P_u \cup P_v \subseteq Z.$$

This is a loose seven-cycle, containing its old edge as a $d$-closed subobject by Lemma A.26. The old edge is also $d$-closed in $X$, since $e$ is an actual $W$-arrow. The induced subhypergraph $X \cup F_e$ is precisely the free amalgam of $X$ and $F_e$ over that edge. Thus $F_e \leq_d X \cup F_e$. Adjoin all remaining paths $P_x$ one at a time. Their endpoints $x, c$ already exist, and Lemma A.25 makes every step $d$-closed. Transitivity gives $F_e \leq_d Z$.

This proof allows $p$ to be a binary endpoint of another support edge. In that case its remaining path joins two existing points of $F_e$ and still has strictly positive relative predimension. This is why paths of length three are used.

Extend the order of $X$ as follows. Place all old points first. Next place the two-point blocks $r_x < s_x$, in increasing order of $x \in B$. Place $c$ after all those blocks. Finally place the three-point blocks $a_x < b_x < d_x$, again in increasing order of $x$. All new hyperedges then have

their private point last, so $P_x$ is an increasing directed binary path. The order induced on every $F_e$ has the same pattern:

$$u < v < p < r_u < s_u < r_v < s_v < c < a_u < b_u < d_u < a_v < b_v < d_v.$$

Hence all $F_e$ are copies, by increasing induced embeddings, of one fixed ordered seven-cycle object $P$. Its distinguished cycle

$$\kappa = e + P_v - P_u$$

has augmentation $1 + 3 - 3 = 1$. Every point and edge of $P$ is $d$-closed by Lemma A.26. The maps $P \cong F_e \hookrightarrow Z$ are therefore arrows in $A$. If $A$ is the point-and-edge-closed subcategory, $Z$ also lies in it: its old atoms are closed by $X \leq_d Z$, and every new atom lies in some $F_e \leq_d Z$, because each $x \in B$ is incident with a support edge.

Let $z_e = e + P_v - P_u \in K(Z)$ be the image of $\kappa$ for edge $e$. The coefficient of $P_x$ in $\sum_e c_e z_e - z$ is the coefficient of $x$ in $\partial z$, which is zero. Thus in the actual cycle group of the one target $Z$,

$$z = \sum_e c_e z_e.$$

Passing to the colimit yields

$$[z] = \left(\sum_e c_e\right)[\kappa] = \epsilon(z)[\kappa].$$

This holds for every cycle at every object. Since $\epsilon(\kappa) = 1$, it proves that augmentation on the colimit is an isomorphism.

Finally, the exact sequence

$$0 \to K \to \mathbb{Z}[A(W,-)] \xrightarrow{v_2 - v_1} \mathbb{Z}[A(V,-)] \to \mathbb{Z}[H] \to 0$$

and projectivity of the two representable modules give

$$H_2(B\,\mathrm{El}_A\,H;\mathbb{Z}) \cong \ker\bigl(\mathrm{colim}_A\,K \xrightarrow{\epsilon} \mathbb{Z}\bigr) = 0.$$

This is the same natural binary comparison as in Proposition C.5. □

**Scope of the binary calculation.** The cone calculation proves vanishing for this one binary action in degree two. Totality requires the separate all-action argument below; the all-ring triple obstruction of Proposition A.24 remains nonzero. The construction also works for cycles with coefficients in any commutative ring, giving the corresponding degree-two vanishing.

## A.12 A functorial contraction of the binary and triple actions

**Proposition A.28** (Proved: the path cone is a functor)**.** *For either category $A$ of Proposition A.23, there is an endofunctor $F : A \to A$ with natural transformations*

$$\eta : \mathrm{id}_A \longrightarrow F, \qquad \gamma : \mathrm{const}_V \longrightarrow F.$$

*The map $\eta_X$ includes $X$, and $\gamma_X$ selects a new centre $c_X$. For every $V$-arrow $x : V \to X$, the two vertices $\eta_X x$ and $\gamma_X$ are joined by a directed binary path of three actual $W$-arrows in $F(X)$.*

*Proof.* Work first with a concrete category of finite structures, with a chosen small set of labels closed under the following finite tagged constructions; passage to a skeleton transports the functor and its natural transformations. Use a tagged disjoint copy for the old vertices as well, so all the following labels are disjoint functorially. For every raw vertex $x$ of $X$, adjoin five tagged points $r_x, s_x, a_x, b_x, d_x$, together with one new centre $c_X$, and the three hyperedges

$$\{x, r_x, a_x\}, \qquad \{r_x, s_x, b_x\}, \qquad \{s_x, c_X, d_x\}.$$

Give $F(X)$ the block order used in Proposition A.27: old points first; then $r_x < s_x$ blocks ordered by $x$; then $c_X$; then $a_x < b_x < d_x$ blocks ordered by $x$. The old object is $d$-closed in $F(X)$ by first adjoining the centre disjointly, then adjoining the paths using Lemma A.25. The centre is $d$-closed in the disjoint union and hence in $F(X)$.

For an arrow $f : X \to Y$, define $F(f)$ on old points by $f$, send $c_X$ to $c_Y$, and send each tagged point indexed by $x$ to the point with the same tag indexed by $f(x)$. This is an increasing induced embedding, and the formulas give functoriality exactly. To prove it $d$-closed, identify $X$ with its image in $Y$. The subhypergraph $Y \cup_X F(X)$ of $F(Y)$ is a free amalgam over the $d$-closed object $X$. Therefore

$$F(X) \leq_d Y \cup_X F(X).$$

Adjoin the remaining paths, indexed by $y \in Y \setminus X$, one at a time. Their two endpoints already exist; the path lemma makes each extension $d$-closed. Thus $F(X) \leq_d F(Y)$. The displayed inclusion and centre maps are natural by their tagged definitions.

It remains to verify that the required paths are paths of actual morphisms, rather than merely raw hyperedges. A loose path has every point and every edge $d$-closed. This follows by the same calculation as Lemma A.26: a chosen set of binary vertices induces a forest, and its minimum predimension after selecting private points is $2|B'| - e(B') \geq |B'| + 1$ when $B'$ is nonempty. In particular, a proper extension of a point has predimension at least three, and a proper extension of an edge has predimension at least four.

Suppose now that $x : V \to X$ is an actual arrow, so $\{x\} \leq_d X$. Let $Q_x$ be its loose path together with its two endpoints $x, c_X$. The union $X \cup_{\{x\}} Q_x$ is a free amalgam over a $d$-closed point. Consequently $Q_x$ is $d$-closed in that union. Adjoining all the other paths by the path lemma proves $Q_x \leq_d F(X)$. All its edges are therefore actual $W$-arrows. Their directions are increasing by the chosen block order.

If $X$ belongs to the point-and-edge-closed subcategory, this last argument applies to every raw $x$. Every new point and edge then lies in one of the $d$-closed paths $Q_x$; the old atoms remain closed by $X \leq_d F(X)$. If the empty object is included, its image is just the one-point object. Thus $F$ preserves that subcategory too. □

**Theorem A.29** (Proved: contractibility of the binary and triple element categories)**.** *Let $A$ be either category of Proposition A.23. The categories of elements of each of the two actions*

$$H_{12} = A(V, -)/(ev_1 \sim ev_2), \qquad H_{123} = A(V, -)/(ev_1 \sim ev_2 \sim ev_3)$$

*have contractible classifying spaces. More generally this holds for every pointwise quotient $H$ of $H_{12}$. In particular all their positive-degree integral homology groups vanish. For the binary cycle functor $K$ of Proposition A.27,*

$$T_i^A(K) = 0 \qquad (i > 0).$$

*Proof.* Use $F, \eta, \gamma$ from Proposition A.28. The natural transformations $\mathrm{id}_A \to F \leftarrow \mathrm{const}_V$ make the identity of $BA$ homotopic to a constant map. The category is nonempty, so $BA$ is contractible.

Let $H_{12} \twoheadrightarrow H$ be the given quotient. Write $h_X \in H(FX)$ for the image of the centre arrow $\gamma_X$. These elements are natural in $X$. Every element $h \in H(X)$ is represented by an actual

point arrow $x : V \to X$. The length-three path from $\eta_X x$ to $\gamma_X$ consists of actual $W$-arrows, so its binary relations give

$$H(\eta_X)(h) = h_X. \tag{CC}$$

This equality also holds in any further quotient, including $H_{123}$.

Let $p : \mathrm{El}_A H \to A$ be the projection, and define

$$J : A \longrightarrow \mathrm{El}_A H, \qquad J(X) = (F(X), h_X), \qquad J(f) = F(f).$$

Naturality of the centre makes $J$ a functor. Equation (CC) says that $\eta_X$ is an arrow

$$(X, h) \longrightarrow (F(X), h_X) = Jp(X, h)$$

in the category of elements. These arrows form a natural transformation $\mathrm{id}_{\mathrm{El}\, H} \to Jp$. After taking nerves, the identity therefore factors up to homotopy through the contractible space $BA$. The element category is nonempty, for example because $H(FV)$ contains its centre class. Its identity is homotopic to a constant map, proving the asserted contractibility.

Finally the binary presentation by two projective representables, split at the image of its middle map, gives dimension shifting

$$T_i^A(K) \cong T_{i+2}^A(\mathbb{Z}[H_{12}]) \cong H_{i+2}(B\,\mathrm{El}_A\, H_{12}; \mathbb{Z}) = 0 \qquad (i > 0).$$

□

**An equivalent homological factorization.** The cone also supplies a natural map

$$\mathbb{Z}[A(W, -)] \longrightarrow F^*K, \qquad e \longmapsto e + P_{ev_2} - P_{ev_1}.$$

Its restriction along $K \hookrightarrow \mathbb{Z}[A(W, -)]$ is exactly $K \xrightarrow{K\eta} F^*K$, because the path coefficients cancel on a cycle. The standard derived-colimit comparison $T_i(F^*K) \to T_i(K)$, induced by the indexing functor $F$, composes with $T_i(K\eta)$ to the identity: the natural transformation $\eta$ gives the usual bar-chain homotopy. For $i > 0$ the factorization through a projective representable therefore gives the same vanishing. The direct nerve contraction above additionally proves the homotopy statement.

**Scope of the cone argument.** The contraction uses an action generated by point arrows, together with the binary hyperedge relation. An arbitrary $A$-set need not be such a quotient. For a general quotient of a representable $A(P, -)$, common fixed substructures of its presenting relations can retain information which the point cone does not remove. The passage to every action is supplied by the separate finite-support argument culminating in Theorem A.34.

## A.13 Alternating amalgams and finite supports

**Lemma A.30** (Proved: a finite alternating bridge)**.** *In either category $A$ of Proposition A.23, let $B, C \leq_d D$ and put $F = B \cap C$. Empty subsets are permitted as formal amalgamation bases. Suppose $d_0, d_1 : D \to E$ are arrows agreeing on $F$. There is an arrow $E \to Z$ and a finite sequence of arrows*

$$a_0, a_1, \ldots, a_{2N} : D \longrightarrow Z,$$

*whose endpoints are the given $d_0, d_1$ followed by $E \to Z$, such that*

$$a_{2j}|_B = a_{2j+1}|_B, \qquad a_{2j+1}|_C = a_{2j+2}|_C \quad (0 \leq j < N).$$

*Proof.* If $D$ is empty, all asserted equalities are automatic and one may take $Z = E$. Assume henceforth that $D$ is nonempty. We construct an actual finite quotient and then prove positivity, closedness, and orderability. Choose $N \geq 4$, to be increased below. Take one copy of $E$ and $2N-1$ copies $D_i$ of $D$, for $1 \leq i \leq 2N-1$. Regard the $E$-copy as carrying $D_0$ by $d_0$ and $D_{2N}$ by $d_1$. Identify corresponding points of $B$ between $D_i, D_{i+1}$ when $i$ is even, and corresponding points of $C$ when $i$ is odd. At the two ends use the indicated maps into $E$. The index graph is the circle of $s = 2N$ factors

$$E, D_1, \ldots, D_{2N-1}.$$

Let $Z$ be this strict set quotient, with exactly the union of the copied hyperedges.

**The fibers and induced subobjects.** Each point of $F$ has one occurrence in every factor, connected around the circle; the two endpoint identifications agree by hypothesis. Every other point of an interior copy belongs to at most one of $B, C$, so it crosses at most one consecutive overlap. Its visibility away from $E$ has at most two indices. A point of $E$ can lie in both $d_0(B)$ and $d_1(C)$, joining its two neighbors through $E$; its visibility then has the three cyclically consecutive indices $\{s-1, 0, 1\}$. Further propagation is impossible: a point in $B \setminus C$ cannot cross a $C$-overlap, and conversely. A point mapped to $F$ at an endpoint was already that same $F$-point, by injectivity of the endpoint maps.

Thus every non-$F$ fiber is a proper cyclic interval of at most three indices, and every $F$-fiber is the full circle. Every factor map is injective, and the consecutive overlaps are exactly the prescribed $B$- or $C$-images. Since $s \geq 8$, these proper intervals are shorter than half the circle. If the three vertices of a copied edge are visible in another factor, they are visible along the unique shortest index path between the two factors, unless all three are full-visibility points, in which case any index path works. At each overlap inducedness of its base propagates the edge to the next factor. The bases are induced both in $D$ and in their endpoint images in $E$. Hence every factor is induced, and an edge's visibility is exactly the intersection of its vertex visibility intervals. All edge identifications, too, are generated by the neighboring relations.

**Predimension and closedness.** Put $m = \max(|B|, |C|)$ and increase $N$ so that

$$s = 2N > 3\big(\delta(E) + 2m\big). \tag{AB}$$

Cut any overlap to produce a linear amalgam $L$. Every factor $E, D_i$ is $d$-closed in $L$, by repeated free amalgamation. The end bases are closed in $E$ by transitivity; all the other bases are closed in the adjacent $D$-copies. The quotient $q : L \to Z$ restores at most $m$ vertex-pair identifications. For $U \subseteq Z$ and $\widetilde{U} = q^{-1}(U)$, it follows that

$$\delta_Z(U) \geq \delta_L(\widetilde{U}) - 2m. \tag{3}$$

As in (1), the exact difference is $2d_v - 3d_e$, with $d_v \leq m$ and $d_e \geq 0$.

If the proper visibility intervals of the vertices in $U$ miss an index $r$, cut the overlap $(r-1, r)$. Every such vertex then has a unique lift, and each full-circle fiber becomes a connected path. Induced edges also have unique lifts. Thus $Z[U] \cong L[\widetilde{U}]$. This gives positivity for every nonempty such $U$, and strict relative positivity whenever $U$ properly contains any of the factors.

If those proper intervals cover all $s$ indices, let $J$ be the set of indices in the linear amalgam where at least one lifted non-$F$ vertex first appears. All vertices first appearing at $j$ are visible only among $j, j+1, j+2$. Therefore $3|J| \geq s$. At each such step a selected vertex lies outside the incoming closed overlap. Submodularity and strict closedness give an integral predimension contribution of at least one, exactly as in (2). All other contributions are nonnegative; the first nonempty factor subset likewise has predimension at least one. Consequently

$$\delta_L(\widetilde{U}) \geq \lceil s/3 \rceil, \qquad \delta_Z(U) \geq \lceil s/3 \rceil - 2m > \delta(E) \geq \delta(D) > 0.$$

The weak inequality uses $D \leq_d E$ at either endpoint. Both cases prove that $Z$ is positive-predimension and that $E$ and every $D_i$ are $d$-closed in it.

**An actual ordered cocone.** Embed $E$ increasingly in $\mathbb{R}$ and view $d_0, d_1$ as coordinate maps on $D$. The linear interpolation

$$p_t(x) = (1-t)d_0(x) + td_1(x), \qquad 0 \leq t \leq 1,$$

is strictly increasing for each $t$ and is constant on $F$. Let $\Delta > 0$ be a common lower bound for its adjacent-coordinate gaps; the minimum of the two endpoint gap bounds suffices. For $|D| \leq 1$ choose any positive $\Delta$ instead. Put $M_0 = \max_x |d_1(x) - d_0(x)|$, and further increase $N$ until $2M_0/N < \Delta$.

Set $b_{2j} = p_{j/N}$. Define $b_{2j+1}$ by keeping the $B$-coordinates at $p_{j/N}$ and changing all coordinates outside $B$ to $p_{(j+1)/N}$. This map differs coordinatewise from $p_{(j+1)/N}$ by at most $M_0/N$, so each adjacent gap is at least $\Delta - 2M_0/N > 0$. Hence it is increasing and agrees with $b_{2j}$ on $B$. Now set $b_{2j+2} = p_{(j+1)/N}$. The intermediate map already has the new coordinates on $C \setminus B$; those on $B \cap C = F$ have not changed. Therefore it agrees with $b_{2j+2}$ on $C$.

These maps, together with the coordinate map on $E$, satisfy exactly the quotient's overlap equations, and have the required endpoint values. They give a genuine function $Z \to \mathbb{R}$. Order $Z$ by these values, breaking ties arbitrarily. Every factor remains strictly increasing, since its coordinate map is strictly increasing. This orders the already constructed hypergraph while preserving the prescribed orders on all factors.

All lower bounds on $N$ are simultaneous lower bounds, so the construction exists. In the point-and-edge-closed subcategory, every atom of $Z$ lies in a closed old factor and is closed there; transitivity puts $Z$ in the same subcategory. The factor maps are the asserted sequence of arrows. □

**Proposition A.31** (Proved: the stabilizer intersection formula)**.** *Let $\mathbb{M}$ be a countable strong Fraïssé limit for either category $A$ of Proposition A.23, and let $G = \mathrm{Aut}(\mathbb{M})$, preserving the order. For a finite $d$-closed subset $B$, write $G_B$ for its pointwise stabilizer. Then for all finite closed $B, C$,*

$$\langle G_B, G_C \rangle = G_{B \cap C}.$$

*The equality is algebraic and in particular implies the corresponding topological-generation assertion.*

*Proof.* The category has countably many isomorphism types, joint embedding, and strong amalgamation. The usual rich-sequence construction therefore gives a countable strong limit; these are the hypotheses and conclusion of Evans et al. (2019, Theorem 2.6). Every finite subset of $\mathbb{M}$ lies in a finite closed subobject, every isomorphism between finite closed subobjects extends to an automorphism, and every finite closed extension of a finite closed subobject embeds in $\mathbb{M}$ over that subobject. For the last assertion, one may first freely amalgamate the proposed extension with a sufficiently large member of the rich sequence over the given subobject, and then use richness.

The generated subgroup plainly fixes $B \cap C$. Conversely let $g \in G_{B \cap C}$. Choose a finite closed $D$ containing $B \cup C$, and a finite closed $E$ containing $D \cup gD$. Apply Lemma A.30 to the two embeddings $\mathrm{id}_D, g|_D : D \to E$. Embed its closed extension $Z$ into $\mathbb{M}$ over $E$. Every intermediate $D$-copy is closed, so its isomorphism from $D$ extends to an automorphism $g_i$ of $\mathbb{M}$. Choose $g_0 = 1$ and $g_{2N} = g$, which extend the stipulated endpoint copies. At each step the two restrictions agree on $B$ or on $C$. Therefore

$$g_i^{-1} g_{i+1} \in \begin{cases} G_B, & i \text{ even}, \\ G_C, & i \text{ odd}. \end{cases}$$

Multiplication telescopes to $g$, proving the reverse inclusion. □

**Corollary A.32** (Proved: classification of open subgroups)**.** *Every open subgroup of $G$ is the pointwise stabilizer of a finite d-closed subset of $\mathbb{M}$, possibly the empty subset. Such a subset can be chosen contained in every finite closed subset whose pointwise stabilizer is contained in the given group.*

*Proof.* An open subgroup $U$ contains the pointwise stabilizer of a finite set, hence that of a finite closed superset. Choose a finite closed $B$ with $G_B \leq U$ of least cardinality. If $g \in U$, then $G_{gB} = gG_Bg^{-1} \leq U$. Proposition A.31 gives $G_{B\cap gB} \leq U$. Minimality forces $gB = B$. An increasing permutation of the finite ordered set $B$ is the identity, so $g \in G_B$. This proves $U = G_B$. If $G_C \leq U$ for another finite closed $C$, the same intersection formula gives $G_{B\cap C} \leq U$, and minimality forces $B \subseteq C$. □

## A.14 Reduction of all actions to open stabilizers

We first display the exact stabilizer hypothesis needed for arbitrary actions. The preceding finite-support argument verifies that hypothesis, and the final theorem combines the two proofs.

**Proposition A.33** (Proved: an all-action stabilizer criterion)**.** *Let $A$ be either ordered predimension category of Proposition A.23. Let $M$ be its countable strongly homogeneous rich limit, let $G = \mathrm{Aut}(M)$ with the topology of pointwise convergence, and for a finite d-closed substructure $B \subseteq M$ put*

$$G_B = \{g \in G : g|_B = \mathrm{id}_B\}.$$

*Assume that every open subgroup of $G$ equals $G_F$ for some finite d-closed $F \subseteq M$, allowing $F = \varnothing$. Then $A$ is total. This conclusion concerns every covariant $H : A \to \mathrm{Set}$, with no finiteness or injectivity assumption on $H$.*

*Proof.* We give all changes of indexing category and all comparisons. We use Quillen's Theorem A in its covariant-comma form: a functor $u : C \to D$ with every $d \downarrow u$ weakly contractible induces a weak equivalence on classifying spaces. This is also the immediate contractible-fiber case of the opened Goerss and Jardine (2009, Theorem IV.5.6): the comma transition maps are then weak equivalences, and the displayed homotopy-cartesian square has contractible fiber and contractible lower-left corner.

*The rich limit and the concrete indexing category.* There are countably many finite objects and arrows up to their finite labels. The closure calculus, joint embedding, and amalgamation of Lemma A.21 permit the usual rich-chain construction: successively include each finite object and serve every finite strong extension of every earlier stage, repeating the countable enumeration. The union $M$ has the extension property for any finite closed $B \subseteq M$: if $B \leq_d C$, first amalgamate $C$ with a stage containing $B$, then use the served extension of that stage. Thus $C$ strongly embeds into $M$ over $B$. The alternating back-and-forth extension of finite closed partial isomorphisms shows that every isomorphism between finite closed substructures of $M$ extends to an automorphism. This also follows from the opened strong Fraïssé theorem, Evans et al. (2019, Theorem 2.6); the construction just given specifies its hypotheses here.

Let $A_M$ have as objects the nonempty finite $d$-closed substructures of $M$, and as arrows *all* increasing induced $d$-closed embeddings between them, not merely inclusions. Every finite object of $A$ is isomorphic to an object of $A_M$, and the hom-sets are unchanged by this identification. Hence $A_M$ is equivalent to $A$. Transport along this equivalence induces equivalences of the corresponding categories of elements, so it suffices to prove the assertion for $A_M$.

We also need the fixed-point identity

$$\mathrm{Fix}_M(G_B) = B. \tag{F}$$

If $x \notin B$, take a finite closed $D$ containing $B \cup \{x\}$. Freely amalgamate two copies of $D$ over $B$, extend their common order, and embed this finite strong extension into $M$ over the first $D$. The two copies of $x$ are distinct. Their closed-copy isomorphism extends to an automorphism fixing $B$, and moves $x$. This proves (F), also for $B = \varnothing$.

*Nonempty cosieves in slices are contractible.* Fix $P \in A_M$, and let $\mathcal{U}$ be a nonempty full cosieve in $P/A_M$. Choose one of its objects $P \to Z$. For $P \to X$ freely amalgamate $X$ and $Z$ over $P$. In each gap of the order on $P$, put the $X$-side points before the $Z$-side points, retaining the order within each side. On a slice morphism $X \to Y$, use its given map on the $X$-side and the identity on $Z$. This gives an increasing induced embedding. It is $d$-closed because the target is the free amalgam of $Y$ and the source over $X$, whose image is closed in both. Thus this construction is a functor on $P/A_M$, with natural maps from the identity functor and from the constant object $P \to Z$. Its image on $\mathcal{U}$ remains in $\mathcal{U}$, since it receives a map from $P \to Z$. These natural transformations contract $B\mathcal{U}$.

For precision about labels, this construction may first be made in the concrete category of all finite tagged structures. Transporting the result along an equivalence with $A_M$, using chosen isomorphisms, gives a functor and natural transformations on the slice in question. No functorial choice of an embedding into $M$ is being assumed. The same disjoint-union construction over the empty structure, with one fixed nonempty object, contracts $BA_M$.

*Removal of noninjective transitions.* For an arbitrary $H : A_M \to \mathrm{Set}$ define an equivalence relation on $H(a)$ by

$$h \sim_a h' \quad \Longleftrightarrow \quad \text{some } f : a \to b \text{ satisfies } H(f)h = H(f)h'.$$

Reflexivity and symmetry are immediate. For transitivity, complete two witnessing arrows with source $a$ to a commuting square by I. Its two composites give one arrow witnessing transitivity. The same square argument proves that $H(u)$ preserves this equivalence relation for every $u : a \to a'$. Hence the quotients form a functor $H^{\mathrm{inj}}$, with a natural surjection $q : H \to H^{\mathrm{inj}}$. Every transition of $H^{\mathrm{inj}}$ is injective: a witness after $u$ composes with $u$ to witness equality before it.

The induced functor

$$\mathrm{El}(q) : \mathrm{El}_{A_M} H \longrightarrow \mathrm{El}_{A_M} H^{\mathrm{inj}}$$

is homotopy final. To prove this, fix $(a, [h])$ in its target and let $\mathcal{D} = (a, [h]) \downarrow \mathrm{El}(q)$. There is a functor $J : a/A_M \to \mathcal{D}$ sending $f : a \to x$ to $((x, H(f)h), f)$. For $d = ((x, k), f) \in \mathcal{D}$, its comma category $d \downarrow J$ is exactly the full cosieve in $x/A_M$ consisting of arrows $g : x \to y$ satisfying

$$H(g)k = H(gf)h.$$

It is nonempty by $[k] = [H(f)h]$. It is contractible by the preceding slice-cosieve construction. Quillen's Theorem A therefore makes $BJ$ a weak equivalence. Since $a/A_M$ has an initial object, $B\mathcal{D}$ is weakly contractible. A second application of Theorem A proves that $B\,\mathrm{El}(q)$ is a weak equivalence. It thus suffices to consider actions with injective transitions.

*Saturation of an injective action.* Let $H$ now have injective transitions. The finite closed substructures of $M$, with inclusion maps only, form a directed poset $\mathcal{I}$: every finite union is contained in one of the finite closed stages. Put

$$S = \mathrm{colim}_{B \in \mathcal{I}} H(B).$$

All canonical maps $i_B : H(B) \to S$ are injective. An automorphism $g \in G$ acts by

$$g \cdot i_B(h) = i_{gB}(H(g|_B)h).$$

Naturality with respect to inclusions makes this well-defined, and functoriality gives a group action. Every $i_B(h)$ is fixed by $G_B$, so $S$ is a continuous discrete $G$-set.

For any such $G$-set $S$, define

$$T_S(B) = S^{G_B}.$$

For an arrow $f : B \to C$, extend $f$ to $g \in G$ and set $T_S(f)(s) = g \cdot s$. This is independent of the extension: two extensions differ on the right by an element of $G_B$. Its value is fixed by $G_C$, since $G_C \subseteq G_{fB}$. Extending composites proves functoriality. The maps $i_B$ above give a natural monomorphism $i : H \hookrightarrow T_S$.

For $(a, s) \in \mathrm{El}\, T_S$, the comma category $(a, s) \downarrow \mathrm{El}(i)$ identifies with the full cosieve of $a/A_M$ on those $f : a \to b$ for which $T_S(f)(s)$ belongs to $i_b(H(b))$. The corresponding element of $H(b)$ is unique by injectivity of $i_b$, so this identification also preserves all arrows. The cosieve is nonempty: represent $s$ by some $h \in H(c)$ and choose a finite closed $b$ containing $a \cup c$. For the inclusion $a \hookrightarrow b$, the identity automorphism of $M$ is an extension, and $s$ is represented in $H(b)$ by the inclusion from $c$. The slice-cosieve contraction and Theorem A now give a weak equivalence

$$B\,\mathrm{El}\, H \longrightarrow B\,\mathrm{El}\, T_S.$$

*The open-subgroup hypothesis finishes the argument.* Decompose $S$ into its $G$-orbits. Since every stabilizer is open, the stated hypothesis identifies each orbit with $G/G_F$ for a finite closed $F$, possibly empty. Taking fixed subsets preserves this orbit decomposition. If $F \neq \varnothing$, there is a natural isomorphism of $A_M$-sets

$$T_{G/G_F}(B) \;\cong\; A_M(F, B).$$

Indeed, a coset $gG_F$ is fixed by $G_B$ precisely when every element of $G_B$ fixes $gF$ pointwise. By (F), this is equivalent to $gF \subseteq B$. The map $g|_F : F \to B$ is a closed embedding by restriction of closure, every such embedding extends to $G$, and two extensions give the same coset exactly when their restrictions to $F$ agree. Postcomposition agrees with the definition of $T_S$. The category of elements of this representable has the initial object $(F, \mathrm{id}_F)$. If $F = \varnothing$, the orbit is a singleton, $T_{G/G_F}$ is constant one, and its category of elements is the already contractible category $A_M$.

Thus every component of $B\,\mathrm{El}\, T_S$ is contractible. The two proved weak equivalences transfer this assertion first to every injective $H$ and then to every $H$. Nonemptiness and connectedness of $A_M$ follow already from joint embedding. This is totality. □

**Theorem A.34** (Proved: a total category with an all-ring triple obstruction)**.** *Both $\mathcal{C}_\delta$ and its full subcategory whose raw points and hyperedges are all d-closed are total. For each of these categories and every covariant $H : A \to \mathrm{Set}$, every component of $\mathrm{El}_A H$ is weakly contractible. In each category the three point arrows $V \rightrightarrows W$ of Proposition A.24 have no affine equalizer over any nonzero unital ring $R$, and the displayed column cokernel $Q_R$ satisfies $T_1(Q_R) \cong R$. Thus both categories are separators, simultaneously for integral, rational and every prime-field coefficient system.*

*Proof.* The closure and amalgamation lemma constructs the required rich homogeneous limit for each of the two classes. The alternating bridge of Lemma A.30 proves the stabilizer-generation identity of Proposition A.31. Minimal finite supports and rigidity of finite linear orders then give Proposition A.32: every open subgroup is the pointwise stabilizer of a finite closed subset, including the empty subset. This is precisely the hypothesis of Proposition A.33, whose two homotopy-final comparisons apply to all actions. It proves totality. The objects $V, W$ belong to both classes, their three point maps are closed embeddings, and Proposition A.24 proves the remaining assertions over every stated coefficient ring. No restriction to finite actions or to injective transition maps remains. □

# B Triples and the algebraic leads

**Corollary B.1** (Refuted: the triple implication, Part B.1)**.** *In the total category $\mathcal{C}_\delta$ of Theorem A.1, the three ordinary point arrows $v_1, v_2, v_3 : V \to W$ have no affine equalizer over any nonzero unital ring. In particular, for integral coefficients,*

$$T_1\Big(\operatorname{coker}\Big(P_W \longrightarrow P_V^2, \quad t \longmapsto (t(v_2 - v_1), t(v_3 - v_1))\Big)\Big) \cong \mathbb{Z}.$$

*Three is the smallest possible obstructing ordinary family size in a total category.*

*Proof.* Totality and the explicit obstruction are Theorem A.34 and Proposition A.24. Every ordinary pair has an affine equalizer in a total category by Lemma B.2 and condition I. Thus the triple is minimal. The algebraic identities below remain valid; the example realizes their column obstruction while totality annihilates the row obstruction. □

## B.1 Algebraic conventions and the audited input

All categories in this subsection are small. We write $P_x = \mathbb{Z}[A(x, -)] \in \mathrm{Ab}^A$ and $T_i(F) = L_i \operatorname{colim}_A F$. An ordinary arrow $f : x \to y$ induces the morphism of covariant modules $P_y \to P_x$, $t \mapsto tf$. The map $\epsilon_x : P_x \to \underline{\mathbb{Z}}$ is the natural augmentation. For $L \subseteq P_x$, the notation $\epsilon(L)$ means the image of $\operatorname{colim}_A L \to \operatorname{colim}_A P_x = \mathbb{Z}$; it does not assert that one object realizes every element of this image. Every $P_x$ is projective and has colimit $\mathbb{Z}$, by the linear Yoneda lemma and the initial object $(x, 1_x)$ of $x\backslash A$. The comparison $T_i(\mathbb{Z}[H]) \cong H_i(B \operatorname{El}_A H; \mathbb{Z})$ is the isomorphism of the normalized bar and nerve chain complexes sending $(a_0 \to \cdots \to a_i, h)$ to the corresponding simplex of the category of elements. In particular, totality implies the acyclicity of these permutation modules, including when $H$ is empty or has many components.

We read the proofs of TOTEX A.1–A.4 and B.1–B.4, rather than using their stated conclusions without their maps. The part of A.1 used here is precisely the preceding chain-complex identification. The part of A.4 used here is that a total category has common successors and satisfies I and IV. Its common-successor argument identifies the two values of the constant two-element action on the union of two disjoint principal cosieves; Mayer–Vietoris gives nonzero first homology of the resulting double. Applying this argument in the category of elements of $A(p, -)$ completes a span. Its pair argument applies the two short exact sequences below to the quotient by one family of binomial relations. These are the exact steps required here. We also opened Isbell and Mitchell (1976), Lemma 2.2, Proposition 2.3, Theorem 3.1 and Proposition 3.3. Lemma 2.2 reduces a finite zigzag in an indecomposable action by successive commuting squares. Proposition 2.3, for IV, then follows by equalizing the two arrows in the resulting cospan. Theorem 3.1 identifies exactness with componentwise filtered affinization, and Proposition 3.3 reduces its pair condition for affine arrows to a finite family of ordinary arrows. These are the versions, with covariant actions, used below.

**Lemma B.2** (Proved: the row calculation used below)**.** *Let $a_0, \dots, a_k : p \to q$ be a finite family of ordinary arrows in an arbitrary small category, and let $z_i = a_i - a_0$ for $1 \le i \le k$. Put*

$$d : P_q^k \longrightarrow P_p, \qquad (t_i) \longmapsto \sum_i t_i z_i, \qquad K = \ker d,$$

*and let $H$ be $A(p, -)$ modulo all relations $ca_i \sim ca_0$. Then $\mathbb{Z}[H] = \operatorname{coker} d$ canonically, and*

$$T_1(\mathbb{Z}[H]) \cong \operatorname{coker}(\operatorname{colim} K \xrightarrow{\epsilon^k} \mathbb{Z}^k),$$
$$T_2(\mathbb{Z}[H]) \cong \ker(\operatorname{colim} K \xrightarrow{\epsilon^k} \mathbb{Z}^k),$$
$$T_j(\mathbb{Z}[H]) \cong T_{j-2}(K) \quad (j \ge 3).$$

*Consequently, for total $A$, $\epsilon^k$ is an isomorphism and $K$ is colimit-acyclic. For $k = 1$ the module $J_i = P_q z_i \subseteq P_p$ is colimit-acyclic and has zero colimit.*

*Proof.* At each object, the kernel of the map from the free group on a set to the free group on its quotient is generated by the differences defining the equivalence relation; a path of elementary relations expresses each other difference as their sum. This identifies the displayed cokernel without any injectivity assumption on the action maps. Set $J = \operatorname{im} d$. Apply the long exact derived-colimit sequences to $0 \to K \to P_q^k \to J \to 0$ and $0 \to J \to P_p \to \mathbb{Z}[H] \to 0$. The map $\operatorname{colim} J \to \mathbb{Z}$ is zero, because its composite with the epimorphism $\mathbb{Z}^k \to \operatorname{colim} J$ is $\operatorname{colim} d = 0$. Exactness and the acyclicity of the representables give all three formulas. For total $A$ the left sides vanish in positive degrees by the bar identification. In the one-row case, the same two exact sequences give $\operatorname{colim} J_i = 0$ and $T_j(J_i) = 0$ for $j > 0$. □

**Proposition B.3** (Proved: Lead 1)**.** *Let $A$ be total and $a_0, a_1, a_2 : p \to q$. Put $K_i = \ker(P_q \xrightarrow{z_i} P_p)$ and $L = K_1 \cap K_2$. Then the surjection in TOTEX B.1 is a canonical isomorphism*

$$T_1(P_q/(K_1 + K_2)) \;\cong\; \mathbb{Z}/\epsilon(L).$$

*Proof.* Lemma B.2 with one relation gives canonical augmentation isomorphisms $\operatorname{colim} K_i \cong \mathbb{Z}$. Let $S = K_1 + K_2 \subseteq P_q$. The exact sequence

$$0 \longrightarrow L \xrightarrow{\ell \mapsto (\ell, -\ell)} K_1 \oplus K_2 \xrightarrow{(u,v) \mapsto u+v} S \longrightarrow 0$$

identifies $\operatorname{colim} S$ with $\mathbb{Z}^2/\epsilon(L)(1, -1)$. The augmentation to $\operatorname{colim} P_q = \mathbb{Z}$ is induced by $(m, n) \mapsto m + n$. The sequence $0 \to S \to P_q \to P_q/S \to 0$ identifies its kernel with $T_1(P_q/S)$. Sending the class of $(m, -m)$ to $m \bmod \epsilon(L)$ gives the stated isomorphism. This is exactly the map defining the prior surjection, so the result proves its injectivity, rather than giving an unrelated abstract isomorphism.

For completeness, the second derivation in the prompt also type-checks, with its necessary sign made explicit. Put $J_i = P_q z_i \subseteq P_p$ and let $D = \{(tz_1, tz_2) : t \in P_q\} \subseteq J_1 \oplus J_2$. The map

$$P_q \longrightarrow (J_1 \oplus J_2)/D, \qquad t \longmapsto [(tz_1, 0)]$$

is onto: $[(uz_1, vz_2)] = [((u - v)z_1, 0)]$. Its kernel is $K_1 + K_2$, because $(tz_1, 0) = (wz_1, wz_2)$ means $w \in K_2$ and $t - w \in K_1$. Also $D \cong P_q/L$. Both $J_i$ have zero colimit and are acyclic by Lemma B.2. The exact sequence $0 \to D \to J_1 \oplus J_2 \to (J_1 \oplus J_2)/D \to 0$ therefore gives $T_1(P_q/S) \cong \operatorname{colim} D = \mathbb{Z}/\epsilon(L)$. With the usual connecting-map convention, this second description using $t \mapsto [(tz_1, 0)]$ is the negative of the first augmentation description: a lift $u_1 + u_2 \in S$ maps to the diagonal element represented by $u_2$, whose augmentation is $-\epsilon(u_1)$. Using $t \mapsto [(-tz_1, 0)]$ makes the two specified isomorphisms agree. □

**Proposition B.4** (Proved: Lead 2)**.** *Let $A$ satisfy I; neither connectedness nor nonemptiness is required in this proposition. The following are equivalent:*

1. $\operatorname{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ *is exact;*
2. *for every $p, r \in A$ and every $z \in \mathbb{Z}[A(p, r)]$ with $\operatorname{aug} z = 0$, some affine $t : r \to s$ satisfies $tz = 0$.*

*For every small $A$, every such $p, r, z$, and $L_z = \ker(P_r \xrightarrow{t \mapsto tz} P_p)$, one has canonically*

$$T_1(P_p/P_r z) \cong \mathbb{Z}/\epsilon(L_z).$$

*Proof.* Set $J = P_r z$. The morphism $P_r \to P_p$ has zero map on colimits. The sequences $0 \to L_z \to P_r \to J \to 0$ and $0 \to J \to P_p \to P_p/J \to 0$ therefore give $T_1(P_p/J) \cong \operatorname{colim} J = \mathbb{Z}/\epsilon(L_z)$, just as in Lemma B.2. This formula does not use I.

If colimits are exact, the displayed quotient is zero. I makes the objects in each connected component have common successors: successively complete the spans along a finite connecting zigzag. All objects at which $L_z$ is nonzero are in the component of $r$. A finite expression for $1 \in \epsilon(L_z)$ can therefore be transported to one object and added there, producing an affine annihilator.

Conversely, if $\alpha, \beta : p \to r$ are affine arrows, their difference is such a $z$, so (2) equalizes every parallel affine pair. Each component of $A$ has common successors by I, and hence so does its affinization. The pair condition makes every component of $\mathrm{Aff}_{\mathbb{Z}}(A)$ filtered. To verify the implication to exactness directly, extend an abelian diagram affinely. Its ordinary and affine colimits are canonically equal, since for $t = \sum n_i f_i$ of augmentation one, $x - tx = \sum n_i(x - f_i x)$ in the direct-sum presentation of the colimit. Each affine component has exact filtered colimits, and the colimit over all components is their direct sum, an exact functor in abelian groups. The empty category gives the zero functor and satisfies both conditions vacuously. □

**Proposition B.5** (Proved: Lead 3)**.** *For every small category $A$, every integer $k \geq 0$ and every family $a_0, \dots, a_k : p \to q$, put*

$$d^{\mathrm{t}} : P_q \longrightarrow P_p^k, \quad t \longmapsto (tz_i)_{i=1}^k, \qquad Q_a = \operatorname{coker} d^{\mathrm{t}}, \quad L = \ker d^{\mathrm{t}}.$$

*Then canonically*

$$\begin{aligned} T_1(Q_a) &\cong \mathbb{Z}/\epsilon(L), \\ T_2(Q_a) &\cong \ker(\operatorname{colim} L \to \mathbb{Z}), \\ T_i(Q_a) &\cong T_{i-2}(L) \qquad (i \geq 3). \end{aligned}$$

*If objects in the component of $q$ have common successors, the family has an affine equalizer if and only if $T_1(Q_a) = 0$. In the covariant-module presentation, $Q_a$ has $k$ generators $x_i$ at $p$ and one relation $\sum_i (a_i - a_0)x_i = 0$ at $q$.*

*Proof.* Let $D = \operatorname{im} d^{\mathrm{t}}$. In the sequence $0 \to D \to P_p^k \to Q_a \to 0$, the map on colimits from $D$ is zero: precompose it with the epimorphism $\mathbb{Z} = \operatorname{colim} P_q \to \operatorname{colim} D$, and obtain the zero column $\operatorname{colim} d^{\mathrm{t}}$. Thus $T_1(Q_a) = \operatorname{colim} D$ and $T_i(Q_a) = T_{i-1}(D)$ for $i \geq 2$. The exact sequence $0 \to L \to P_q \to D \to 0$ gives the three asserted formulas. The common-successor argument in the preceding proof converts augmentation image $\mathbb{Z}$ into one affine equalizer; the reverse implication is immediate. The presentation follows by sending the generator $1_q \in P_q(q)$ to the indicated vector in $P_p(q)^k$. For $k = 0$, $Q_a = 0$ and $L = P_q$, so every formula still holds and $1_q$ is the equalizer of the one-arrow family. The transposed row $P_q^k \to P_p$ has the permutation cokernel of Lemma B.2; no identification of these two cokernels is being used when $k > 1$. □

**Proposition B.6** (Proved: the precise assertions of Lead 4)**.** *Suppose $A$ is nonempty and connected and satisfies I and IV. For $M : A \to \mathrm{Ab}$ and $\alpha, \beta \in M(q)$, equality of their classes in $\operatorname{colim}_A^{\mathrm{Set}} UM$ implies that an affine $t : q \to s$ satisfies $t\alpha = t\beta$ in $M(s)$. Moreover $A$ is exact if and only if, for every such $M, q, \alpha, \beta$, equality in $\operatorname{colim}_A^{\mathrm{Ab}} M$ implies the same affine equalizability.*

*Proof.* The first equality puts the two elements in one connected component $X \subseteq UM$. We verify the part of Isbell and Mitchell (1976), Lemma 2.2 and Proposition 2.3, being used. Condition I reduces a zigzag in $X$ to a cospan: at each reduction, a span of ordinary arrows is replaced by a commuting square. Thus some $b, b' : q \to r$ satisfy $b\alpha = b'\beta$ in $X(r)$. Condition

IV gives an affine $c : r \to s$ with $cb = cb'$ in $\mathbb{Z}[A(q, s)]$. Put $t = cb = cb'$. Linearity of $M$ yields $t\alpha = c(b\alpha) = c(b'\beta) = t\beta$.

If $A$ is exact, Theorem 3.1 of the opened paper makes its affinization filtered. The equality of ordinary and affine abelian colimits was proved in Proposition B.4. In a filtered colimit of groups, two elements at the same object become equal under one transition, which here is an affine arrow. Conversely apply the asserted property to $M = P_p$ and to any two affine arrows $\alpha, \beta : p \to q$. Their classes in $\operatorname{colim} P_p = \mathbb{Z}$ are both 1, so the affine category equalizes every parallel pair. Common successors and Proposition B.4's argument prove exactness. □

**Remark B.7** (Proved: scope of the final sentence of Lead 4)**.** The comparison map

$$\operatorname{colim}_A^{\mathrm{Set}} UM \longrightarrow U \operatorname{colim}_A^{\mathrm{Ab}} M$$

is surjective under the common-successor hypothesis: transport the finite sum representing an abelian colimit element to one object and add there. The two assertions above precisely locate the distinction in its fibres: set equality is already sufficient, while abelian equality may identify additional components. This does not identify exactness with finite-product preservation of the set-colimit functor. For example, for $A = \Delta_{\mathrm{inj}}$, $p = [0]$, and $M = P_p$, the elements $2[a] - [b]$ for distinct $a, b : p \to q$ and $[a]$ have the same abelian-colimit class. Ordinary increasing injections preserve the number and coefficients of nonzero basis terms, so these two elements lie in different set-colimit components. Nevertheless $A$ is exact, by the already audited Isbell example. Thus the final phrase of Lead 4 is valid as motivation for the difference between the two colimits, and would be false if read as the asserted equivalence with siftedness or with injectivity of this comparison.

**Lemma B.8** (Proved: a strengthened normal form for zero classes)**.** *Let $A$ be a nonempty category in which every pair of objects has a common successor, and let $F : A \to \mathrm{Ab}$. For $\kappa \in F(s)$, its colimit class is zero if and only if there are an ordinary arrow $u : s \to y$, finitely many ordinary parallel pairs $e_j, e'_j : x_j \rightrightarrows y$, and $k_j \in F(x_j)$, such that*

$$F(u)\kappa = \sum_j (F(e_j) - F(e'_j))k_j \quad \text{in } F(y).$$

*In particular Lead 5(ii) holds; its condition I can be weakened to common successors.*

*Proof.* Define the subgroup

$$N_y = \sum_{x \in A} \sum_{e, e' \in A(x,y)} \operatorname{im}(F(e) - F(e')) \subseteq F(y).$$

Postcomposition sends $N_y$ into $N_z$, so $N$ is a subfunctor. Any parallel arrows $y \rightrightarrows z$ induce the same map $F(y)/N_y \to F(z)/N_z$, by the definition of $N_z$. Consequently $F/N$ factors through the reachability preorder $D$: its objects are those of $A$, and it has exactly one arrow $y \to z$ when $A(y, z) \neq \varnothing$. Common successors make $D$ filtered. Every element of every $N_y$ dies in $\operatorname{colim}_A F$, and the universal properties therefore identify

$$\operatorname{colim}_A F \cong \operatorname{colim}_A (F/N) \cong \operatorname{colim}_D \overline{F}$$

canonically. If $\kappa$ dies, its image in $\overline{F}(s)$ becomes zero at one object $y$ of the filtered diagram $\overline{F}$. Choose any arrow $u : s \to y$ in $A$. Then $F(u)\kappa \in N_y$, and membership in a sum of subgroups supplies the finite expression claimed; coefficients can be absorbed into $k_j$. Conversely every summand on the right has zero colimit class, and an arrow preserves that class. The empty sum is allowed. □

**Proposition B.9** (Proved: Lead 5(i), (iii), and (iv)). *Let $A$ be total and let $a_0, a_1, a_2 : p \to q$. Use $z_i, K_i, K, L$ as in Lemma B.2 and Proposition B.3, and put $K_3 = \ker(P_q \xrightarrow{a_2-a_1} P_p)$. Choose affine pair equalizers $\rho_1 \in K_1(s)$, $\rho_2 \in K_2(s)$ and $\rho_3 \in K_3(s)$ at a common object $s$. Then*

$$\kappa = (\rho_1 - \rho_3, \rho_3 - \rho_2) \in K(s)$$

*has zero class in $\operatorname{colim} K$. If an affine $t : s \to r$ kills $\kappa$, then $t\rho_1 = t\rho_2 = t\rho_3$ is an affine equalizer of the triple. Such a $t$ exists whenever the finitely many pairs in a normal form of Lemma B.8 have a common affine equalizer. In particular, a normal form with at most one summand implies the existence of such a $t$. Finally the natural morphism*

$$\psi : K_1 \oplus K_2 \oplus K_3 \longrightarrow K, \qquad (k_1, k_2, k_3) \longmapsto (k_1 - k_3, k_2 + k_3)$$

*has kernel canonically isomorphic to $L$, and*

$$T_1(K/\operatorname{im}\psi) \cong \mathbb{Z}/\epsilon(L).$$

*Proof.* The row applied to $\kappa$ is $\rho_1 z_1 - \rho_2 z_2 + \rho_3(z_2 - z_1) = 0$. Both coordinates of $\kappa$ have augmentation zero, and Lemma B.2 identifies $\operatorname{colim} K$ with $\mathbb{Z}^2$ by those augmentations. Hence its colimit class vanishes, proving (i). If $t\kappa = 0$, equality of its first and second coordinates gives $t\rho_1 = t\rho_3 = t\rho_2$; the common affine arrow belongs to $K_1(r) \cap K_2(r)$, proving the first assertion in (iii). For a normal form $u\kappa = \sum(e_j - e'_j)k_j$, a common affine equalizer $v$ of the pairs gives $(vu)\kappa = 0$. Condition IV gives such a $v$ for one pair. No assertion is made that IV alone equalizes a finite list of pairs.

The displayed formula for $\psi$ has image in $K$, because $-k_3 z_1 + k_3 z_2 = k_3(a_2 - a_1) = 0$. Its kernel consists precisely of $(\ell, -\ell, \ell)$ with $\ell \in L$; membership in $K_3$ follows from membership in $K_1 \cap K_2$. Write $S = K_1 \oplus K_2 \oplus K_3$, $I = \operatorname{im}\psi$ and $C = K/I$. Each $K_i$ is colimit-acyclic with augmentation colimit $\mathbb{Z}$, by the one-relation case of Lemma B.2, and $K$ is colimit-acyclic with colimit $\mathbb{Z}^2$. Right exactness applied to $0 \to L \to S \to I \to 0$ gives

$$\operatorname{colim} I = \mathbb{Z}^3/\epsilon(L)(1, -1, 1).$$

The map to $\operatorname{colim} K = \mathbb{Z}^2$ is induced by $(a, b, c) \mapsto (a - c, b + c)$. Its kernel is $\mathbb{Z}(1, -1, 1)/\epsilon(L)(1, -1, 1)$. The exact sequence $0 \to I \to K \to C \to 0$ identifies this kernel with $T_1(C)$, because $T_1(K) = 0$.

For the graph interpretation, at each $x$ take a bipartite multigraph with vertex set $A(q, x) \amalg A(p, x)$ and, for each $g : q \to x$, three labelled edges $E_{g,i} : g \to ga_i$, $i = 0, 1, 2$. Distinct edges remain distinct even when their endpoints coincide. The homomorphism from $K(x)$ to the integral graph cycle group is

$$(t_1, t_2) \longmapsto \sum_g \left(-(t_{1,g} + t_{2,g})E_{g,0} + t_{1,g}E_{g,1} + t_{2,g}E_{g,2}\right).$$

The boundary at a vertex $g$ is zero, and the boundaries at the other vertices are exactly $t_1 z_1 + t_2 z_2$. Conversely the boundary equation at each $g$ expresses its edge-zero coefficient in terms of the other two, proving this is an isomorphism. It is natural under postcomposition, including maps that identify vertices or edges. The three two-colour cycle groups give respectively $(k_1, 0)$, $(0, k_2)$ and $(-k_3, k_3)$, whose sum is exactly $\operatorname{im}\psi$. Thus the graph description is literal over $\mathbb{Z}$, not just an analogy over a field. □

**Corollary B.10** (Proved: what the mixed-cycle obstruction says). *Under the hypotheses of Proposition B.9, suppose the triple has no affine equalizer. Its class $[\kappa]$ in $\operatorname{colim} I$ is the nonzero generator of $\ker(\operatorname{colim} I \to \operatorname{colim} K) \cong \mathbb{Z}/\epsilon(L)$. It cannot have a normal form in which every $k_j$ belongs to $I(x_j)$, and it cannot have a normal form whose pairs have a common affine equalizer. In particular every one-term normal form is impossible.*

*Proof.* The element $\kappa$ is $\psi(\rho_1, -\rho_2, \rho_3)$; its three augmentations are $(1, -1, 1)$. This gives the asserted generator. The absence of an affine equalizer means $\epsilon(L) \neq \mathbb{Z}$, using common successors, so that generator is nonzero. A normal form with coefficients in $I$ would kill its class already in $\operatorname{colim} I$, a contradiction. The second assertion follows from Proposition B.9(iii). Thus "carried around a 1-cycle" must mean a bar chain with coefficients in $K$, or equivalently the connecting class in $T_1(K/I)$, rather than a nontrivial ordinary homology class of $BA$: the latter is contractible for a total category. □

# C The category construction and finite tests

**Theorem C.1** (Proved: universal repaired AKR survival, Part C.1)**.** *For the repaired AKR construction and every admissible choice of its parameters and targets,*

$$H_2(\mathrm{El}_{K^*}(K^*(V,-)/(cv_1 \sim cv_2)); \mathbb{Z}) \neq 0.$$

*Every attachment preserves the whole old colimit of the binary cycle functor. In particular, the first-stage class $\kappa_+ - \kappa_-$ survives the infinite union, and every such category $K^*$ is not total.*

*Proof.* Theorem C.27 proves that every actual arrow is injective on binary graph components. Its simultaneous induction uses the prescribed C.1 bounds only: component intersections of size at least two descend to an actual commuting span with older source, and component paths have at most one more vertex than the point-path bound. The pointwise equalizer descends to one actual arrow image by Proposition C.22. Hence its graph meets each source component in a connected set, and the exact fixed-path kernel of Proposition C.10 vanishes. Corollary C.28 proves the stated first-stage calculation and passage to the union. □

The earlier selected large-target proof is retained as a separate geometric argument and an explicit deterministic witness. The universal proof does not impose its additional diameter or ordering bounds. The all-coefficient triple obstruction in Proposition C.4 is a separate nonexactness result.

## C.1 The link comparison in Lead 7

**Proposition C.2** (Proved: the link comparison, with its necessary hypothesis)**.** *Let $A$ and $D$ be small categories, let $u : D \to A$ be a functor, and let*

$$R = \operatorname*{colim}_{d \in D} A(-, u(d)) : A^{\mathrm{op}} \longrightarrow \mathrm{Set}.$$

*Assume that every connected component of every comma category $x \downarrow u$ is weakly contractible. Form $A'$ by adjoining one object $Z$, by setting $A'(x, Z) = R(x)$ for $x \in A$, by allowing no arrows from $Z$ to $A$, and by setting $A'(Z, Z) = \{1_Z\}$; composition with arrows of $A$ is the presheaf action of $R$. Let $H' : A' \to \mathrm{Set}$ be any covariant functor, and put $H = H'|_A$.*

*Define the two-sided category $E(R, H)$ to have objects $(x, r, k)$ with $x \in A$, $r \in R(x)$ and $k \in H(x)$. An arrow $(x, r, k) \to (y, s, l)$ is an arrow $b : x \to y$ of $A$ satisfying $R(b)s = r$ and $H(b)k = l$. Then the functor*

$$F : \mathrm{El}_D(u^*H) \longrightarrow E(R, H), \qquad (d, k) \longmapsto (u(d), [d, 1_{u(d)}], k)$$

*which sends an arrow $\delta : d \to d'$ to $u(\delta)$ is a weak equivalence on nerves. For every $h \in H'(Z)$ the old comma link*

$$L_h = (\mathrm{El}_A H) \downarrow (Z, h)$$

*is a full union of connected components of $E(R,H)$. If $e = \dim ND < \infty$, then $H_j(BL_h;\mathbb{Z}) = 0$ for $j > e$ and the inclusion induces an injection*

$$H_j(B\,\mathrm{El}_A\,H;\mathbb{Z}) \longrightarrow H_j(B\,\mathrm{El}_{A'}\,H';\mathbb{Z}) \quad (j > e).$$

*The assertion also holds with arbitrary constant coefficients. If $D$ is empty, $R$ and all links are empty and adjoining $Z$ adds only isolated objects to the category of elements.*

*Proof.* For any $x \in A$, the set $R(x)$ is canonically $\pi_0(x \downarrow u)$: its representatives are pairs $(d, b : x \to u(d))$, and its generating identifications are $(d,b) \sim (d', u(\delta)b)$ for $\delta : d \to d'$. Fix $w = (x,r,k) \in E(R,H)$. An object of $w \downarrow F$ is specified by $d \in D$, $l \in H(u(d))$ and $b : x \to u(d)$ with $[d,b] = r$ and $H(b)k = l$. Thus $l$ is forced. Sending this object to $(d,b)$ gives an isomorphism of categories

$$w \downarrow F \;\cong\; (x \downarrow u)_r,$$

where the right side is the connected component labelled by $r$. Indeed, an arrow on either side is exactly a $\delta : d \to d'$ with $u(\delta)b = b'$. The assumed weak contractibility proves the nerve comparison as follows. For a general functor $F : X \to Y$ with these comma nerves weakly contractible, form the bisimplicial set

$$B_{p,q} = \{(y_0 \to \cdots \to y_p \to F(x_0),\; x_0 \to \cdots \to x_q)\}.$$

For a fixed $q$-simplex of $NX$, its horizontal fiber is $N(Y \downarrow F(x_0))$, a nerve with terminal object $(F(x_0), 1)$. For a fixed $p$-simplex of $NY$, its vertical fiber is $N(y_p \downarrow F)$, weakly contractible by hypothesis. Realizing first in either direction therefore shows that the projections from the diagonal realization to $BX$ and $BY$ are weak equivalences. Here the levelwise statement passes to realization because degeneracy inclusions of simplicial sets are cellular inclusions; the usual skeletal filtration proves the assertion by successive gluing along those inclusions. The two maps to $BY$, one by its projection and one by $BF$ after projection to $BX$, are homotopic: the joining arrow $y_p \to F(x_0)$ is the joining edge of the displayed chain and supplies the simplicial mapping-cylinder homotopy. Thus $BF$ is a weak equivalence. Applying this argument to the comma categories just computed proves the claimed comparison.

The function

$$\lambda : \mathrm{Ob}\,E(R,H) \longrightarrow H'(Z), \qquad (x,r,k) \longmapsto H'(r)k$$

is constant on every arrow, by functoriality. The full subcategory $\lambda^{-1}(h)$ is exactly $L_h$, including its arrows. It is consequently a union of connected components. The restriction of $F$ to the union of source components mapping into $L_h$ remains a weak equivalence: the same comma-category computation applies separately on these components. The projection $\mathrm{El}_D(u^*H) \to D$ is a discrete opfibration. A nonidentity arrow cannot project to an identity, so a nondegenerate simplex in its nerve has dimension at most $e$. This gives the homology vanishing for every link.

For completeness, the attachment assertion is a chain-level statement, not an assumption that a link embeds as a subcategory of the old category. Every nondegenerate simplex of $N\,\mathrm{El}_{A'}\,H'$ that is not old ends at exactly one $(Z,h)$; deleting this final vertex gives a simplex of the corresponding comma link, including the final arrow. Hence the normalized relative chain complex is canonically

$$C_*(N\,\mathrm{El}_{A'}\,H', N\,\mathrm{El}_A\,H) \cong \bigoplus_{h \in H'(Z)} \widetilde{C}_{*-1}(NL_h).$$

Its long exact sequence contains

$$\bigoplus_h H_j(BL_h) \longrightarrow H_j(B\,\mathrm{El}_A\,H) \longrightarrow H_j(B\,\mathrm{El}_{A'}\,H'),$$

and the first group is zero for $j > e$. All chains have finite support, so the direct sum and the argument remain valid for infinite sets of new elements and for arbitrary small indexing categories. □

**Proposition C.3** (Proved: the equalizing-precomposition obstruction)**.** *Consider a cyclic quasi-filter indexing graph with copies of $S$ at its vertices and copies of $T$ on its edges, the two maps on each edge being chosen from a parallel pair $f, g : T \rightrightarrows S$. If an arrow $c : x \to T$ satisfies $fc = gc$, the component of $x \downarrow u$ containing these lifts has nonzero first integral homology. If $f$ and $g$ are monomorphisms, this component is precisely a subdivided circle. Consequently the comma hypothesis in Proposition C.2 fails. In the repaired first-stage AKR quasi-filter for $v_1, v_2$, the last and first maps satisfy $\alpha_{4k} v_1 = \alpha_0 v_1$, so this obstruction occurs when quasi-filtering $(\alpha_0, \alpha_{4k})$ at the next stage.*

*Proof.* At every vertex choose the arrow $fc = gc : x \to S$, and at every edge choose $c : x \to T$. These choices form a copy of the subdivided indexing circle in $x \downarrow u$. This comma category has no nonidentity composable pair, because its projection to the bipartite indexing graph has none. Its nerve is therefore a graph; the displayed cycle cannot be the boundary of a two-chain and has nonzero first homology. If $f$ and $g$ are monomorphisms, either map $A(x, T) \to A(x, S)$ is injective. The vertices of the displayed cycle then have no additional incident edges, and its component is exactly that circle. The final application is the last neighboring equality in the repaired formula, with $c = v_1$. In a category where $f, g$ are not monomorphisms the component can have further branches or cycles; only the noncontractibility, not literal equality with a circle, is automatic. □

**What this does not prove about the infinite AKR category.** The dimension-one conclusion applies only when the strict colimit of representables has the stated comma-category comparison. The preceding proposition supplies explicit AKR attachments where that hypothesis fails. Thus it cannot, by itself, establish persistence of the first-stage two-dimensional class. In particular we do not infer $H_2(L_h) = 0$ for those exceptional links from the dimension of the quasi-filter indexing graph.

**Proposition C.4** (Proved: all-ring consequence of the known counting invariant)**.** *For every admissible choice in the repaired AKR Construction C.1, for every nonzero unital commutative ring $R$, and for every object $X$ of $K^*$, the $R$-linear map*

$$R[K^*(W, X)] \longrightarrow R[K^*(V, X)]^2, \qquad t \longmapsto \big(t(v_2 - v_1), t(v_3 - v_1)\big)$$

*is injective. In particular, the three initial point maps have no $R$-affine equalizer. The same injectivity holds for finitely supported coefficients in an arbitrary abelian group, with the evident meaning of the two differences.*

*Proof.* We use the counting invariant of the opened AKR Theorem C.2, pp. 385–389, with the repaired maps satisfying its required (9)–(11). We checked the induction, not merely its conclusion. Its statements are: for an object $X$ created by stage $m$ and an arrow $g : Y \to X$,

1. if a $W$-arrow $h : W \to X$ has at least two image points in the image of $g$, it factors through $g$;
2. if $h_1, \dots, h_r : W \to X$ are distinct and each meets the image of $g$ in at most one point, then their union outside that image has at least $a_m r$ points.

Here $3/2 < a_m \leq 2$. To check the source's inductive use of these statements, write every incoming map at a new quasi-filter target as $\alpha_i$ followed by an old map. Property (10) moves any overlap of two copies through consecutive copies, and (11) identifies each consecutive

overlap with the image of one of the old pair. The first inductive statement then routes the entire $W$-arrow through the neighbor, proving the new first statement. For the second statement, partition the $W$-arrows by their first available copy, discard a block of $3n$ consecutive indices of least cardinality, and sweep the remaining indices from the two ends toward that block. Every newly counted $W$-arrow meets the preceding overlap in at most one point: otherwise the just-proved routing statement would have assigned it to an earlier copy. The old second statement thus adds at least $a_m$ points per retained arrow. The discarded fraction is at most $3n/s \le 1 - a_{m+1}/a_m$, giving $a_{m+1}r$. A disjoint-sum target separates the two sets of arrows; the same inequalities apply separately to its summands. These are precisely the induction steps on pp. 386–389, and all use of overlaps is justified by the repaired (10)–(11), not the invalid restriction assertion.

For clarity, the source's displayed set $L = \{1, \dots, n\} \setminus \widetilde{H}_{j_0}$ on p. 387 must read $\{1, \dots, r\}$; all surrounding cardinality formulas use the latter set. This is an index typo, not an additional hypothesis.

Fix now a nonempty finite set $S$ of $r$ distinct $W$-arrows into $X$. Two distinct such arrows meet in at most one point: otherwise the first statement would factor one through the other via an endomorphism of $W$, and $K^*(W, W) = \{1_W\}$. Applying the second statement with one member of $S$ as $g$ therefore gives

$$\left| \bigcup_{h \in S} h(W) \right| \ge a_m(r-1) + 3 > \frac{3}{2} r.$$

This is also valid for $r = 1$. There are exactly $3r$ incidences $(h, i)$ with $h \in S$ and $i \in \{1, 2, 3\}$. If each image point had at least two incidences, the displayed union would have at most $3r/2$ points. Hence some image point has exactly one incidence.

Let $t = \sum_{h \in S} c_h h$ have all $c_h \neq 0$, and choose that private point, say $h_0 v_i$. Its coefficient in $tv_i$ is $c_{h_0}$, whereas its coefficient in $tv_j$ is zero for either $j \neq i$. Thus the three vectors $tv_1, tv_2, tv_3$ cannot be equal. This uses no division and no restriction on the characteristic or on zero divisors. It proves the claimed injectivity for coefficients in any abelian group and hence for every specified ring. In particular, it applies to $\mathbb{Q}$ and every $\mathbb{F}_p$ simultaneously. □

**Proposition C.5** (Proved: an exact cycle-space test for one attachment)**.** *Let $A'$ be obtained from $A$ by adjoining one new maximal object $Z$ with identity endomorphism only. Suppose a family $\alpha_i : T_i \to Z$ ($i \in I$, possibly infinite) generates all incoming arrows: every $b : X \to Z$, $X \in A$, can be written $\alpha_i c$ with $c : X \to T_i$ in $A$. For any covariant functor $F' : A' \to \mathrm{Ab}$, write $F = F'|_A$ and $Q = \operatorname{colim}_A F$. Define homomorphisms*

$$a : \bigoplus_{i \in I} F(T_i) \longrightarrow F'(Z), \quad (x_i) \longmapsto \sum_i F'(\alpha_i) x_i, \qquad b : \bigoplus_{i \in I} F(T_i) \longrightarrow Q, \quad (x_i) \longmapsto \sum_i [x_i].$$

*Then, canonically,*

$$\operatorname{colim}_{A'} F' \cong (Q \oplus F'(Z)) / \{(b(x), -a(x)) : x \in \bigoplus_i F(T_i)\},$$

$$\ker(Q \to \operatorname{colim}_{A'} F') = b(\ker a).$$

*For the AKR action $H = h_V/(cv_1 \sim cv_2)$, put $K(X) = \ker(\mathbb{Z}[A(W, X)] \xrightarrow{\cdot v_2 - v_1 \cdot} \mathbb{Z}[A(V, X)])$. Thus $K(X)$ is the integral cycle group of the graph whose vertices are $V$-arrows and whose edges are $W$-arrows, with edge $c$ directed from $cv_1$ to $cv_2$. For a quasi-filter attachment all $T_i = T$. Its old-to-new map on $H_2(B \operatorname{El} H)$ is injective if and only if*

$$\sum_i \alpha_i \kappa_i = 0 \text{ in } K(Z) \quad \Longrightarrow \quad \left[ \sum_i \kappa_i \right] = 0 \text{ in } \operatorname{colim}_A K$$

*for every finitely supported family $\kappa_i \in K(T)$.*

*Proof.* A cocone from $F'$ to an abelian group $M$ is an old cocone $Q \to M$ together with a map $F'(Z) \to M$ agreeing along all incoming arrows. Agreement along $\alpha_i$ suffices: for $b = \alpha_i c$, naturality of the old cocone gives agreement along $c$ first, hence along $b$. This proves the displayed pushout presentation. The class of $(q, 0)$ is zero precisely when $(q, 0) = (b(x), -a(x))$ for some $x$, which proves the kernel formula. No ordinary single-object representative for a general colimit element is assumed here; the direct sum explicitly allows finite sums at several objects.

For the last assertion, linearizing the binary equivalence relation gives

$$0 \longrightarrow K \longrightarrow P_W \xrightarrow{,v_2 - v_1,} P_V \longrightarrow \mathbb{Z}[H] \longrightarrow 0.$$

Split this at the image of the middle map. The projectivity of $P_W$ and $P_V$, together with $\operatorname{colim} P_W = \operatorname{colim} P_V = \mathbb{Z}$ and the zero induced middle map, gives

$$H_2(B \operatorname{El}_A H; \mathbb{Z}) \cong \ker(\operatorname{colim}_A K \xrightarrow{\text{aug}} \mathbb{Z}).$$

This is the binary instance of the formula whose proof is given in TOTEX Part B; we checked it by these two short exact sequences and by the identification of the simplicial replacement of $\mathbb{Z}[H]$ with the nerve chain complex of $\operatorname{El}_A H$. These identifications are natural under the stated extension, since the old representables and their relation map are its restrictions. The augmentation of an element in the kernel of the old-to-new map on $\operatorname{colim} K$ is necessarily zero. Hence that whole kernel is precisely the kernel on $H_2$. Applying the first part with $F' = K$ proves the criterion. □

**Example C.6** (Refuted literal wording: without monomorphisms the component need not be a circle)**.** The unqualified phrase "the component is a circle" in Lead 7 is false if it is read without the monomorphism hypothesis of the AKR application. Let $A$ have objects $x, T, S$, arrows

$$A(x,T) = \{c, d\}, \qquad A(T,S) = \{f, g\}, \qquad A(x,S) = \{h\},$$

identity endomorphisms only, no other arrows, and the four compositions $fc = fd = gc = gd = h$. Composition with identities is prescribed in the usual way. This is a complete finite category: there are no three composable nonidentity arrows, and the identity laws settle every remaining associativity equation.

Take the five-vertex cyclic quasi-filter indexing graph ($k = 1$). It has copies $S_i$ of $S$ and copies $T_i$ of $T$, with two arrows from $T_i$ to $S_i, S_{i+1}$ (cyclic indices), mapped by $u$ to $(g, f)$ for $i = 0, 1$, to $(g, g)$ for $i = 2$, to $(f, g)$ for $i = 3$, and to $(f, f)$ for $i = 4$. The comma category $x \downarrow u$ has five $S$-vertices $(S_i, h)$ and ten $T$-vertices $(T_i, c), (T_i, d)$, and twenty nonidentity arrows. It is connected. Its nerve is a graph, with $H_1 \cong \mathbb{Z}^{20-15+1} = \mathbb{Z}^6$; each $S$-vertex has valence four. Thus its component is neither isomorphic nor homotopy equivalent to a circle, although $fc = gc$ and its noncontractibility is exactly as asserted in the corrected proposition. The complete composition table, all typed composable triples, and these graph counts are checked by `akr_nonmonic_comma_circle.py` and its supplied output.

**Proposition C.7** (Refuted: pointwise free split injections do not suffice for acyclicity)**.** *There exists a small nonempty connected quasi-filtered category $A$ and a covariant functor $Q : A \to \mathrm{Ab}$ such that every $Q(X)$ is finitely generated free abelian, every $Q(g)$ is split injective with free abelian cokernel, and nevertheless*

$$T_1(Q) \cong \mathbb{Z}, \qquad T_i(Q) = 0 \quad (i \geq 2).$$

*One may take any repaired AKR category $A = K^*$ and*

$$Q = \operatorname{coker}(D : P_W \longrightarrow P_V^2), \qquad D(t) = \big(t(v_2 - v_1), t(v_3 - v_1)\big).$$

*Thus pointwise freeness and split injectivity do not by themselves close the cycle-module colimit obstruction.*

*Proof.* Proposition C.4 says that $D$ is injective, so $0 \to P_W \xrightarrow{D} P_V^2 \to Q \to 0$ is a projective resolution of length one. Its induced colimit map is the zero map $\mathbb{Z} \to \mathbb{Z}^2$, because each component is a difference of two augmentation-one representable maps. Therefore $T_1(Q) = \ker(0 : \mathbb{Z} \to \mathbb{Z}^2) = \mathbb{Z}$ and all higher $T_i(Q)$ vanish.

We verify the asserted pointwise and transition properties integrally. For an arrow $g : Y \to X$, identify the images of $P_V(Y)$ and $P_W(Y)$ inside their counterparts at $X$, using monicity. The induced relative map

$$\overline{D} : P_W(X)/gP_W(Y) \longrightarrow P_V(X)^2/gP_V(Y)^2$$

has one column for every $W$-arrow $h : W \to X$ not factoring through $g$. By AKR C.2($c_m$), its three-point image meets $g(Y)$ in at most one point. For any nonempty collection of these columns, C.2($d_m$) supplies more than $3r/2$ points outside $g(Y)$, where $r$ is the number of columns. Hence one such outside point is private to one column. At that point the column has coordinate pair $(-1,-1)$, $(1,0)$ or $(0,1)$, according to its position in the triple. Choose a unit entry from that pair as pivot. Its row has zero entries in all other columns. Eliminating this column therefore leaves all other columns unchanged. Repeating gives a full unit-pivot reduction. All these matrices are finite because each object's incoming hom-sets are frozen at its finite stage. Thus $\overline{D}$ is injective and its cokernel is free abelian. The same argument without the relative quotient proves that $Q(X)$ is free abelian.

The exact diagram comparing the two short exact sequences at $Y$ and $X$ now gives, by the kernel–cokernel sequence,

$$0 \longrightarrow Q(Y) \xrightarrow{Q(g)} Q(X) \longrightarrow \operatorname{coker} \overline{D} \longrightarrow 0.$$

The final group is free, so this sequence splits. This verifies every claimed transition property while retaining the nonzero derived colimit calculated above. □

**Corollary C.8** (Proved: the same obstruction with free groups and free-factor maps)**.** *For the same quasi-filtered category $A = K^*$ there is a covariant functor $\Gamma : A \to \mathrm{Grp}$ such that every $\Gamma(X)$ is a finitely generated free group and every transition map identifies its source with a free factor of its target, but the abelianization functor $\Gamma_{\mathrm{ab}} : A \to \mathrm{Ab}$ has $T_1(\Gamma_{\mathrm{ab}}) \cong \mathbb{Z}$.*

*Proof.* For each $X$, take group generators $x_{v,1}, x_{v,2}$ indexed by $v \in A(V,X)$ and impose, for each $h \in A(W,X)$, the relation

$$x_{hv_2,1}\, x_{hv_1,1}^{-1}\, x_{hv_3,2}\, x_{hv_1,2}^{-1} = 1.$$

Postcomposition on $v$ and $h$ defines a homomorphism on these presentations for every arrow of $A$, and makes $\Gamma$ a covariant functor. Its abelianization is exactly the functor $Q$ of Proposition C.7.

For any nonempty family of remaining relations the private-vertex argument supplies a vertex occurring in just one of them. One of the two generators indexed by that vertex occurs exactly once in that relation and nowhere in the others. At $hv_1$ we may choose the coordinate-one generator; the coordinate-two occurrence is a different generator. A Tietze elimination solves for the chosen generator and removes that sole relation, leaving all other relations unchanged. Repeating removes every relation and proves that $\Gamma(X)$ is free.

For an arrow $g : Y \to X$, retain the entire old presentation at $Y$. Apply the relative private-vertex argument only to the new relations, those indexed by $W$-arrows not factoring through $g$. Each chosen private vertex lies outside the old image, so the eliminated generator is new and occurs in no old relation. Eliminating all new relations leaves the old presentation

together with additional free generators. Thus the natural map identifies $\Gamma(Y)$ with a free factor of $\Gamma(X)$. No compatible choices of free bases or of retractions are required. Finally the preceding proposition computes the stated nonzero derived colimit of its abelianization. □

**Proposition C.9** (Proved: a free-factor nerve-colimit obstruction)**.** *There is a diagram of finitely generated free groups, with every transition a free-factor inclusion and with free group colimit, whose ordinary simplicial-set colimit of nerves has second integral homology* $\mathbb{Z}$*.*

*Proof.* Use the category consisting of a parallel pair $f, g : 0 \rightrightarrows 1$ and the diagram

$$G_0 = F(a, c), \qquad G_1 = F(a, c, b), \qquad f = (a, c), \quad g = (ab, cb^2).$$

The automorphism $a \mapsto ab$, $c \mapsto cb^2$, $b \mapsto b$ has inverse $a \mapsto ab^{-1}$, $c \mapsto cb^{-2}$, $b \mapsto b$. Thus both maps have free-factor image.

We claim that their image subgroups intersect only in the identity. For a nonempty reduced word $w$ in $a, c$, consider $g(w)$ in the free product $F(a, c) * F(b)$. A negative first letter gives a nonzero initial $b$-block, and a positive last letter gives a nonzero terminal $b$-block. In the remaining case, $w$ starts positively and ends negatively, so it has a positive-to-negative transition. At such a transition the two underlying generators differ, since $w$ is reduced. Its $b$-exponents are therefore $1 - 2$ or $2 - 1$, both nonzero. These are the only places where $b$-blocks from distinct letters can merge. No intervening $a, c$ block reduces to the identity: it is a subword of the original reduced word. Hence at least one nonzero $b$-block survives, and $g(w) \notin F(a, c)$. This proves the claim for arbitrary word length.

Consequently $BG_0 \vee BG_0 \to BG_1$, induced by $f, g$, is a simplicial monomorphism. Indeed its two images intersect in each nerve degree only in the all-identity simplex. The ordinary coequalizer $Y$ of $Bf, Bg$ is the pushout of this monomorphism along the fold $BG_0 \vee BG_0 \to BG_0$. This is a homotopy pushout: realizations of simplicial monomorphisms are inclusions of CW subcomplexes, or one may factor the fold through its mapping cylinder before pushing out.

Free groups have classifying spaces homotopy equivalent to graphs. The Mayer–Vietoris sequence therefore gives

$$H_2(Y; \mathbb{Z}) \cong \ker(f_{\mathrm{ab}} - g_{\mathrm{ab}} : \mathbb{Z}^2 \to \mathbb{Z}^3) = \mathbb{Z}(2a - c),$$

since $(f_{\mathrm{ab}} - g_{\mathrm{ab}})(a) = -b$ and its value on $c$ is $-2b$. On the other hand,

$$\operatorname{colim} G = \langle a, c, b \mid a = ab,\ c = cb^2 \rangle \cong F(a, c).$$

The first relation kills $b$, and the second is redundant. Thus the comparison $Y \to B(\operatorname{colim} G)$ cannot be a weak equivalence. □

**Scope of the obstruction.** The parallel-pair indexing category is not quasi-filtered and is not total. This example does not refute totality-to-exactness or establish anything about the totality of AKR. It rules out deriving asphericity of an ordinary nerve colimit solely from free values, free-factor transitions and a free group colimit.

The program `finite_freefactor_nerve.py` checks the Nielsen inverses, exact word identities certifying the group quotient, and the integer Mayer–Vietoris matrix. The latter has a unit $3 \times 3$ minor, all $4 \times 4$ minors zero, and primitive kernel generator $(2, -1, -2, 1)$. Its additional check of all 39,364 reduced nonempty words of length at most nine is supplementary; the proof above is unbounded.

## C.2 An exact fixed-path test for a repaired attachment

The following statement strengthens the cycle-space criterion of Proposition C.5. It concerns the repaired path construction, with its prescribed target covered by the images of the generating maps. Relabelling that target does not affect the statement. In particular it applies with every admissible numerical choice in that repaired construction.

**Proposition C.10** (Proved: the kernel is generated by fixed-endpoint paths)**.** *Let $A$ be a finite stage, or a full finite substage containing $V, W$ and every prescribed incoming generator and factorization for its retained targets, of the repaired AKR construction. Adjoin a single quasi-filter target $Z$ for $f, g : S \rightrightarrows T$ and write $A'$ for the resulting category. Put $s = 4k + 1$, where $k \geq n$, and write $\alpha_i : T \to Z$, $i \in \mathbb{Z}/s$, for its generating maps. All old objects and arrows remain unchanged. For an object $X$ let $G_X$ have vertices $A(V, X)$ and one directed edge $b$ from $bv_1$ to $bv_2$ for each $b \in A(W, X)$, using $A'$ when $X = Z$. Let $K(X) = H_1(G_X; \mathbb{Z})$ and $Q = \operatorname{colim}_A K$.*

*Let $F \subseteq G_S$ be the equalizer subgraph: its vertices are those $c : V \to S$ with $fc = gc$, and its edges are those $b : W \to S$ with $fb = gb$. Then*

$$\ker(Q \longrightarrow \operatorname{colim}_{A'} K) = \langle [(g_\# - f_\#)\sigma] \ : \ \sigma \in C_1(G_S; \mathbb{Z}), \quad \partial\sigma = w - v, \quad v, w \in F_0 \rangle \,. \quad \text{(FP)}$$

*Only pairs $v, w$ lying in the same component of $G_S$ occur. A path may be chosen for $\sigma$; reversing edges is represented by negative coefficients. The displayed difference is a cycle in $G_T$, and its class in $Q$ is independent of the chosen path. It is enough to choose one vertex from each component of $F$ and paths joining those representatives which lie in a common component of $G_S$.*

*Consequently this attachment preserves $Q$, and hence the old second homology of the binary category of elements, if*

$$\pi_0(F) \longrightarrow \pi_0(G_S) \quad \textit{is injective.} \qquad \text{(FC)}$$

*More generally, preservation holds if and only if all classes in (FP) are zero in $Q$; injectivity in (FC) is sufficient but is not asserted to be necessary.*

*Proof.* We first identify the cell fibers of the attachment. Every underlying point of every prescribed repaired target is the value of a $V$-arrow. This holds for $V$ and $W$. It is preserved by a disjoint-sum target. For a quasi-filter target it follows because its fixed points, private points, and each point of every repaired path target are covered by the generating copies: the interval formulas in Proposition C.1 of the supplied corrected construction give a nonempty visibility interval for every path-target point. Induction gives the assertion. Thus the vertices of each $G_X$ may be identified with its underlying points. No assertion about arbitrary extra, unused points adjoined to a target is needed here.

For a point $z \in Z$, put $I_z = \{i : z \in \alpha_i(T)\} \subseteq \mathbb{Z}/s$. Property (10) makes $I_z$ geodesically convex in the odd index cycle. Indeed, between any two indices there is a unique shortest cyclic arc of length at most $2k \leq s - 2n$. Starting at either end, (10) puts the next index of that arc in $I_z$, and iteration puts the entire arc in $I_z$. A nonempty proper connected subset of a cycle is an interval, so $I_z$ is either an interval or the whole cycle.

For an edge $e : W \to Z$, its visibility set is

$$I_e = \{i : e = \alpha_i b \text{ for some } b : W \to T\} = \bigcap_{r=1}^{3} I_{ev_r}.$$

The nontrivial inclusion follows from the established AKR routing invariant $(c_{m+1})$: if $e(W) \subseteq \alpha_i(T)$, its three image points, in particular at least two, lie in that image, so $e$ factors through $\alpha_i$. Each factorization is unique because $\alpha_i$ is injective. Intersections of geodesically convex

sets are geodesically convex. Consequently each nonempty $I_e$ is also an interval or the whole cycle.

Define the neighboring legs $\lambda_i, \mu_i : S \to T$ by

$$(\lambda_i, \mu_i) = \begin{cases} (g, f), & 0 \le i < 2k, \\ (g, g), & i = 2k, \\ (f, g), & 2k < i < 4k, \\ (f, f), & i = 4k. \end{cases}$$

The repaired identities say $\alpha_i \lambda_i = \alpha_{i+1} \mu_i$. Property (11) says that the image of this composite is exactly $\alpha_i(T) \cap \alpha_{i+1}(T)$. Therefore the fiber graph of a vertex $z$ has one vertex at each index of $I_z$ and one edge between each neighboring pair of such indices. For an edge $e$, the same assertion follows by one further application of the old routing invariant: the unique lift $b : W \to T$ in copy $i$ has its image in $\lambda_i(S)$ whenever $i, i+1 \in I_e$, so it factors uniquely through $\lambda_i$. Its factor $d : W \to S$ gives the neighbor lift $\mu_i d$ in copy $i+1$. Conversely every such source edge gives this neighbor relation. Thus, in either cell dimension, every fiber graph is a path or a circle, including a one-vertex path.

The circular fibers are precisely the equalizing cells. Here is the explicit verification, which also identifies their generators. Suppose a cell lies in all copies and let $d_i$ be its source lift on the neighboring overlap at $i$. Its lift in copy $i$ is simultaneously $\lambda_i d_i = \mu_{i-1} d_{i-1}$, with cyclic indices. The two legs are injective on vertices and on edges. At the upper junction these equalities imply $d_{2k+1} = d_{2k-1}$, and successive cancellation gives

$$d_{2k+j} = d_{2k-j} \qquad (1 \le j \le 2k).$$

In particular $d_{4k} = d_0$. The lower junction then gives $g d_0 = f d_{4k} = f d_0$. Working forward through the first half, and then through the second half, forces every $d_i = d_0$. Conversely an equalizing cell $d$ gives a circular fiber with every $d_i = d$.

Write $C_*(G)$ for the cellular chain complex of an oriented graph, concentrated in degrees zero and one. Define a chain map

$$\delta : C_*(G_S)^s \longrightarrow C_*(G_T)^s, \qquad (\delta x)_i = (\lambda_i)_\# x_i - (\mu_{i-1})_\# x_{i-1},$$

and define $\epsilon : C_*(G_T)^s \to C_*(G_Z)$ by $\epsilon(y) = \sum_i (\alpha_i)_\# y_i$. The cell-fiber graphs just computed show, integrally and degree by degree, that

$$0 \longrightarrow C_*(F) \xrightarrow{j} C_*(G_S)^s \xrightarrow{\delta} C_*(G_T)^s \xrightarrow{\epsilon} C_*(G_Z) \longrightarrow 0 \tag{CF}$$

is exact, where $j$ is diagonal inclusion. For a path fiber the incidence map has zero kernel and cokernel $\mathbb{Z}$; for a circle fiber its kernel is the diagonal $\mathbb{Z}$ and its cokernel is $\mathbb{Z}$. These descriptions use bases and hence introduce no torsion. They commute with the boundary because all displayed maps are graph maps.

For completeness, $F$ is the subgraph induced on its fixed vertices. If the endpoints of $b$ are fixed, then $fb$ and $gb$ share their first two image points. Invariant $(c_m)$ factors one through the other. The only endomorphism of the initial object $W$ is its identity, since the construction adds only new maximal targets and leaves old hom-sets unchanged. Hence $fb = gb$. This induced-subgraph description is useful for testing (FC), but the chain argument only uses the equalizer definition.

Let $D_* = \operatorname{im} \delta$ be the image chain complex in $C_*(G_T)^s$. Splitting (CF) into two short exact sequences of complexes gives

$$\ker\big(K(T)^s \longrightarrow K(Z)\big) = \operatorname{im}\big(H_1(D_*) \longrightarrow K(T)^s\big),$$

and

$$K(S)^s \longrightarrow H_1(D_*) \longrightarrow H_0(F) \xrightarrow{H_0(j)} H_0(G_S)^s. \tag{HC}$$

The last map is diagonal on the component-inclusion homomorphism. Its kernel is therefore exactly $\ker(H_0(F) \to H_0(G_S))$. That kernel is generated by differences $[w]-[v]$ of components of $F$ lying in one component of $G_S$.

Choose a graph path $\sigma$ from $v$ to $w$. Its diagonal copy in $C_1(G_S)^s$ has boundary $j(w-v)$, so $\delta(\sigma,\dots,\sigma)$ is a cycle in $D_*$ lifting that component difference, up to the immaterial common boundary sign. Its $i$th component

$$\kappa_i = (\lambda_i)_\#\sigma - (\mu_{i-1})_\#\sigma$$

is separately a cycle in $G_T$, because $f$ and $g$ agree at $v, w$. The sum of these components is

$$\sum_i \kappa_i = \sum_i ((\lambda_i)_\# - (\mu_i)_\#)\sigma = (g_\# - f_\#)\sigma.$$

Here there are $2k$ occurrences of $(g,f)$, $2k-1$ of $(f,g)$, and two zero differences. On the other hand, the image in $Q$ of every element coming from $K(S)^s$ in (HC) is zero: each of its terms is the difference of the images of a source cycle along two old arrows. Proposition C.5 now gives exactly (FP), in both directions.

If $\sigma'$ has the same boundary as $\sigma$, their difference is a cycle $\tau \in K(S)$, and $[(g_\# - f_\#)\tau] = [\tau] - [\tau] = 0$ in $Q$. Moving either endpoint within its component of $F$ also leaves the class unchanged, since $f_\# = g_\#$ on chains of $F$. These observations establish all path and representative assertions. Finally, the dimension-shifting formula identifies the old second homology with the kernel of the augmentation $Q \to \mathbb{Z}$; every generator in (FP) has augmentation zero. Thus the criterion for preserving $Q$ is also the criterion for preserving that old second homology. □

**Resolution for every repaired C.1 stage.** The kernel formula is exact without a fixed-component connectedness hypothesis. In the repaired C.1 construction, Proposition C.22 identifies each nonempty fixed vertex set with the image of an actual old arrow, and Theorem C.27 proves that every such arrow is injective on binary graph components. Thus (FC) holds at every stage and every kernel in (FP) is zero, for all admissible target choices. Corollary C.28 deduces survival of the first-stage class in the infinite union. The formula still distinguishes the general issues correctly: outside these hypotheses, a disconnected fixed subgraph alone would not establish a nonzero kernel without computing its path-difference class in the old colimit, and killing one old class would not by itself determine the whole later homology.

**Lemma C.11** (Proved: exact fixed subsets of the repaired copy maps)**.** *Consider a repaired attachment for injections $f, g : S \rightrightarrows T$ between finite sets. Assume that the partial relation $f(x) \mapsto g(x)$ has no nontrivial directed cycles. Let $F = \{f(x) : f(x) = g(x)\}$ and write the nonconstant directed paths of the partial relation $f(x) \mapsto g(x)$ as*

$$p_{\nu,0} \longrightarrow \cdots \longrightarrow p_{\nu,l_\nu-1}.$$

*Here $l_\nu$ is the number of vertices of the path, not its number of edges. Put $L = \max_\nu l_\nu$, or $L = 1$ when there are no such paths. For $s = 4k+1$ and $k \geq n \geq L$, let $\alpha_0, \dots, \alpha_{4k} : T \to Z$ be the repaired copy maps. For $0 \leq i < j \leq 4k$, their set of fixed source vertices is exactly:*

1. *if $(i,j) = (r, 4k-r)$ with $0 \leq r \leq L-2$, it is*

$$F \ \cup \ \{p_{\nu,t} : 0 \leq t \leq l_\nu - r - 2\};$$

2. *if* $(i,j) = (2k-r, 2k+1+r)$ *with* $0 \le r \le L-2$, *it is*

$$F \ \cup \ \{p_{\nu,t} : r+1 \le t \le l_\nu - 1\};$$

3. *for all other pairs it is* $F$.

*An empty index range contributes no points. Thus at most* $2L-1$ *distinct fixed subsets have to be tested, independently of* $s$.

*Proof.* Fixed points in $F$ are shared by every copy; private points are shared by no two distinct copies. On a path of length $l$, write $t$ for its zero-based position. Its repaired coordinate is

$$\beta_i(t) = \begin{cases} t+i, & i \le 2k,\ t+i \le 2k, \\ t+i+2k-l+1, & i \le 2k,\ t+i > 2k, \\ t-i+6k-l+2, & i > 2k,\ t-i+6k-l+2 > 2k, \\ t-i+4k, & i > 2k,\ t-i+6k-l+2 \le 2k. \end{cases}$$

For fixed $t$, this expression is strictly increasing on the first arm and strictly decreasing on the second arm: within each branch its step is respectively 1 or $-1$, and at a branch transition the jump has the same sign because $l \le k$. Thus an equality between distinct copies must use $i \le 2k < j$.

If both values lie in the lower branch, equality says $i+j = 4k$. Writing $i = r$, the branch inequalities reduce exactly to $t \le l-r-2$. If both lie in the upper branch, equality says $i+j = 4k+1$. Writing $i = 2k-r$, the inequalities reduce to $t \ge r+1$. For the first-lower/second-upper case, equality would give $i+j = 6k-l+2$; the lower inequality forces $j \ge t+4k-l+2$, contrary to the upper inequality. For the first-upper/second-lower case, equality would give $i+j = 2k+l-1$, whereas the lower inequality implies $j \ge t+4k-l+2$, impossible since $l \le k$. These calculations give precisely the two displayed families, path by path. Different paths have different tags in the repaired target. Only $l \le k$ was used, so the calculation applies to every admissible common bound $n$ and every actual path size $l \le n \le k$; no restriction to the tested size-three examples is involved. □

**Corollary C.12** (Proved: the small-overlap case is harmless)**.** *Assume that* $f$ *and* $g$ *induce injections on binary graph components and that their underlying point images meet in at most one point. Then every equalizer subgraph of a pair of copy maps* $\alpha_i, \alpha_j : T \to Z$ *is connected in each component of* $G_T$. *Thus the fixed-path criterion applies to a subsequent attachment for any such copy pair, even when* $K(T)/F(T)$ *is nonzero.*

*Proof.* An internal vertex of a nonconstant point path belongs to $f(S) \cap g(S)$. Hence every path has length at most three, at most one has length three, and a fixed point cannot coexist with a length-three path. In Lemma C.11, the first prefix and suffix are respectively $f(S)$ and $g(S)$, including the fixed points. All remaining subsets have at most one point. The image of each source graph component is connected, and injection on components prevents two distinct source components from lying in one ambient component. This proves the claim for the two full images; the singleton and empty cases are immediate. The equalizer graph is induced on its fixed vertices by the established AKR two-point routing property, as in Proposition C.10. □

**Theorem C.13** (Proved: protected first-stage cycles and a split-monic quotient)**.** *In any repaired AKR Construction C.1, let* $P_+$ *and* $P_-$ *be the fresh first-stage quasi-filter targets for* $(v_1, v_2)$ *and* $(v_2, v_1)$. *Write* $s_+, s_-$ *for their respective numbers of quasi-filter maps. In the binary graph* $G_{P_\pm}$, *let* $S_\pm$ *be the* $s_\pm$ *vertices of its unique circular component, and choose its integral cycle* $\kappa_\pm$ *with augmentation one. For* $X \in K^*$ *define*

$$F(X) = \mathbb{Z}[K^*(P_+, X)] \oplus \mathbb{Z}[K^*(P_-, X)], \qquad \iota_X(b) = b\kappa_\pm \in K(X) \quad (b : P_\pm \to X).$$

*Then $\iota : F \to K$ is a natural monomorphism. Its quotient $M = K/F$ is pointwise a finitely generated free abelian group, and every arrow of $K^*$ induces a split injection on $M$ with free abelian cokernel. The splitting is an existence statement for each individual arrow; no compatible family of splittings is asserted.*

*More precisely, if $X \in K_m$, $m \geq 1$, $g : Y \to X$ is any arrow, and $h_i : P_{\epsilon_i} \to X$ is a finite family of distinct typed arrows, then:*

1. *$|h_i(S_{\epsilon_i}) \cap g(Y)| \geq 3$ implies that $h_i$ factors through $g$;*
2. *if every displayed intersection has at most two points, then*

$$\left| \bigcup_i h_i(S_{\epsilon_i}) \setminus g(Y) \right| \geq \frac{a_m}{a_1} \sum_i (s_{\epsilon_i} - 2).$$

*These assertions hold for mixed families with either source $P_+$ or $P_-$.*

*Proof.* We first verify the marked-set versions of the source induction. At stage one, a map from $P_+$ or $P_-$ to an old object is its identity, if the target is that same object, and does not otherwise exist. Every proper incoming map to $P_\pm$ has source $V$ or $W$. Its image meets the marked circle in at most two points: a $W$-map supplies one edge of that circle and its third point lies outside it. Therefore the first assertion holds at stage one, and the second has bound $s_\pm - 2$ for the only possible nonempty family. There are no arrows between $P_+$ and $P_-$, and the only endomorphisms of either are identities; this remains so because all stages are full extensions.

We give the quasi-filter step explicitly. Write the old pair as $\phi_1, \phi_2 : R \rightrightarrows T$ and its new maps as $\alpha_j : T \to Z$, with $j \in \mathbb{Z}/s$. An identity comparison map $g = 1_Z$ makes the first assertion trivial and the second vacuous. Otherwise write $g = \alpha_0 g^*$ after a cyclic renumbering. Every marked arrow into $Z$ factors as $h = \alpha_j h^*$.

Suppose first that $h(S_\epsilon)$ meets $g(Y)$ in at least three points. If $1 \leq j \leq s - 2n$, (10) gives $\alpha_j(T) \cap \alpha_0(T) \subseteq \alpha_j(T) \cap \alpha_{j-1}(T)$. If $s - 2n < j < s$, move instead toward $j + 1$; its forward distance to 0 is less than $2n \leq s - 2n$, so the other containment in (10) applies. In either case choose the adjacent identity

$$\alpha_j \phi_p = \alpha_{j'} \phi_q, \qquad j' = j - 1 \text{ or } j + 1, \quad p, q \in \{1, 2\}.$$

By (11) the intersection of the two copies is exactly $\alpha_j \phi_p(R)$. Injectivity of $\alpha_j$ therefore gives $|h^*(S_\epsilon) \cap \phi_p(R)| \geq 3$. The old first assertion supplies $l : P_\epsilon \to R$ with $h^* = \phi_p l$. Hence $h = \alpha_{j'} \phi_q l$: the whole typed arrow, not merely its marked set, has moved to the adjacent copy. Its image and its marked image are unchanged. If we started with $j \leq s - 2n$, repeat downward to 0; otherwise repeat upward through $s - 1$ to 0. Each relevant distance decreases, so (10) remains applicable. Once $h = \alpha_0 h_0^*$, injectivity of $\alpha_0$ and the old first assertion applied to $h_0^*, g^*$ show that $h$ factors through $g$. This proves the new first assertion without any use of the number of points of $P_\epsilon$.

For the second assertion let the family be $h_1, \ldots, h_r$, give member $i$ the positive weight $w_i = s_{\epsilon_i} - 2$, and put $\beta_m = a_m/a_1$. Define disjoint sets

$$H_j = \{i : h_i \text{ factors through } \alpha_j\} \setminus \bigcup_{t<j} H_t, \qquad 0 \leq j < s.$$

They partition the family. Partition the indices $0, \ldots, s - 1$ into $s/(3n)$ consecutive blocks of length $3n$. Choose a block $B = \{j_0, \ldots, j_0 + 3n - 1\}$ minimizing $\sum_{j \in B} \sum_{i \in H_j} w_i$, and discard its assigned family members. For the retained index set $L = \{1, \ldots, r\} \setminus \bigcup_{j \in B} H_j$ we have

$$\sum_{i \notin L} w_i \leq \frac{3n}{s} \sum_{i=1}^{r} w_i, \qquad \sum_{i \in L} w_i \geq \frac{a_{m+1}}{a_m} \sum_{i=1}^{r} w_i.$$

Now order the remaining copy indices by

$$0, 1, \dots, j_0 - 1, \quad s-1, s-2, \dots, j_0 + 3n,$$

omitting the first arm if $j_0 = 0$. Assign each retained member to the first available copy in this order. Denote by $L_j$ the family assigned to $j$, write $h_i = \alpha_j h_i^*$ for $i \in L_j$, and put $D_j = \bigcup_{i \in L_j} h_i(S_{\epsilon_i})$.

If the first index is 0, the old second assertion, applied to $g^*$ and the $h_i^*$ assigned there, yields

$$|D_0 \setminus g(Y)| \geq \beta_m \sum_{i \in L_0} w_i.$$

For every later index $j$ in the first arm, all intersections with earlier copies and with $g(Y)$ lie, by (10), in $\alpha_j(T) \cap \alpha_{j-1}(T)$. Indeed all backward distances are at most $j_0 - 1 < s - 2n$. For an index $j$ in the second arm, every earlier index, and also 0, is reached in the forward cyclic direction in at most $s - 3n$ steps, so those intersections lie in $\alpha_j(T) \cap \alpha_{j+1}(T)$. This includes the first index $s - 1$ when $j_0 = 0$.

Let $j'$ be this adjacent comparison index. No member of $L_j$ factors through $\alpha_{j'}$. Usually $j'$ is earlier in the sweep, so this follows from the assignment rule. In the only exceptional case $j = s - 1, j_0 = 0$, a factorization through $\alpha_0$ would put that member in $H_0$, hence in the discarded block, again impossible. Choose $\alpha_j \phi_p = \alpha_{j'} \phi_q$ as above. If $h_i^*(S_{\epsilon_i})$ met $\phi_p(R)$ in three points, the old first assertion would factor $h_i^*$ through $\phi_p$ and hence $h_i$ through $\alpha_{j'}$, a contradiction. Therefore the old second assertion applies to the family $h_i^*$ and comparison arrow $\phi_p$, and (11) gives

$$\begin{aligned} \left| D_j \setminus \left( g(Y) \cup \bigcup_{t \text{ earlier}} D_t \right) \right| &\geq |D_j \setminus \alpha_j \phi_p(R)| \\ &= \left| \bigcup_{i \in L_j} h_i^*(S_{\epsilon_i}) \setminus \phi_p(R) \right| \\ &\geq \beta_m \sum_{i \in L_j} w_i. \end{aligned}$$

Summing these disjoint increments over the sweep and using the retained-weight estimate proves

$$\left| \bigcup_i h_i(S_{\epsilon_i}) \setminus g(Y) \right| \geq \beta_m \sum_{i \in L} w_i \geq \beta_{m+1} \sum_{i=1}^{r} w_i.$$

Each factorization preserves the source and its weight; injectivity of a copy map preserves distinctness of the typed arrows. Thus the argument is valid for mixed protected source types. It uses only (10), (11), monicity, the old threshold-three assertion, and the old weighted bound. The original special cardinality $|W| = 3$ is not used in this inductive step.

At a disjoint-sum target, the two summands have disjoint point sets. If the comparison arrow is the identity of the new target, the factorization assertion is immediate and the nonfactorization bound is vacuous. Otherwise every nonempty marked overlap places the marked arrow and the comparison arrow in the same summand; the old assertion then gives the required factorization. Apply the old bound in the comparison summand. In the other summand one may use the following unconditional bound. Choose one marked arrow $h_r$ as comparison. Every other member has at most two marked points in its full image, since otherwise it would factor through $h_r$. Such a factorization between the two fixed source objects is impossible unless both sources and arrows coincide. Applying the old second assertion to the other members and then adding all $s_{\epsilon_r}$ points of $h_r(S_{\epsilon_r})$ gives at least $(a_m/a_1) \sum_i w_i$, because $(a_m/a_1)(s_{\epsilon_r} - 2) \leq s_{\epsilon_r}$. This proves both marked assertions for all stages.

Every admissible first-stage length satisfies $s_\pm \geq 45$. Indeed its active path has two vertices, divisibility $3n \mid s = 4k + 1$ forces $n$ odd, hence $n \geq 3$, and $s/(3n) \geq (1 - a_1/2)^{-1} > 4$ is an integer. Also $a_m/a_1 > 3/4$. Consequently

$$\frac{a_m}{a_1}(s_\pm - 2) > \frac{s_\pm}{2}.$$

The unconditional bound from the preceding paragraph shows that every nonempty finite family of marked circles has a vertex belonging to exactly one of them: otherwise the union would have at most half the total number of marked incidences. At such a vertex, an incident edge of that circle belongs to no other circle in the family. Its coefficient detects the coefficient of that circle in a proposed linear relation. This proves injectivity of $\iota_X$, over $\mathbb{Z}$ and indeed with arbitrary abelian-group coefficients. Naturality follows from composition of arrows.

For the quotient assertion, attach a two-cell to $G_X$ along each marked embedded circle $b\kappa_\pm$ and denote the resulting finite CW complex by $C_X$. The preceding private-vertex argument provides a boundary edge of one two-cell that belongs to no other two-cell. Collapsing that cell across that edge and repeating removes all two-cells. Thus $C_X$ is homotopy equivalent to a graph, and

$$H_1(C_X; \mathbb{Z}) = K(X)/F(X) = M(X)$$

is free abelian.

An arrow $g : Y \to X$ embeds $C_Y$ cellularly into $C_X$: it is injective on vertices, edges and typed protected cells. The cells not in its image correspond exactly to protected arrows that do not factor through $g$. For any nonempty family of those cells the marked assertions give

$$\left| \bigcup_i h_i(S_{\epsilon_i}) \setminus g(Y) \right| > \frac{1}{2} \sum_i s_{\epsilon_i}.$$

Hence there is a private marked vertex outside $g(Y)$. A boundary edge incident to it is free among the new two-cells and is not an edge of $g(C_Y)$. We can therefore perform all collapses relative to $g(C_Y)$. The resulting relative CW complex has only additional zero- and one-cells. Adding such cells injects each old first homology group and has free abelian cokernel, as follows directly by extending a maximal forest. Thus $M(g)$ is injective with free cokernel and, over $\mathbb{Z}$, is split injective. □

**The separate colimit obstruction.** The theorem is pointwise and relative to individual arrows. It does not assert injectivity after taking a colimit. The long exact sequence of $0 \to F \to K \to M \to 0$ contains

$$T_1(M) \longrightarrow \mathbb{Z}^2 \longrightarrow \operatorname{colim}_{K^*} K.$$

The two standard basis vectors are the two protected cycles. Their augmentation difference $(1, -1)$ is the first-stage two-dimensional class. Corollary C.28 separately proves that this difference lies outside the image of $T_1(M)$.

## C.3 Sharper intersections of the protected cycles

**Proposition C.14** (Proved: a proper overlap lies on one boundary edge)**.** *Use the protected objects $P_+, P_-$, their marked circles $S_+, S_-$, and their cycle generators from Theorem C.13. For every object $X$ of the repaired construction, every protected arrow $h : P_\epsilon \to X$, and every arrow $g : Y \to X$, the following stronger routing assertion holds. If $h(S_\epsilon) \cap g(Y)$ contains the images of two vertices which are not adjacent in $S_\epsilon$, then $h$ factors through $g$. Consequently, if $h$ does not factor through $g$, that intersection is empty, a single vertex, or two adjacent vertices. The boundary graphs of two distinct protected cells in $G_X$ meet in an empty set, one vertex, or one edge.*

*Proof.* We prove the routing assertion by stage induction. At the first stage there is no arrow between the two distinct protected objects and no nonidentity endomorphism of either. A proper incoming $V$-arrow meets the marked circle in at most one vertex; a $W$-arrow meets it in the endpoints of its one boundary edge. Thus the assertion holds at the birth stage.

Suppose first that $X$ is a new quasi-filter target, with neighboring maps $\alpha_i : T \to X$ and old pair $\phi_0, \phi_1 : S \rightrightarrows T$. The case $g = 1_X$ is immediate. Otherwise cyclically rename the copies so that $g = \alpha_0 g^*$ and write $h = \alpha_j h^*$. Choose two nonadjacent marked vertices $p, q \in S_\epsilon$ whose images lie in $g(Y)$. If $j \neq 0$, take the next index $j'$ on the shortest arc from $j$ to 0. Its total length is at most $2k \leq s - 2n$, so (10) puts $h(p), h(q)$ in the overlap of copies $j, j'$. By (11) that overlap is $\alpha_j \phi_r(S) = \alpha_{j'} \phi_t(S)$ for appropriate $r, t \in \{0, 1\}$. Injectivity of $\alpha_j$ puts $h^*(p), h^*(q)$ in $\phi_r(S)$. The old induction hypothesis, applied to the same two nonadjacent vertices in the fixed source $P_\epsilon$, gives an old arrow $l : P_\epsilon \to S$ with $h^* = \phi_r l$. Hence $h = \alpha_{j'} \phi_t l$. The entire protected arrow has moved one step towards copy zero. Repeating at most $2k$ times writes $h = \alpha_0 \widetilde{h}$. Injectivity then puts $\widetilde{h}(p), \widetilde{h}(q)$ in $g^*(Y)$, and the old assertion factors $\widetilde{h}$ through $g^*$. This factors $h$ through $g$.

At a disjoint-sum target, $g = 1_X$ is again immediate. Otherwise any nonempty overlap puts $h$ and $g$ in the same summand, where the old assertion applies. This completes the induction for every stage and every numerical choice. A subset of a cycle of length at least four with no nonadjacent pair has at most two vertices, and a two-vertex such subset is an edge's pair of endpoints.

Finally suppose two distinct protected arrows have two common marked vertices. If these were nonadjacent in either source circle, the assertion just proved would factor that protected arrow through the other. There are no arrows between $P_+$ and $P_-$, and their only endomorphisms are identities; such a factorization would therefore make the arrows equal. Thus the common vertices must be adjacent in both circles. Their two boundary $W$-arrows share two image points, so the original AKR routing invariant and $\mathrm{End}(W) = \{1\}$ make those $W$-arrows equal. A further common marked vertex would give a nonadjacent pair. The graph intersection is therefore exactly one edge in this case, proving the final assertion. □

**Proposition C.15** (Proved: protected chains supported on two arrow images)**.** *Let $P(X) \subseteq K(X) \subseteq C_1(G_X; \mathbb{Z})$ be the subgroup freely generated by the protected cycles in Theorem C.13. For any two arrows $f : Y \to X$ and $g : Z \to X$, let $U \subseteq G_X$ be the union of their image subgraphs. Then*

$$P(X) \cap C_1(U; \mathbb{Z}) = f_\# P(Y) + g_\# P(Z). \tag{PI}$$

*The same assertion holds with coefficients in an arbitrary abelian group. Thus a protected-cycle combination whose edge support lies in the union uses only protected cells factoring through one of those two arrows.*

*Proof.* The inclusion from right to left follows from functoriality. For the other inclusion, use the unique expansion in the protected-cell basis, and consider the nonempty family, if any, of its cells which factor through neither $f$ nor $g$. Index those cells by $i = 1, \ldots, r$, write their marked images as $S_i$, and let $s_i = |S_i| \geq 45$. Each meets either full arrow image in at most two points, by the protected routing theorem. At a stage $m$ containing all the data, put $q = a_m / a_1 > 3/4$. The weighted counting inequality gives

$$\left| \bigcup_i S_i \setminus f(Y) \right| \geq q \sum_i (s_i - 2).$$

Deleting the second image removes at most $2r$ further points. Consequently

$$\left| \bigcup_i S_i \setminus (f(Y) \cup g(Z)) \right| \geq q \sum_i (s_i - 2) - 2r > \frac{3}{4} \sum_i s_i - \frac{7}{2} r > \frac{1}{2} \sum_i s_i.$$

The final strict inequality uses $s_i \geq 45$. If every point of this outside union belonged to at least two of the indicated marked circles, the number of marked incidences would be at least twice that union's size, contradicting the bound by $\sum_i s_i$. There is therefore a marked vertex outside both images belonging to exactly one of the indicated cells. All other cells in the full expansion factor through $f$ or $g$ and have every marked vertex inside the union, so this vertex belongs to no other cell in the expansion.

Every boundary vertex of a protected cell is marked. Either boundary edge incident to this private vertex consequently belongs to that cell alone, and lies outside $U$. Its coefficient in the given combination is the nonzero cell coefficient, with sign $\pm 1$. This contradicts the assumed edge support in $U$. Thus no such cell occurs, proving (PI). The argument only uses that changing sign preserves a nonzero coefficient, so it works for coefficients in any abelian group. □

**Limitation of this local conclusion.** Proposition C.15 is a statement about chains at one object. It does not assert that the protected subgroup remains injective after taking the colimit of $K$. A path-difference class from Proposition C.10 could agree with a protected combination only after old colimit relations have been used, rather than as chains at its target. Corollary C.28 controls those prior relations for every admissible C.1 construction.

## C.4 A source-specific local convexity bound

The counting invariant of C.2 gives a uniform local geometric statement which is stronger than inducedness and uses no extra choice of target size. It also identifies a class of source objects for which the fixed-path kernel vanishes at every later stage.

**Proposition C.16** (Proved: the shortest possible outside excursion)**.** *Let $X \in K_m$, let $h : Y \to X$ be an arrow of the repaired C.1 construction, and let $J = h_{\#}G_Y \subseteq G_X$. If a simple graph path has distinct endpoints in $J$, all its internal vertices outside $J$, and all its edges outside $J$, its length $r$ satisfies*

$$(2 - a_m)r \geq 1.$$

*For $m \geq 1$ put*

$$R_m = \left\lceil \frac{1}{2 - a_m} \right\rceil - 1.$$

*Every simple path of length at most $R_m$ with endpoints in $J$ is contained in $J$. In particular every arrow image is* two-convex*: every geodesic of length at most two whose endpoints lie in that image stays in the image. This holds at all stages and for every admissible numerical choice.*

*Proof.* The graph $G_X$ is simple by $(c_m)$ and $\mathrm{End}(W) = \{1\}$. Its vertices are the underlying points of $X$, as in Proposition C.10. Write $b_1, \ldots, b_r : W \to X$ for the distinct arrows supplying the edges of the displayed simple excursion. None factors through $h$, because its binary edge is outside $J$. Invariant $(c_m)$ therefore gives $|b_i(W) \cap h(Y)| \leq 1$ for every $i$. The relative counting invariant $(d_m)$ applies and yields

$$\left| \bigcup_{i=1}^{r} b_i(W) \setminus h(Y) \right| \geq a_m r.$$

The binary endpoints of these edges contribute exactly the $r - 1$ internal path vertices outside $h(Y)$. Each $b_i$ has only one further point, namely $b_i v_3$, and these contribute at most $r$ additional outside points. Consequently the same union has size at most $2r - 1$. Combining the bounds proves $a_m r \leq 2r - 1$.

If a simple path with endpoints in $J$ leaves $J$, take a maximal consecutive excursion between two successive visits to $J$. Its endpoints are distinct, because the entire path is simple. Its edges are outside $J$ and its internal vertices are outside $J$. Thus its length is at least $1/(2-a_m)$, and the length of the original path is at least that number as well. This proves the bound with $R_m$. Since $a_m > 3/2$, one has $1/(2-a_m) > 2$, hence $R_m \geq 2$. At stage zero the assertion is immediate from the singleton, edge, and isolated-vertex graphs; equivalently $a_0 = 2$ makes the displayed excursion inequality impossible. □

**Corollary C.17** (Proved: small-diameter sources never kill a class)**.** *Let $f, g : S \rightrightarrows T$ be parallel arrows at an old stage $K_m$, where $m \geq 1$. If each component of $G_S$ has diameter at most $R_m$, then its equalizer subgraph $F$ is connected in every component of $G_S$ that it meets. A repaired quasi-filter attachment for this pair therefore preserves the whole old colimit of $K$, and in particular the whole old binary second homology.*

*The same conclusion holds at any later stage whenever every component of the fixed source $G_S$ has diameter at most two. It does not require prior component injectivity of $f$ or $g$.*

*Proof.* Let $u, v$ be fixed vertices in one source component and put $d = d_{G_S}(u, v) \leq R_m$. Each embedding carries a source shortest path to a target path of length $d$. A target shortest path between the image endpoints therefore has length at most $d$, and Proposition C.16 puts every such path in each arrow image. Inducedness and injectivity then lift it to a source path. In particular the target distance equals $d$.

Let $I_S(u, v)$ be the finite set of all source vertices on shortest $u$–$v$ paths, and write $I_T$ for the corresponding target interval between the common images of $u, v$. The preceding argument gives

$$f(I_S(u, v)) = I_T = g(I_S(u, v)).$$

Hence $g^{-1}f$ is a permutation of $I_S(u, v)$. A nonidentity cycle of this permutation would give a nontrivial directed cycle of the point-transition relation $f(x) \mapsto g(x)$, contrary to the established AKR bounded-chain property (4). This permutation is therefore the identity. Every vertex of every shortest source path from $u$ to $v$ is fixed. The equalizer graph is induced on its fixed vertices by Proposition C.10, so these paths lie in $F$.

Thus $\pi_0(F) \to \pi_0(G_S)$ is injective. The exact fixed-path kernel formula proves the preservation assertion. Finally $R_m \geq 2$ at every stage, giving the uniform diameter-two case. □

**Theorem C.18** (Proved: every first-stage source remains harmless)**.** *In any repaired C.1 construction, let $S$ be an object already in $K_1$. At any later finite stage, every repaired quasi-filter attachment for any pair $f, g : S \rightrightarrows T$ preserves the whole old colimit of the binary cycle functor $K$. It therefore preserves every old binary second-homology class, independently of the target $T$, the chosen pair, and all admissible numerical parameters.*

*Consequently an attachment with a nonzero fixed-path kernel, if one occurs, must have a source object born at stage two or later.*

*Proof.* We first classify the binary graphs of the first-stage objects, allowing different admissible necklace lengths at their births. The original objects, the targets for identity pairs, and all first-stage disjoint sums have graph components of diameter at most two. More explicitly, the equal pair $(v_1, v_1)$ or $(v_2, v_2)$ produces a star together with isolated third points; the equal pair $(v_3, v_3)$ produces disjoint edges and one isolated fixed point. The identity pairs on $V$ or $W$ merely reproduce their underlying graphs on a fresh target.

For a distinct pair involving $v_3$ and one binary endpoint, every binary edge has one active endpoint on a repaired two-vertex point path and one private endpoint. The formulas for $\beta_i^2(0)$ and $\beta_i^2(1)$ show that each active point is used as that binary endpoint at most twice: one coordinate is used twice, one is omitted, and all remaining coordinates are used once.

Accordingly all nontrivial binary components are edges or two-edge stars. They have diameter at most two. The only remaining objects are the two protected targets $P_+, P_-$: each has its unique circle of length $s_\pm \geq 45$, with all other vertices isolated.

Corollary C.17 handles every case except $S = P_+$ or $S = P_-$. In one of these two cases, if $f = g$ the entire graph is fixed. Suppose $f \neq g$. Two fixed marked vertices which were nonadjacent on the protected circle would lie in $f(S_\pm) \cap g(S)$. Proposition C.14 would factor $f$ through $g$. The factor would be an endomorphism of $S$, and the only such endomorphism is the identity. This would contradict $f \neq g$.

Thus the fixed subset of the circular component is empty, a singleton, or a pair of adjacent vertices. In the last case its edge is fixed as well, since the equalizer graph is induced. Every other source component is an isolated vertex. The equalizer graph is therefore connected in each source component that it meets. Proposition C.10 again proves that the attachment kernel is zero. □

**Scope.** The local bound and the first-stage safe-source theorem apply to the arrows actually generated by C.1 and use the source's relative counting invariant. They do not by themselves prove convexity for arbitrary long paths, or connectedness of fixed sets for all later source objects. The separate birth-cover and component-path induction in Theorem C.27 supplies the latter conclusion, and Corollary C.28 completes the universal-choice assertion in C.1.

## C.5 Descent through the generating cover at birth

For a new object $P$ of the repaired C.1 construction, its *generating cover at birth* consists of its prescribed incoming copy maps. These are the two summand maps for a sum, or the maps $\alpha_i : T \to P$ for a quasi-filter. Every nonidentity arrow into $P$ factors through a member of this cover. For the original object $W$, use the three singleton maps $v_i : V \to W$ as its generating cover. The original singleton $V$ will be treated separately.

Two distinct points $p, q \in P$ are called a *forbidden pair* if no member of this cover contains both. This terminology refers only to the fixed source $P$ and its actual generating cover.

**Lemma C.19** (Proved: routing a forbidden pair)**.** *Let $P \neq V$ be an object with its generating cover at birth, and let $p, q \in P$ be a forbidden pair. At every later stage, if $h : P \to X$ and $g : Y \to X$ satisfy*

$$h(p), h(q) \in g(Y),$$

*then $h$ factors through $g$ by an arrow of the constructed category.*

*Proof.* We induct on the stage of $X$, starting at the birth stage of $P$. At that stage the only possible arrow with source $P$ is its identity. If the comparison arrow into $P$ is proper, it factors through one generating image, which cannot contain the forbidden pair. The identity comparison is immediate. There are no arrows from $P$ to distinct objects born at the same stage, or to earlier objects. This also proves the base case $P = W$ in the original category.

Let $X$ now be a new quasi-filter target, with old target $T$, old pair $\phi_0, \phi_1 : R \rightrightarrows T$, and generating maps $\alpha_i : T \to X$. The identity comparison is immediate. Otherwise write

$$h = \alpha_j h^*, \qquad g = \alpha_i g^*.$$

When $j \neq i$, choose the next index $j'$ on the shortest cyclic arc from $j$ to $i$. Its length is at most $2k \leq s - 2n$. Property (10) puts the two displayed points in the overlap of copies $j, j'$. Property (11) identifies that overlap with

$$\alpha_j \phi_r(R) = \alpha_{j'} \phi_t(R)$$

for suitable $r, t \in \{0, 1\}$. Injectivity of $\alpha_j$ places $h^*(p), h^*(q)$ in $\phi_r(R)$. The old induction hypothesis, applied to the same forbidden pair in the fixed source $P$, supplies $l : P \to R$ with $h^* = \phi_r l$. Thus the entire arrow $h$ also factors through the next copy: $h = \alpha_{j'} \phi_t l$.

Repeat along the chosen shortest arc. Once $h = \alpha_i \widetilde{h}$, injectivity places $\widetilde{h}(p), \widetilde{h}(q)$ in $g^*(Y)$. The old assertion factors $\widetilde{h}$ through $g^*$, giving the required factorization through $g$.

At a disjoint-sum target, any nonempty intersection puts the two proper arrows in the same summand, where the old assertion applies. This completes the induction. The argument uses the actual prescribed incoming factorizations, not arbitrary graph embeddings. □

**Lemma C.20** (Proved: pairwise visibility has a common copy)**.** *Let $P \neq V$ have its generating cover at birth. A nonempty subset $U \subseteq P$ with no forbidden pair is contained in one member of that generating cover.*

*Proof.* For $W$, the assertion says that a set with at most one point is contained in one singleton image. For a disjoint sum, any pair of points in different summands is forbidden, so all of $U$ lies in one summand.

For a quasi-filter target, let

$$I_x = \{i : x \in \alpha_i(T)\} \subseteq \mathbb{Z}/s$$

be the visibility set of a point. The repaired coordinate formulas, as recorded in Proposition C.32, make each $I_x$ either the full index cycle or a circular interval with at most

$$w = \max(1, 2n - 2)$$

indices. The extra 1 covers private points even when $n = 1$. For every C.1 attachment,

$$\frac{s}{3n} \geq \left(1 - \frac{a_{m+1}}{a_m}\right)^{-1} > 4.$$

The left side is an integer, so $s \geq 15n > 3w$.

The hypothesis on $U$ says exactly that its visibility sets meet pairwise. Full-circle sets impose no restriction. If a proper set occurs, select one and lift it to an integer interval $I = [0, a]$ with $a \leq w - 1$. Every other proper visibility interval has a unique lift meeting $I$. Every such lift is contained in $[-(w-1), a + (w-1)]$, whose span is at most $3w - 3 < s$. Thus two of these lifts intersect whenever their circular images do: distinct points in that common window cannot differ by $s$. They are therefore pairwise intersecting ordinary intervals. Their largest left endpoint is at most their smallest right endpoint, so they have a common index. That index also belongs to every full-circle visibility set. If all the sets were full circles, any index would work. Hence some generating copy contains every point of $U$. □

**Proposition C.21** (Proved: intersections are existing arrow images)**.** *For any two arrows $h : P \to X$ and $g : Q \to X$ of a repaired C.1 stage, their nonempty point-image intersection is the image of a commuting span already in that stage: there are an object $R$ and arrows $u : R \to P$, $v : R \to Q$ such that*

$$hu = gv, \qquad (hu)(R) = h(P) \cap g(Q).$$

*Consequently every nonempty finite intersection of arrow images in one object is itself an existing arrow image.*

*This statement asserts equality of point sets and the existence of the displayed factorization span. It does not assert the categorical pullback universal property.*

*Proof.* If $h$ factors through $g$, take $R = P$, $u = 1_P$, and the given factor as $v$. Suppose it does not. By Lemma C.19, the nonempty set

$$h^{-1}(g(Q)) \subseteq P$$

contains no forbidden pair. When $P \neq V$, Lemma C.20 places that whole set in a single incoming generating image $e : E \to P$. Replacing $h$ by $he$ does not change its image intersection with $g$:

$$(he)(E) \cap g(Q) = h(P) \cap g(Q).$$

The source has moved strictly earlier in its generation history; for the original object $W$, this step moves to $V$.

Continue until the current arrow factors through $g$. This must happen: otherwise the strict descent reaches $V$. A nonempty intersection at $V$ says that its point is $g(q)$ for some underlying point $q \in Q$. Every point is represented by a $V$-arrow, so the corresponding map $V \to Q$ gives a factorization through $g$. Composing the source-descent maps produces $u$, and this final factorization produces $v$.

For finite intersections, repeat the two-image assertion. At each step the new intersection image is represented by an actual arrow, so the argument applies again; all earlier factorizations remain available by composition. □

**Proposition C.22** (Proved: pointwise equalizers descend to one arrow)**.** *For every parallel pair $f, g : S \rightrightarrows T$ in a repaired C.1 stage, its pointwise fixed subset is empty or is exactly the image of a single arrow $e : R \to S$ already in that stage:*

$$\{x \in S : f(x) = g(x)\} = e(R), \qquad fe = ge.$$

*No categorical equalizer universal property is claimed.*

*Proof.* For $f = g$, use $e = 1_S$. Assume $f \neq g$ and that the fixed subset $F$ is nonempty. If $F$ contained a forbidden pair in the source $S$, Lemma C.19, with $h = f$ and comparison $g$, would factor $f$ through $g$. That factor would be an endomorphism of $S$. All endomorphisms of every constructed object are identities, so this would force $f = g$, a contradiction.

For $S \neq V$, the Helly lemma therefore places $F$ in one generating image $e_0 : E \to S$. Its preimage is exactly $\mathrm{Eq}(fe_0, ge_0)$. If these two composites are equal, then $F = e_0(E)$ and the proof is finished. Otherwise repeat the argument with the strictly earlier source $E$.

This descent terminates. At $V$, a nonempty fixed subset means that the two maps of its singleton agree, and hence are the same concrete arrow. Composing the descent maps gives $e : R \to S$. Injectivity at every descent step gives the asserted exact equality $F = e(R)$, rather than merely containment. □

**Corollary C.23** (Proved: a finite certificate for all future targets)**.** *Fix an object $S$ of a repaired C.1 construction. Suppose every arrow $e : R \to S$ is injective on binary graph components. Then every parallel pair $f, g : S \rightrightarrows T$ at every later stage has a fixed subgraph connected in each component of $G_S$. Every later quasi-filter attachment whose source is this $S$ therefore preserves the whole old cycle colimit and the whole old binary second homology.*

*The hypothesis is a finite condition which can be checked at the birth of $S$, provided the retained finite substage contains all its incoming generators and their ancestors. No subsequent stage adds an object with an arrow into $S$, or adds an arrow between its old objects.*

*Proof.* Proposition C.22 writes any nonempty fixed subset as $e(R)$ for an actual arrow into $S$. Its induced graph is exactly $e_{\#}G_R$. The stated hypothesis gives $\pi_0(F) \to \pi_0(G_S)$ injective, so the exact fixed-path kernel is zero. The construction's full-stage and new-maximal-object rules make the entire incoming category of $S$ permanent from its birth onward. It is finite, and the binary component maps are finite maps, which proves the certificate assertion. □

**Corollary C.24** (Proved: the separate component-injectivity obligation)**.** *Suppose every arrow of an old repaired C.1 stage induces an injection on binary graph components. Then every quasi-filter attachment for a pair in that stage has zero old cycle-colimit kernel. If this component-injectivity assertion holds at every stage, the first-stage binary second-homology class survives the whole infinite construction.*

*Proof.* By Proposition C.22, the fixed subset of any defining pair is empty or $e(R)$ for an existing arrow $e : R \to S$. Arrow images induce their source graphs, by $(c_m)$, and the equalizer graph is induced on its fixed vertices. Consequently its graph is precisely $e_{\#}G_R$. Component injectivity of $e$ is exactly the sufficient condition $\pi_0(F) \to \pi_0(G_S)$ in Proposition C.10. The fixed-path kernel therefore vanishes. Disjoint-sum attachments have zero kernel as well, and filtered unions preserve the resulting injections on homology. □

**Resolution of the component obligation.** The descent results above do not by themselves make arrows injective on binary graph components: a bounded point-transition relation does not exclude component cycles merely by passage to components. The simultaneous induction in Theorem C.27 supplies this separate step. It proves actual-arrow routing for forbidden pairs of components, represents component-image intersections of size at least two by older actual spans, and bounds every component path by the corresponding point-path bound plus one. Consequently Corollary C.24 applies at every stage, and Corollary C.28 completes the universal-choice assertion in C.1.

## C.6 A simultaneous induction on binary components

The following induction separates two levels of structure. Statements about $H(X) = \pi_0 G_X$ concern sets of graph components; every factorization asserted in the argument is nevertheless an actual factorization of arrows in the original concrete category. In particular, equality of component maps is never substituted for equality of arrows.

For a pair of injections $f, g : S \rightrightarrows T$ whose partial point-transition relation has no nontrivial directed cycle, let $L_U(f, g)$ be the maximum number of vertices in a nonconstant directed path, or 1 if there is no such path. Once the analogous component relation is known to be acyclic, define $L_H(f, g)$ in the same way. Actual distinct arrows have $L_U(f, g) \geq 2$, because equal concrete functions are the same arrow.

**Lemma C.25** (Proved: component intersections of size at least two)**.** *Consider a finite generated stage in which:*

1. *every arrow is injective on $H$;*
2. *each source's generating cover at birth, interpreted on $H$, has the Helly property of Lemma C.20;*
3. *the forbidden-pair routing assertion of Lemma C.19 holds for pairs of source* components, *and yields factorizations of actual arrows.*

*If $h : P \to X$, $g : Q \to X$ have at least two components in the intersection of their $H$-images, there are actual arrows $u : R \to P$, $v : R \to Q$ with*

$$hu = gv, \qquad H(hu)(H(R)) = H(h)(H(P)) \cap H(g)(H(Q)). \tag{CI}$$

*If $h$ does not factor through $g$, $R$ is strictly earlier than $P$ in its generation history, allowing the step $W \to V$ at stage zero.*

*Proof.* If $h$ factors through $g$, use that factorization. Otherwise the preimage under the injective map $H(h)$ of the displayed intersection contains no forbidden pair of source components: such a pair would factor the actual arrow $h$ through $g$. The component-cover Helly property puts this whole preimage in one incoming image $H(e)(H(E))$, where $e : E \to P$ is an actual generating arrow. Replacing $h$ by $he$ preserves the entire component-image intersection and descends to an older source.

This descent cannot reach $V$, because its component set has one element whereas the retained intersection has at least two. At $W$, the two components are the binary edge component and its isolated third point. They are a forbidden pair for the singleton generating cover, so a retained two-component intersection would already force factorization. Therefore the descent must stop at an actual factorization through $g$. Composing its descent maps and its final factorization gives $u, v$ in (CI). If $h$ did not initially factor through $g$, at least one descent step was performed. Notice why the restriction to intersections of size at least two is essential: membership of one component in an image does not force equality of its chosen underlying points. □

**Lemma C.26** (Proved: component paths inflate by at most one)**.** *Under the three hypotheses of the preceding lemma, every actual parallel pair $f, g : S \rightrightarrows T$ has an acyclic component transition relation and*

$$L_H(f, g) \le L_U(f, g) + 1. \tag{PB}$$

*Here the absence of nontrivial point-transition cycles for actual arrows is the established AKR property (4).*

*Proof.* Induct on the generation rank of the source $S$, with the claim simultaneously for all targets in the given stage. If $f = g$, the component relation is fixed and the assertion is immediate. Assume $f \neq g$, so $L_U(f, g) \ge 2$.

Put $I = H(f)(H(S)) \cap H(g)(H(S))$. If $|I| \le 1$, a nontrivial directed cycle is impossible, since all its at least two vertices would lie in $I$. A nonconstant directed path of $r$ vertices has $r - 2$ internal vertices in $I$, so $r \le 3$. This proves the claim in this case, including the source $V$.

Suppose $|I| \ge 2$. The actual arrow $f$ cannot factor through $g$: its factor would be an endomorphism of $S$, hence the identity, contradicting $f \neq g$. The preceding lemma gives actual arrows $u, v : R \rightrightarrows S$ with strictly earlier source $R$ and

$$fu = gv, \qquad H(fu)(H(R)) = I. \tag{RS}$$

Injectivity of $H(f)$ and $H(g)$ then gives

$$H(u)(H(R)) = H(f)^{-1}(I), \qquad H(v)(H(R)) = H(g)^{-1}(I).$$

The reduced component transition relation is $H(v)(r) \mapsto H(u)(r)$. If the original component relation has a nonconstant path

$$D_0 \longrightarrow D_1 \longrightarrow \cdots \longrightarrow D_{r-1},$$

write its source components as $C_0, \ldots, C_{r-2}$, so that $H(f)(C_i) = D_i$ and $H(g)(C_i) = D_{i+1}$. Each internal match $H(g)(C_i) = H(f)(C_{i+1})$ lies in $I$. Equation (RS) and injectivity give an element of $H(R)$ whose $H(v)$-image is $C_i$ and whose $H(u)$-image is $C_{i+1}$. Thus

$$C_0 \longrightarrow \cdots \longrightarrow C_{r-2}$$

is a path for the reduced pair $(v, u)$, with one fewer vertex. The same argument turns any nontrivial original component cycle into a reduced component cycle of the same length. All

the $C_i$ are distinct in either construction, by injectivity of $H(f)$. The source induction rules out a reduced component cycle, so the original relation is acyclic.

For comparison with the point-level path bound, any nonconstant point path $x_0 \to \cdots \to x_{p-1}$ for $(v, u)$ satisfies $g(x_i) = f(x_{i+1})$ by $fu = gv$. It therefore extends to the point path

$$f(x_0) \longrightarrow f(x_1) \longrightarrow \cdots \longrightarrow f(x_{p-1}) \longrightarrow g(x_{p-1})$$

for $(f, g)$. The first $p$ points are distinct because $f$ is injective. The last cannot equal an earlier one: a return before the last gives a nontrivial point-transition cycle, and equality to $f(x_{p-1}) = g(x_{p-2})$ would contradict injectivity of $g$. Thus it has $p + 1$ distinct vertices. It follows that

$$L_U(v, u) \leq L_U(f, g) - 1,$$

also when the reduced pair has no nonconstant path, because its left side is then 1 and $L_U(f, g) \geq 2$.

For an original component path with $r \geq 3$, the reduced path is nonconstant, and source induction now gives

$$r - 1 \leq L_H(v, u) \leq L_U(v, u) + 1 \leq L_U(f, g).$$

Hence $r \leq L_U(f, g) + 1$. Paths with at most two vertices already satisfy the bound. This proves the induction. □

**Theorem C.27** (Proved: universal component injectivity)**.** *Every arrow in every repaired AKR C.1 construction is injective on the component sets $H(X) = \pi_0 G_X$. At every finite stage the component transition relation of every actual parallel pair is acyclic and satisfies $L_H(f, g) \leq L_U(f, g) + 1$.*

*Proof.* We maintain simultaneously the three structural hypotheses of Lemma C.25. Once these hold at a stage, Lemma C.26 supplies the path assertion for every parallel pair in that stage.

At stage zero, $H(V)$ is a singleton and $H(W)$ consists of the binary edge component and the isolated third point. Every component map is injective. The singleton generating cover of $H(W)$ has the stated Helly property, and its sole distinct component pair cannot lie in the image of a proper incoming arrow. This proves the forbidden-pair routing base case.

Assume the structural hypotheses at stage $m$, and consider one new quasi-filter target for an actual pair $f, g : S \rightrightarrows T$. Its specified point-path bound is $n$, so $L_U(f, g) \leq n$. The path lemma gives acyclic component paths of size at most $n + 1$. The C.1 numerical inequalities imply

$$s = 4k + 1 \geq 15n, \qquad k \geq n + 1.$$

The second inequality follows, for every integer $n \geq 1$, from $k \geq (15n - 1)/4 > n + 1 - 1$ and integrality; in fact the numerical gap is larger than needed.

The cell-fiber calculation in Proposition C.10 gives the strict graph colimit here, not only a homological comparison. Every target vertex and edge is in a generating copy; its copy-index fiber is a connected path or circle; and every neighboring edge identification comes from an actual $W$-arrow into the overlap source, by $(c_m)$ and (11). Thus there are no additional vertex or edge identifications beyond the neighboring relations. Passing to graph components preserves this colimit, so the component set is the set necklace for $H(f), H(g)$. These old maps are injections. Their paths are acyclic and have at most $n + 1 \leq k$ vertices. The strict set-necklace result therefore identifies the new component set and its copy maps with the repaired actual-path-size beta construction on these component paths. In particular each new generating map is injective on $H$. All other new arrows are composites through one such copy, so they too are injective on $H$.

On these component sets the usual repaired properties (10), (11) hold with the common bound $n+1$. Equivalently, their visibility sets are circular intervals or full circles, adjacent overlaps are the prescribed component-source images, and shortest-index arcs route intersections. The shortest-arc range is valid since $k \geq n+1$. Every proper component visibility set has at most

$$2(n+1) - 2 = 2n$$

indices, including private singleton fibers. Since $s \geq 15n > 6n$, the short-arc Helly proof applies to this new component cover.

The proof of Lemma C.19, now with source components as its marked pair, extends the component forbidden-pair routing assertion to the new target. At each step it uses the old assertion to factor an *actual* arrow through an old neighboring leg; the component maps are used only to locate the two components in that overlap. Thus no faithfulness of $H$ is being assumed. For the newly born source itself, its forbidden pairs cannot be contained in a proper incoming component image, which is the required new base case.

A disjoint-sum target preserves component injectivity and has a disjoint component cover. Its Helly assertion is immediate, and any nonempty overlap of two proper component images lies in one summand, where the old routing assertion applies. Distinct new targets have no arrows between them, so these verifications apply to every new object independently. All three structural hypotheses therefore hold at stage $m+1$. Applying the path lemma there completes the simultaneous induction. □

**Target choices and unused underlying points.** The repaired construction of the supplied Proposition C.1 uses the actual size of each directed point path, one common $k$, shared fixed points, and private copies of the remaining old points. The preceding proof allows all its admissible parameters, all path enumerations, and arbitrary relabelings of these targets. It also remains valid if a target is enlarged by unused underlying points, while retaining exactly the prescribed generating arrows. Here is the required verification for that optional enlargement.

Call an underlying point *represented* if it is the value of a $V$-arrow. Every generated arrow preserves and reflects this property. This holds initially, when all points are represented. Assume it for the old arrows. In a defining transition $f(x) \mapsto g(x)$, both endpoints are represented exactly when $x$ is represented. Consequently every entire point-transition path has a constant represented status. The repaired beta maps therefore never mix represented and unrepresented path pieces. Fixed pieces and private pieces have the same property, and a new point outside all generating images is unrepresented. This proves preservation and reflection for the new copy maps; the sum case and composition are immediate.

Restriction to represented points consequently retains whole actual-size point paths, not truncated paths of the kind excluded by the repair. Its target points are covered by the generating copies and have the same visibility and neighboring-overlap formulas. The actual binary graph has exactly these represented vertices; its edge maps are still the actual $W$-arrows. Thus the strict graph-colimit argument and every component-cover argument above are unchanged.

For the final fixed-subgraph step, apply the point birth-cover descent to fixed *represented* points. These point sets are covered at birth, and the base factorization at $V$ is valid by the definition of representedness. Forbidden-pair routing still factors actual arrows. If two current composites are distinct, descend through an older generating copy containing all the fixed represented points; terminate when the actual composites agree, which must occur by the singleton $V$. This gives an actual arrow whose represented-point image is exactly the fixed vertex set. Its induced graph is the required fixed subgraph. The point-path bound $L_U$ in the simultaneous induction is still computed on the full underlying point sets, so distinct actual

arrows still have $L_U \geq 2$. Hence none of the argument assumes that restriction to represented points is faithful on arrows.

**Corollary C.28** (Proved: universal survival in C.1)**.** *In every repaired C.1 construction, every attachment preserves the old colimit of the binary cycle functor $K$. The first-stage class $\kappa_+ - \kappa_-$ consequently survives to a nonzero element of*

$$H_2(B\,\mathrm{El}_{K^*}(h_V/(cv_1 \sim cv_2)); \mathbb{Z})\,.$$

*Thus every such repaired C.1 category is not total.*

*Proof.* The fixed vertex set of any defining pair is empty or the represented-point image of an actual old arrow. In the covered construction this is Proposition C.22; for optionally enlarged targets it is the represented-point descent in the preceding paragraph. Its induced graph is the image of that source graph. Universal component injectivity therefore makes the fixed subgraph connected in each old source component that it meets. The exact fixed-path kernel is zero. Sum attachments have zero kernel as well. There are no arrows between distinct targets added at one stage, so their cycle-colimit kernels combine as the sum of these zero kernels.

At stage one, for arbitrary admissible first-stage lengths, the only nonzero cycle groups are the independent cyclic groups of the two protected targets. Choosing generators of edge augmentation one gives the augmentation map $\mathbb{Z}^2 \to \mathbb{Z}$, $(a, b) \mapsto a + b$. Its kernel is generated by $\kappa_+ - \kappa_-$. All finite-stage transition maps on this binary second homology are injective by the preceding paragraph. Every finite nerve chain in the countable union occurs in one finite stage, so filtered unions preserve this nonzero class. The binary category of elements therefore has nonzero second integral homology and cannot have contractible components. □

## C.7 A convexity reduction for the exceptional AKR links

For a concrete AKR object $X$, let $G_X$ be the graph whose vertices are the arrows $V \to X$ and whose edges are the arrows $b : W \to X$, joining $bv_1$ to $bv_2$. The functor $h_V/(bv_1 \sim bv_2)$ is $\pi_0 G_X$. In the repaired C.1 construction, the opened counting invariant implies that $G_X$ is simple: distinct $W$-arrows cannot have the same two endpoints. It also implies that an arrow image induces the corresponding source graph, since a $W$-arrow whose two endpoints lie in another arrow image factors through that arrow.

**Lemma C.29** (Proved: fixed sets of convex partial isometries)**.** *Let $S, T$ be finite simple graphs and let $f, g : S \to T$ be induced graph embeddings. Suppose each connected component of each image is geodesically convex in $T$, and that both embeddings induce injections on connected components. Suppose also that the directed graph with edges $f(x) \to g(x)$ whenever $f(x) \neq g(x)$ has no directed cycle. Then the induced subgraph on*

$$F = \{x \in S : f(x) = g(x)\}$$

*is geodesically convex within each connected component of $S$. In particular its intersection with every component of $S$ is empty or connected.*

*Proof.* Convex induced graph embeddings preserve distances within each connected component. Fix $u, v \in F$ in one component of $S$ and put $a = f(u) = g(u)$, $b = f(v) = g(v)$. Let $I_S(u, v)$ be the finite set of all vertices lying on a shortest path from $u$ to $v$, and define $I_T(a, b)$ similarly. Isometry and convexity give

$$f(I_S(u, v)) = I_T(a, b) = g(I_S(u, v)).$$

Thus $g^{-1}f$ is a permutation of the finite set $I_S(u, v)$. If it had a cycle of length greater than one, the corresponding values under $g$ would give a nontrivial directed cycle of the relation $f(x) \to g(x)$, with the orientation reversed. This is excluded. Consequently this permutation is the identity, so $f(x) = g(x)$ for every $x \in I_S(u, v)$. Every shortest $u$–$v$ path therefore lies in $F$, as required. □

**Historical scope of the convexity argument.** The convexity lemma supplies one route to connected fixed subgraphs, and Theorem C.37 carries that route through a controlled choice of sufficiently large targets. That selected geometric construction is retained. The universal C.1 assertion is settled by the separate component descent in Theorem C.27 and Corollary C.28, which use the original numerical bounds and the exact fixed-path comparison of Proposition C.10. The finite convexity checks below record additional geometry of the indicated substage examples.

**Proposition C.30** (Proved: a limitation of a generic preservation lemma)**.** *For arbitrary finite graphs, convex image components, injectivity on connected components, and bounded point-transition paths do not suffice to ensure that the repaired beta gluing is injective on connected components of each old copy. This assertion concerns a generic graph construction; the displayed arrow configuration is not asserted to occur in the category generated by C.1.*

*Proof.* Let $S$ have two isolated vertices, let $T$ have vertex set $\{0,1,2,3\}$ and edges $01, 23$, and define induced embeddings

$$f : S \to T, \quad f = (0,2), \qquad g : S \to T, \quad g = (3,1).$$

Each image component is a singleton, hence convex, and each map sends the two components of $S$ to the two distinct components of $T$. The nontrivial point transitions are the disjoint paths $0 \to 3$ and $2 \to 1$.

Apply the actual-path-size beta formula to these two paths, with $k = 3$, $s = 13$, and the common bound $n = 3$. Write the two path-coordinate sets of the target as $L_j, R_j$ for $0 \le j < 13$. If $(u_i, v_i) = (\beta_i^2(0), \beta_i^2(1))$, the image of the two edges of $T$ in copy $i$ is

$$L_{u_i} R_{v_i}, \qquad R_{u_i} L_{v_i}.$$

The thirteen unordered pairs $\{u_i, v_i\}$ are exactly the edges of the cycle

$$0, 1, 2, 3, 4, 5, 6, 12, 11, 10, 9, 8, 7, 0.$$

The resulting graph is therefore its bipartite double cover. Since the cycle has odd length 13, this double cover is a connected 26-cycle. In particular $L_0 = \alpha_0(0)$ and $R_0 = \alpha_0(2)$ have distance 13 in the new graph, although 0 and 2 lie in different components of $T$.

The file `finite_akr_graph_gluing.py` independently builds all thirteen maps, verifies that the target has 26 vertices and 26 edges, checks degree two and connectedness, and prints a length-13 path between these two vertices. The calculation uses the same repaired beta formula but makes no claim that the two displayed embeddings are among the arrows produced from the prescribed C.1 initial category. Thus it invalidates only the generic preservation lemma with these insufficient hypotheses. □

**Additional exact finite checks.** The program `finite_akr_intersections.py` reconstructs the selected second-stage attachment and the critical third-stage attachment specified in the earlier manuscript. It checks that every arrow induces an injection on $\pi_0 G$. For incoming images of source cardinality at least four it then computes the closure under all finite intersections and verifies that each intersection is connected within every ambient $G$-component that it meets. At the selected second-stage target there are 154 distinct images including the identity, 11781 unordered image pairs, and 307 distinct intersections with at least two vertices. At the critical third-stage target the corresponding counts are 892, 397386, and 1189. In both closures every new nontrivial intersection already occurs as an intersection of two generating images. No connectivity or component-injectivity failure occurs.

The separate program `finite_akr_convex.py` checks geodesic convexity directly. For every vertex of every nontrivial component of a proper incoming image it compares intrinsic graph distances with an ambient breadth-first search, also recording whether an equal-length shortest path can leave the image. It checks all 153 proper larger-source images at the selected second-stage target, using 6732 source searches, and all 891 such images at the critical third-stage target, using 52272 searches. All these image components are geodesically convex. The small-source image components in these finite stages are points or edges and so are geodesically convex automatically; component injectivity was checked separately for all arrows. The program would reconstruct an explicit shorter or equal-length path outside an image upon failure. None was found.

These checks concern the two stated finite substages, not complete later stages and not their infinite union. The output files include the exact object and arrow counts and the source-size restriction used in the intersection enumeration. They provide evidence for a possible convexity induction, not its proof.

**Proposition C.31** (Refuted: the diameter condition cannot be dropped for generic connected graphs)**.** *For a generic pair of induced graph embeddings with connected source and target, strong convexity of both source images, injectivity on components, and bounded nontrivial point paths do not guarantee convexity of the old copies in the repaired beta target. This failure can occur at $n = 3$, $s = 45$, $k = 11$, even though these numbers satisfy the numerical C.1 condition for $a_0 = 2$, $a_1 = 8/5$. The graph configuration below is not asserted to arise among the arrows of the prescribed C.1 construction.*

*Proof.* More generally fix an odd $s = 4k + 1$ with $k \geq 2$, and an integer $L \geq s$. Let $S$ be the path with vertices $0, \dots, L$, let $T$ be the cycle with vertices $0, \dots, 2L + 1$ in cyclic order, and define

$$f(r) = r, \qquad g(r) = L + 1 + r \qquad (0 \leq r \leq L).$$

Both image paths have length $L$, strictly less than half the cycle length $L + 1$. Every shortest path in $T$ between two vertices of either image is therefore the unique path in that image. Thus both embeddings are strongly convex. Source and target are connected, so their component maps are injective. The point-transition relation is the disjoint union of the two-vertex paths $r \mapsto L + 1 + r$. In particular any common bound $n \geq 2$ is valid. The two actual point maps are also increasing for the natural orders on the displayed vertex sets.

Use the actual-path-size beta construction. Its vertex set is $\{0, \dots, L\} \times \mathbb{Z}/s$, with copy $i$ given by

$$\alpha_i f(r) = (r, u_i), \qquad \alpha_i g(r) = (r, v_i), \qquad (u_i, v_i) = (\beta_i^2(0), \beta_i^2(1)).$$

The graph has a vertical path of length $L$ in every column, and for every index pair $\{u_i, v_i\}$ it has the two end bridges

$$(L, u_i)(0, v_i), \qquad (0, u_i)(L, v_i).$$

The unordered index pairs are the edges of the cycle

$$0, 1, \dots, 2k, 4k, 4k - 1, \dots, 2k + 1, 0.$$

Because $s$ is odd, starting at $(0, 0)$ and following only the end bridges along this whole index cycle ends at $(L, 0)$ after $s$ edges. The two vertices belong to copy zero, where their intrinsic distance is $L$. This bridge path leaves copy zero. If $L > s$, it is shorter than the old distance; if $L = s$, it has the old distance and is an outside shortest path. To verify the latter assertion, any path that uses an interior vertex of a vertical column and connects distinct column ends must traverse a full column and has length at least $L$. The bridge subgraph on the two ends of each column is the connected double cover of an odd $s$-cycle, hence a $2s$-cycle; the two ends

of column zero are opposite and have bridge distance $s$. Thus the new distance is exactly $\min\{L, s\}$.

For a strict shortening at numerically admissible parameters, take $L = 46$, $n = 3$, $k = 11$, and $s = 45$. The source is a 46-edge path, the old target a 94-cycle, and the repaired target has 2115 vertices and 2160 edges. The indicated old distance 46 becomes 45. The standalone program `akr_connected_convex_counterexample.py` verifies both old convex embeddings, the point-path bound, all neighboring copy identities, the explicit 45-edge bridge path, and its exact new distance. Its JSON certificate specifies every map and every vertex of the shorter path.

The numerical compatibility is not a counterexample to universal C.1. In particular, if the displayed old cycle were one of the protected first-stage circles, either proper half-cycle arrow image would contradict Proposition C.14. The specific realization of such a configuration among later generated objects remains an additional requirement. □

## C.8 Ordering a sufficiently long repaired necklace

We first specify the set-colimit assertion used in the ordering argument. The target in this subsection is exactly the covered repaired target: it has the prescribed fixed points, private points, and actual-length path targets. No additional unused points are required.

**Lemma C.32** (Proved: the repaired target is the strict necklace colimit)**.** *Let $f, g : S \rightrightarrows T$ be injections of finite sets. Suppose the nonidentity part of the relation $f(x) \mapsto g(x)$ has only directed paths. Let $n \geq 2$ bound their numbers of vertices, choose $k \geq n$, and put $s = 4k + 1$. Let*

$$\alpha_i : T \longrightarrow Z_k, \qquad i \in \mathbb{Z}/s,$$

*be the actual-path-length repaired beta maps. Define*

$$(\lambda_i, \mu_i) = \begin{cases} (g, f), & 0 \leq i < 2k, \\ (g, g), & i = 2k, \\ (f, g), & 2k < i < 4k, \\ (f, f), & i = 4k. \end{cases}$$

*Then $Z_k$ is the coequalizer in sets of*

$$\coprod_{i \in \mathbb{Z}/s} S \rightrightarrows \coprod_{i \in \mathbb{Z}/s} T, \qquad (i, x) \longmapsto (i, \lambda_i x), \quad (i, x) \longmapsto (i + 1, \mu_i x). \tag{NC}$$

*In particular, compatible maps from the displayed copies of $T$ to any set induce a unique map from $Z_k$.*

*Proof.* Write a nontrivial relation path as $t_0 \mapsto t_1 \mapsto \cdots \mapsto t_{m-1}$, where $m \geq 2$. Its target consists of the $s$ points $0, \ldots, 4k$, and the restriction of copy $i$ is

$$\beta_i^m(t) = \begin{cases} t + i, & i \leq 2k,\ t + i \leq 2k, \\ t + i + 2k - m + 1, & i \leq 2k,\ t + i > 2k, \\ t - i + 6k - m + 2, & i > 2k,\ t - i + 6k - m + 2 > 2k, \\ t - i + 4k, & i > 2k,\ t - i + 6k - m + 2 \leq 2k. \end{cases}$$

Substitution gives $\alpha_i \lambda_i = \alpha_{i+1} \mu_i$. It also gives the exact consecutive overlap: for $i \leq 2k$ it is the image under $\beta_i^m$ of $\{1, \ldots, m - 1\}$, and for $i > 2k$ it is the image of $\{0, \ldots, m - 2\}$. These are exactly the images of the appropriate source edges of the path under $\lambda_i$.

For clarity, every identification in the target is generated by these neighboring identifications. For a target point $u$, its visibility set $I_u = \{i : u \in \beta_i^m(\{0, \dots, m-1\})\}$ is

| range of $u$ | $I_u$ |
|---|---|
| $0 \le u \le m-2$ | $[4k-u, 4k] \cup [0, u]$ |
| $m-1 \le u \le 2k$ | $[u-m+1, u]$ |
| $2k+1 \le u \le 4k-m+1$ | $[6k-m+2-u, 6k+1-u]$ |
| $4k-m+2 \le u \le 4k$ | $[u-2k, 6k+1-u]$. |

Each row is a nonempty cyclic interval. Because every $\beta_i^m$ is injective, there is one preimage of $u$ at each index of $I_u$. The consecutive-overlap calculation joins exactly the adjacent preimages. Thus every such fiber is connected by the relations in (NC).

A fixed point has one preimage in every copy, joined cyclically by those same relations. A private point has exactly one preimage. The three kinds of targets are disjoint, and the visibility table also proves that all path-target points are covered. Therefore $\coprod_i T \to Z_k$ is onto, and each of its fibers is exactly one equivalence class generated by the two displayed maps. This is the stated strict coequalizer. □

**Proposition C.33** (Proved: all sufficiently long repaired necklaces are orderable)**.** *Let $S, T$ be finite linearly ordered sets and let $f, g : S \rightrightarrows T$ be increasing injections. Fix $n \ge 2$ at least the number of vertices in every nontrivial path of the relation $f(x) \mapsto g(x)$. There is an integer $K \ge n$ such that, for every $k \ge K$, the covered repaired target $Z_k$ admits a linear order making every generating map $\alpha_i : T \to Z_k$ increasing. The prescribed order on $T$ is unchanged. The assertion permits any additional lower bounds on $k$.*

*Proof.* The partial bijection $f(S) \to g(S)$, $f(x) \mapsto g(x)$, is increasing. It has no nontrivial directed cycle: restriction to such a cycle would give a nonidentity increasing permutation of a finite linear order. Its nonfixed components are therefore paths, so Lemma C.32 applies.

Embed $T$ increasingly in $\mathbb{R}$. Extend the increasing partial bijection to an increasing homeomorphism

$$h : \mathbb{R} \longrightarrow \mathbb{R}, \qquad h(f(x)) = g(x) \quad (x \in S). \tag{H}$$

For nonempty $S$, this can be done by linear interpolation through the finitely many points $(f(x), g(x))$, with slope-one translations on the two unbounded complementary intervals. If $S$ is empty, take $h$ to be the identity.

Every component $J$ of $\mathbb{R} \setminus \mathrm{Fix}(h)$ is invariant under $h$. On $J$, either $h(t) > t$ everywhere or $h(t) < t$ everywhere. In the first case $h^r(t)$ tends to the right endpoint of $J$ as $r \to \infty$; in the second it tends to the left endpoint. Indeed the iterates are monotone, and a finite limit in the interior would be fixed by continuity, contrary to the definition of $J$. The assertion includes convergence to an infinite endpoint.

For each $x \in S$, the points $f(x)$ and $g(x)$ either coincide in $\mathrm{Fix}(h)$ or lie in the same such component $J$. In the nonfixed case put

$$B_J = (f(S) \cup g(S)) \cap J.$$

Only finitely many components meet these finite sets. Consequently there is $K \ge n$ such that, for every $k \ge K$,

$$\begin{cases} h^{2k}(g(S) \cap J) > \max B_J, & h|_J > \mathrm{id}, \\ h^{2k}(g(S) \cap J) < \min B_J, & h|_J < \mathrm{id}\,. \end{cases} \tag{S}$$

An inequality for a finite set means that every element satisfies it. Once these inequalities hold, monotonicity of the iterates makes them hold for every larger $k$.

Fix such a $k$. There is an increasing homeomorphism $w$ of $\mathbb{R}$ satisfying simultaneously

$$w(g(x)) = f(x) \quad (x \in S), \qquad w(h^{2k}g(x)) = h^{2k}g(x) \quad (x \in S). \tag{W}$$

To verify this, first consider the prescription on its finite domain. Within a positive component $J$, all points of $h^{2k}g(S)\cap J$ lie strictly above both $f(S)\cap J$ and $g(S)\cap J$; in a negative component they lie strictly below both. Thus the increasing bijection $g(x) \mapsto f(x)$ is compatible with fixing these additional points. Prescriptions in different components preserve their relative order. At fixed points the two prescriptions agree, since $g(x)$ fixed implies $f(x) = g(x)$ by injectivity of $h$. The finite prescription is therefore well-defined and strictly increasing. Linear interpolation, with slope-one tails, extends it to the required homeomorphism $w$.

Define increasing maps $a_i : T \to \mathbb{R}$ by

$$a_i = \begin{cases} h^i|_T, & 0 \le i \le 2k, \\ (w \circ h^{4k+1-i})|_T, & 2k < i \le 4k. \end{cases}$$

They satisfy exactly the prescribed necklace equations. For $0 \le i < 2k$, equation (H) gives

$$a_i g = h^i g = h^{i+1} f = a_{i+1} f.$$

At the upper junction, equation (W) gives

$$a_{2k} g = h^{2k} g = w h^{2k} g = a_{2k+1} g.$$

For $2k < i < 4k$,

$$a_i f = w h^{4k+1-i} f = w h^{4k-i} g = a_{i+1} g.$$

At the lower junction,

$$a_{4k} f = whf = wg = f = a_0 f.$$

All equalities are equalities of actual functions on $S$.

The strict set-colimit property of Lemma C.32 now gives a unique function

$$\rho : Z_k \longrightarrow \mathbb{R}, \qquad \rho\alpha_i = a_i.$$

Order the finite set $Z_k$ first by its $\rho$-values and then by any chosen linear order within each fiber of $\rho$. If $t < t'$ in $T$, then $a_i(t) < a_i(t')$, so $\rho\alpha_i(t) < \rho\alpha_i(t')$. Hence this order makes every $\alpha_i$ strictly increasing. Possible equal values belonging to different copies cause no problem, since no two points of one copy have equal $\rho$-values. This proves the proposition. □

**Remark C.34** (Proved scope and effective parameter selection)**.** The proposition asserts existence for all sufficiently large $k$; it does not assert orderability at every minimally admissible value. Its proof is a strict set-colimit argument and uses no conversion of a homotopy-coherent cocone into a strict one.

The choice can be made effectively. Place $T$ at integer coordinates and choose the piecewise-linear $h$ just described, with rational coefficients. Its fixed intervals and the endpoints of the other components are determined by finitely many affine equations. Iterate the finitely many rational points $g(x)$ and increase $k$ until (S) holds, together with all additional numerical lower bounds. The convergence argument proves termination. The subsequent prescriptions for $w$ and $\rho$ are finite and rational. Thus arbitrary tie-breaking in the last step may also be replaced by a fixed deterministic ordering rule.

**Proposition C.35** (Refuted: every admissible-length ordered necklace is orderable)**.** *The sufficiently-large-$k$ hypothesis in Proposition C.33 cannot be replaced, for arbitrary increasing maps of finite linear orders, by the condition that $k$ bounds the sizes of their nontrivial point paths. The failure can occur at $n = 3$, $s = 45$, $k = 11$, satisfying the numerical C.1 condition for $a_0 = 2$, $a_1 = 8/5$. The pair below is not asserted to occur among the actual generated arrows of C.1, so this does not refute the universal statement of Part C.1 or the selected construction already proved.*

*Proof.* For any integer $k \geq 2$, put

$$d = 2k, \qquad s = 2d + 1 = 4k + 1, \qquad M = d(d + 1).$$

Give the finite sets

$$S = \{0, \dots, M\}, \qquad T = \{0, \dots, 2M + s\}$$

their natural linear orders, and define

$$f, g : S \rightrightarrows T, \qquad f(r) = 2r, \quad g(r) = 2r + s.$$

Both are increasing injections. Their images are disjoint, since $s$ is odd, and their nontrivial point relation consists entirely of two-vertex paths. Thus the repaired target with this $k$ and actual path size two is defined and satisfies the usual neighboring identities.

Write those identities as $\alpha_i \lambda_i = \alpha_{i+1} \mu_i$, with cyclic indices and

$$(\lambda_i, \mu_i) = \begin{cases} (g, f), & 0 \leq i < d, \\ (g, g), & i = d, \\ (f, g), & d < i < 2d, \\ (f, f), & i = 2d. \end{cases}$$

We give an explicit cycle of forced strict inequalities in the repaired target. Begin with $r_{-1} = M$ and define

$$r_i = \begin{cases} r_{i-1} - d, & 0 \leq i \leq d, \\ r_{i-1} + d + 1, & d < i \leq 2d. \end{cases}$$

All these indices lie in $S$: the first part decreases from $M - d$ to zero, and the second increases from $d + 1$ to $M$. In particular $r_{2d} = r_{-1}$, so this is a cyclic list.

For $0 \leq i \leq d$, the incoming and outgoing old points in copy $i$ are $f(r_{i-1})$ and $g(r_i)$, and

$$g(r_i) = 2(r_{i-1} - d) + (2d + 1) = f(r_{i-1}) + 1.$$

For $d < i \leq 2d$, they are $g(r_{i-1})$ and $f(r_i)$, and

$$f(r_i) = 2(r_{i-1} + d + 1) = g(r_{i-1}) + 1.$$

Thus in every copy the incoming point is strictly smaller than the outgoing point, in fact immediately precedes it in $T$. Equivalently,

$$\mu_{i-1}(r_{i-1}) < \lambda_i(r_i) \qquad (0 \leq i < s),$$

where the lower junction uses $\mu_{2d} = f$. If every $\alpha_i$ preserved the old linear order, these inequalities and the neighboring equalities would give

$$\alpha_0 \mu_{s-1}(r_{s-1}) < \alpha_0 \lambda_0(r_0) = \alpha_1 \mu_0(r_0) < \cdots < \alpha_{s-1} \lambda_{s-1}(r_{s-1}) = \alpha_0 \mu_{s-1}(r_{s-1}).$$

This is impossible in a linear order. No order on this repaired target therefore makes all its copy maps increasing.

Taking $k = 11$, $n = 3$ gives $d = 22$, $M = 506$, and $s = 45$. The source has 507 points, the old target has 1058 points, and the repaired target has 24795 points. Its forced precedence cycle has 45 edges. The standalone program `akr_ordering_counterexample.py` constructs the repaired maps, verifies the source path sizes and all neighboring identities, and checks each consecutive-point comparison and equality in that cycle. The output and JSON certificate give all 45 comparisons. Since $45/(3 \cdot 3) = 5 = (1 - (8/5)/2)^{-1}$, these are numerically admissible C.1 parameters for a first-step ratio, though no realization of this particular source pair in the prescribed construction is claimed. □

## C.9 A choice of large repaired targets with persistent second homology

**Lemma C.36** (Proved: convexity after a sufficiently long attachment)**.** *Let $f, g : S \rightrightarrows T$ be a parallel pair at a finite stage of the repaired AKR construction. Suppose that each of the induced graph embeddings $G_S \to G_T$ is injective on connected components and that the image of each component is geodesically convex. Here geodesically convex means that every shortest graph path between two of its vertices lies in that image. Let $D$ be the largest diameter of a component of the finite graph $G_T$, taking the diameter of a singleton to be zero. Choose the repaired path bound $n \geq 2$ and a parameter $s = 4k + 1$, $k \geq n$, such that*

$$s > (2n - 2)(D + 1). \tag{LC}$$

*Then every generating map $\alpha_i : T \to Z$ of the repaired target has geodesically convex image on each component of $G_T$. This lemma alone does not assert injectivity on components of $G_T \to G_Z$.*

*Proof.* We first record the elementary linear-amalgam fact used below. If two graphs are glued along a graph whose maps into both pieces are injective on components and have geodesically convex component images, then each component of either piece is geodesically convex in the amalgam, and the two piece maps are injective on components. Indeed, a component of either piece contains at most one component of the gluing graph. Thus the component incidence graph of this gluing is a matching: each nontrivial resulting component is one component of each piece glued along a single connected convex subgraph. Any excursion of a shortest path from one piece into the other has endpoints in that common subgraph. If the excursion were shortest between its endpoints, convexity in the other piece would place it in the common subgraph; if it were not shortest, replacing it would shorten the original path. Either possibility rules out a shortest excursion outside the original piece. The component and convexity claims follow. Induction applies this fact to a linear sequence of copies of $G_T$, glued along copies of $G_S$ by any of the neighboring pairs $(g, f), (g, g), (f, g), (f, f)$. Every copy is componentwise convex and injective on components in that linear amalgam. At each induction step the image of the next gluing component is convex in the previous amalgam by transitivity of convexity, and is still injective on components.

Now fix a generating copy, two vertices in one of its source components, and a shortest path $P$ between their images in $G_Z$. The path in the old component gives $|P| \leq D$. As in the cell-fiber proof of Proposition C.10, a target vertex is visible either in every copy, or in a proper cyclic interval of indices. The explicit repaired path formulas bound the size of every proper such interval by $2n - 2$; a private point has a singleton interval. Explicitly, on an actual path of $l \geq 2$ vertices the two middle visibility ranges have size $l$, and the two folded ranges have sizes $2(u + 1)$ for $0 \leq u \leq l - 2$ and $2(4k - u + 1)$ for $4k - l + 2 \leq u \leq 4k$. Each is at most $2l - 2 \leq 2n - 2$. There are at most $D + 1$ vertices of $P$. By (LC), the union of all proper visibility intervals of these vertices misses an index $r$. Vertices visible in every copy are deliberately omitted from this union.

Delete the one gluing overlap from copy $r-1$ to copy $r$, with indices taken modulo $s$. The remaining diagram is a linear chain containing all $s$ copies of $G_T$. Let $L$ be its graph colimit. Its canonical map $q : L \to G_Z$ is the quotient which restores the omitted overlap. For every vertex of $P$, its cell-fiber graph remains connected after that overlap is deleted: a proper visibility interval avoids $r$, so it has no edge crossing this cut; a full circular fiber becomes a path. The same is true for every edge of $P$. Its visibility set is an interval or the entire index cycle. If either endpoint has proper visibility, the edge's interval lies in that endpoint's interval and avoids $r$. If both endpoints have full visibility, the routing invariant $(c_{m+1})$ factors the edge through every copy, since its two endpoints lie in that copy's image. Its fiber is circular and again becomes a path.

The preimages of a cell of $G_Z$ in $L$ are exactly the components of that cell's fiber after the indicated gluing edge is removed. Thus each vertex and edge used by $P$ has a unique preimage in $L$. These preimages respect incidence, because the quotient is a graph map and the endpoint preimages are unique. They give a path $\widetilde{P}$ of the same length in $L$, whose endpoints are the original vertices of the selected copy of $G_T$.

The selected copy component is convex in $L$ by the linear-amalgam fact. Its intrinsic distance between those endpoints is therefore their distance in $L$. This distance is at most $|\widetilde{P}|$. On the other hand, mapping an intrinsic shortest path to $G_Z$ shows $|P|$ is at most that same intrinsic distance. The two lengths are equal, so $\widetilde{P}$ is shortest in $L$ and convexity places it entirely in the selected copy. Mapping back by $q$ places $P$ in that copy's image. This proves the claim for every shortest path. □

**Theorem C.37** (Proved: a selected repaired AKR category is not total)**.** *For every sequence $(a_m)$ satisfying the numerical requirements of the repaired AKR Construction C.1, one can choose its finite targets so that all of the following hold at every stage:*

1. *the underlying point sets $X$, and separately the sets $H(X) = \pi_0 G_X$, admit linear orders for which every arrow induces an increasing injection on each of these two sets;*
2. *every arrow image component in $G_X$ is geodesically convex;*
3. *every attachment induces an injection on the second integral homology of the binary category of elements.*

*For the resulting countable union $A = \bigcup_m K_m$, the action*

$$H = h_V/(bv_1 \sim bv_2)$$

*satisfies*

$$H_2(B\,\mathrm{El}_A\, H; \mathbb{Z}) \neq 0.$$

*In particular there exists a quasi-filtered category which is not total. This is an existence theorem about selected sufficiently large targets; it does not assert the universal choice statement of Part C.1.*

*Proof.* Initially $G_V$ is a point and $G_W$ is one edge and one isolated vertex. Give their underlying point sets and their component sets arbitrary linear orders. Every initial arrow is then increasing on points and on components, injective on components, and componentwise convex. The point order and component order are auxiliary independent orders; the quotient map $X \to H(X)$ is not required to preserve them.

Assume these assertions at a finite stage $K_m$, and consider one parallel pair $f, g : S \rightrightarrows T$. By the induction hypothesis both the point maps $f, g : S \rightrightarrows T$ and the component maps $H(f), H(g) : H(S) \rightrightarrows H(T)$ are increasing injections of finite linear orders. Neither partial transition relation has a nontrivial directed cycle: on the finite set of vertices of such a cycle

the induced increasing bijection would be an increasing permutation, hence the identity. Both relations therefore consist of fixed points and finite directed paths. More generally every closed morphism-chain relation is a composition of restrictions of increasing maps and their increasing partial inverses. It is an increasing partial bijection on a finite order and has the same boundedness property. Thus the AKR property (4) holds here directly from these auxiliary orders.

Choose an odd integer $n \geq 3$ at least as large as the number of vertices in every nontrivial point path and every nontrivial component path. Apply Proposition C.33 to the ordered component pair and to the underlying ordered point pair. For each it permits every sufficiently large $k$ and orders the strict set colimit of its repaired necklace so that all canonical copies are increasing injections. Choose $k$ beyond both thresholds, and still larger if necessary so that (LC) holds and so that

$$s = 4k + 1, \qquad s/(3n) \in \mathbb{N}, \qquad \frac{s}{3n} \geq \frac{1}{1 - a_{m+1}/a_m}.$$

This is possible: since $3n$ is odd, the integers $s$ divisible by $3n$ and congruent to 1 modulo 4 form an unbounded arithmetic progression. The ordinary finite repaired target with these parameters is an admissible target of C.1. Its point-set order is the order supplied by Proposition C.33 and Lemma C.32 for the point pair; all $\alpha_i$ are increasing as actual point maps.

We check that the component order just chosen is an order on the actual $H(Z)$. The vertex and edge fiber calculation in Proposition C.10 identifies $G_Z$ with the strict graph colimit of the neighboring diagram of $G_S$ and $G_T$. Graph components are computed by the coequalizer of the two endpoint maps from the edge set to the vertex set. Since colimits of sets commute with colimits, taking components preserves this graph colimit. Hence $H(Z)$ is the strict set colimit of the same necklace of $H(S), H(T)$. Lemma C.32, applied to their actual component path sizes, identifies this with the component target ordered above. Transfer that order along this canonical bijection. The maps $H(\alpha_i)$ are increasing injections.

Lemma C.36 supplies convexity of every component image of $\alpha_i$. Every other incoming arrow is $\alpha_i b$ for an old arrow $b$, so increasing component injectivity follows by composition, actual point monotonicity also follows by composition, and componentwise convexity follows by transitivity. For a weak disjoint sum use the disjoint union graph and concatenate the two old orders, separately for points and for components. Its incoming maps satisfy both assertions. There are only finitely many pairs and sums at a finite stage; make these choices for all of them, using fresh targets as the construction prescribes. No maps between distinct new targets are added. This proves the two geometric assertions at every stage.

**A deterministic instance.** To remove all choices from one counterexample, fix

$$a_0 = 2, \qquad a_m = \frac{3}{2} + \frac{1}{10 \cdot 2^{m-1}} \quad (m \geq 1).$$

These numbers strictly decrease and remain above $3/2$. The required reciprocals are 5 at $m = 0$ and $15 \cdot 2^m + 2$ at $m \geq 1$, hence integers. Initially label $V$ and $W$ by $0, 1$, label their points by integers, order the three points of $W$ by $v_1 < v_2 < v_3$, and order its two graph components with the edge component first. Enumerate all subsequent objects, points, arrows, and ordered pairs lexicographically by their finite construction tags; arrows are stored as their tuples of point values.

At stage $m$, take the least odd $n \geq 3$ bounding every point-path and component-path size for every parallel pair of the finite stage. Let $D$ be the largest diameter of any of its graph components. Test $k = n, n + 1, \ldots$ in order. Retain only those satisfying the three displayed C.1 arithmetic conditions and $4k + 1 > (2n - 2)(D + 1)$. For each retained $k$, form the finite

repaired target for every ordered parallel pair. On each target point set form the precedence graph with arcs

$$\alpha_i(x) \longrightarrow \alpha_i(y) \quad \text{for all } i \text{ and all } x < y \text{ in } T;$$

on its actual component set form the analogous graph with arcs $H(\alpha_i)(C) \to H(\alpha_i)(C')$ for $C < C'$ in $H(T)$. A loop is counted as a directed cycle. Choose the first retained $k$ for which all these finitely many precedence graphs are acyclic. The two orderability applications prove that this search terminates. Assign each resulting point set and component set its lexicographically least topological ordering, always selecting the least available tag at a tie. This is a prescribed finite algorithm. It can use one common $n, k$ for all pairs at the stage, so it also covers that more restrictive parameter convention.

Fixed-point, private-point, and path-coordinate tags specify the covered repaired target uniquely; path components are indexed by their least old point tag. Give every target a fresh tag consisting of its stage, construction type, source and target tags, and ordered pair of arrow tuples. For each ordered pair of old objects also add the freshly tagged disjoint-sum target, with left/right point tags and the concatenated point and component orders. Keep the old hom-sets, insert the prescribed generating maps, and close under composition as actual functions, identifying equal function tuples with the same source and target. These finite rules specify all objects and arrows. Composition is ordinary function composition, so it is associative and has the indicated identity functions. Each stage is finite, hence the union is a countable small category. The following homology argument applies to this fully specified instance as well as to the preceding existence construction.

For every parallel pair at such a stage, Lemma C.29 now applies: its old images are convex, its component maps are injective, and actual point monotonicity excludes a nontrivial point-transition cycle. Its fixed subgraph therefore meets each source component in an empty or connected subgraph. The exact kernel formula of Proposition C.10 shows that adjoining the pair's quasi-filter target kills no old second homology class. A disjoint-sum target also kills none: its cycle group is the direct sum of the cycle groups of its two summands, so the map $a$ in Proposition C.5 is injective. Adding the finitely many new targets one at a time proves injectivity across every full stage.

Finally the first stage already has a nonzero class, independently of how large these first choices were. The two quasi-filter targets $P_+, P_-$ for $(v_1, v_2)$ and $(v_2, v_1)$ each have one binary graph cycle; orient their generators $\kappa_+, \kappa_-$ so that their augmentations are 1, as follows from the signed neighboring count $2k - (2k-1) = 1$. All other first-stage graphs are forests. For a pair involving $v_3$ and exactly one of $v_1, v_2$, every binary edge has its other endpoint private to that copy, so it cannot belong to a cycle. For a repeated point map the binary graph is a star, a disjoint union of edges, and isolated vertices. The identity quasi-filters and the initial weak sums also give forests. There are no arrows between distinct new first-stage targets, and the old cycle groups at $V, W$ are zero. Consequently

$$\operatorname{colim}_{K_1} K = \mathbb{Z}\kappa_+ \oplus \mathbb{Z}\kappa_-, \qquad H_2(B\operatorname{El}_{K_1} H; \mathbb{Z}) = \ker(\mathbb{Z}^2 \xrightarrow{(1,1)} \mathbb{Z}) \cong \mathbb{Z}.$$

The generator $\kappa_+ - \kappa_-$ survives every subsequent stage. Every finite nerve chain of the union occurs in some stage, so its normalized chain complex is the filtered union of the stage chain complexes. Filtered unions of abelian groups are exact; therefore homology commutes with this union, and the persistent generator is nonzero in the displayed $H_2$ of $A$. The repaired construction is quasi-filtered, whereas totality would force this category of elements to have zero positive homology. Hence $A$ has the claimed properties. □

## C.10 Finite tests for totality: Lead 6

**Lemma C.38** (Known, with the reduction proved: finite-poset representatives)**.** *Let $C$ be a small category and let $n \geq 1$. Every free homotopy class of maps $S^n \to |NC|$ has a*

*representative* $|Nv| : |ND| \to |NC|$*, after a homeomorphism* $|ND| \cong S^n$*, where* $D$ *is the poset of nonempty faces of a finite triangulation of* $S^n$ *and* $v : D \to C$ *is a functor.*

*Proof.* We recall the simplicial approximation argument, including the change of domain that is needed here. Choose a finite ordered simplicial complex $K$ triangulating $S^n$. The natural weak equivalence $NC \longrightarrow \mathrm{Ex}^\infty NC$ has Kan target. Consequently a homotopy class $|K| \to |NC|$ is represented by a simplicial map $f : K \to \mathrm{Ex}^\infty NC$. Since $K$ has finitely many nondegenerate simplices, $f$ factors through $\mathrm{Ex}^r NC$ for some finite $r$. The subdivision–extension adjunction gives a simplicial map $g : \mathrm{Sd}^r K \to NC$. Under realization the adjunction compares $|g|$ with the original class by the last-vertex maps $\mathrm{Sd}^r K \to K$; their realizations are homotopic to the subdivision homeomorphisms. We may increase $r$ to ensure $r \geq 1$. Now $\mathrm{Sd}^r K$ is the nerve of the poset $D$ of nonempty faces of the finite complex $\mathrm{Sd}^{r-1} K$. The nerve is fully faithful on categories: the images of vertices and edges of $g$ specify objects and arrows, and its two-simplices impose precisely the composition equalities. Thus $g = Nv$ for a unique functor $v : D \to C$. The facts about $\mathrm{Ex}^\infty$ and last vertices used in this argument are (Goerss and Jardine, 2009, Theorem I.11.4, Section III.4, and Theorem III.4.8); they concern arbitrary simplicial sets and do not require $NC$ to be a simplicial complex. □

**Theorem C.39** (Proved: Lead 6)**.** *Let* $A$ *be a small nonempty connected category. For a functor* $u : D \to A$*, define the covariant* $A$*-set*

$$H_u(a) = \pi_0(u \downarrow a), \qquad H_u(b)([d, f]) = [d, bf]$$

*for* $b : a \to a'$*. Here* $(d, f)$ *has* $f : u(d) \to a$*, and an arrow* $(d, f) \to (d', f')$ *is* $\delta : d \to d'$ *with* $f'u(\delta) = f$*. Define a functor*

$$j_u : D \longrightarrow \mathrm{El}_A(H_u), \qquad d \longmapsto (u(d), [d, 1_{u(d)}]),$$

*whose value on an arrow is its image under* $u$*. The following conditions are equivalent:*

1. *Every component of every* $\mathrm{El}_A(H)$*, for every covariant* $H : A \to \mathsf{Set}$*, is weakly contractible.*
2. *For every finite poset* $D$ *and every* $u : D \to A$*,* $Bj_u$ *is null-homotopic on every connected component of* $BD$*.*
3. *The same conclusion holds when* $D$ *is the face poset of a finite triangulation of* $S^n$*, for every integer* $n \geq 1$*.*

*In particular the tests in (3) suffice; no bound on the dimension of the spheres is being imposed.*

*Proof.* The arrow $u(\delta) : u(d) \to u(d')$ defines an arrow in the displayed category of elements because $(d, u(\delta))$ and $(d', 1)$ are joined by $\delta$ in $u \downarrow u(d')$. This proves that $j_u$ is well-defined.

The essential factorization is strict, not merely a homotopy factorization. Let $H : A \to \mathsf{Set}$ and let $v : D \to \mathrm{El}_A(H)$ be a functor. Write $v(d) = (u(d), h_d)$, where $u = \pi v$ and $\pi : \mathrm{El}_A(H) \to A$ is the projection. Define

$$\eta_v : H_u \longrightarrow H, \qquad (\eta_v)_a([d, f]) = H(f)(h_d).$$

If $f'u(\delta) = f$, then $H(f)(h_d) = H(f')H(u(\delta))(h_d) = H(f')(h_{d'})$, so this formula is constant on components of $u \downarrow a$. It is natural in $a$ because $H(b)H(f)(h_d) = H(bf)(h_d)$. It therefore induces a functor over $A$,

$$E_v : \mathrm{El}_A(H_u) \longrightarrow \mathrm{El}_A(H), \qquad (a, \xi) \longmapsto (a, (\eta_v)_a(\xi)),$$

which is the identity on underlying arrows of $A$. Direct substitution shows $E_v j_u = v$ as functors.

Assume (1). Each component of the target of $Bj_u$ is a weakly contractible CW complex and hence contractible by Whitehead's theorem. The continuous image of a component of $BD$ lies in one such component, so the restricted map is null-homotopic. This proves (2), and (2) implies (3).

Assume (3), fix $H$ and a component $C$ of $\mathrm{El}_A(H)$, and consider any map $f : S^n \to BC$, $n \geq 1$. By Lemma C.38, after identifying the source with $BD$ its free homotopy class is represented by $Bv$, for a functor $v : D \to C$. The preceding factorization gives $Bv = BE_v\, Bj_{\pi v}$. Condition (3) makes this composite null-homotopic. Thus every unbased sphere map into the connected space $BC$ is null. This implies the vanishing of every based homotopy group: for $n = 1$ a free null-homotopy makes a loop conjugate to the identity and hence trivial; for $n > 1$ the change of basepoint acts by automorphisms and fixes the zero class. Therefore $BC$ is weakly contractible. The case $H = \varnothing$ has no components and imposes no extra condition. Empty or disconnected $D$ causes no difficulty in (2), which is expressly componentwise. □

## C.11 The opposite interval-type category: Lead 8

**Proposition C.40** (Proved: precise variance and a homological obstruction)**.** *Let* $\mathsf{Ty}$ *be the category in PRETOT Definition E.1: its objects are finite words* $\tau$ *in* $+, -$*, including the empty word, and a morphism* $\tau \to \rho$ *is an endpoint-preserving functor* $I_\tau \to I_\rho$ *whose function on vertices is nondecreasing. Here* $I_\tau$ *is the free category on the oriented line prescribed by* $\tau$*. Then* $\mathsf{Ty}^{\mathrm{op}}$ *is isomorphic to the category of these words and nondecreasing sign-preserving maps of their letter sets. Under this isomorphism, the colored-word category* $W_2$ *is its wide subcategory of injective letter maps, after identifying its two colors with* $+$ *and* $-$*.*

*Put* $p = (++)$*,* $q = (+-+)$*, and use one-based letter indices. The arrows*

$$e = (1, 3), \quad c = (1, 1) : p \rightrightarrows q$$

*have no completion to a commutative square. For every nonzero unital ring* $R$ *they have no affine equalizer in* $\mathrm{Aff}_R(\mathsf{Ty}^{\mathrm{op}})$*. In fact, for the covariant set quotient*

$$H = \mathsf{Ty}^{\mathrm{op}}(p, -)/(ge \sim gc)$$

*we have* $H_1(B\,\mathrm{El}_{\mathsf{Ty}^{\mathrm{op}}}(H); \mathbb{Z}) \cong \mathbb{Z}$*. Thus* $\mathsf{Ty}^{\mathrm{op}}$ *is neither total nor exact. The phrase "over any ring" in Lead 8 must exclude the zero ring; with that qualification the lead is true.*

*Proof.* For $\tau$ of length $n$ and $\rho$ of length $m$, write a type map on vertices as $\phi : \{0, \dots, n\} \to \{0, \dots, m\}$, with $\phi(0) = 0$, $\phi(n) = m$ and $\phi(i-1) \leq \phi(i)$. The functor condition says exactly that every letter of $\rho$ in the interval $(\phi(i-1), \phi(i)]$ has the sign of the $i$th letter of $\tau$; a collapsed interval imposes no condition. Define

$$f_\phi : \{1, \dots, m\} \longrightarrow \{1, \dots, n\}, \qquad f_\phi(j) = \text{the unique } i \text{ with } \phi(i-1) < j \leq \phi(i).$$

This is nondecreasing and sign-preserving. Conversely such a map $f$ defines $\phi_f(i) = |\{j : f(j) \leq i\}|$. These two formulas are inverse. They also cover the empty words: the map from a nonempty letter set to an empty one does not exist, while the map from an empty letter set is unique. If $\psi : I_\rho \to I_\sigma$ is another type map, the defining interval inequalities give $f_{\psi\phi} = f_\phi f_\psi$. Therefore these bijections form the asserted isomorphism of categories with the opposite variance.

Let $g : q \to \sigma$ be any arrow in the opposite category. Since adjacent letters of $q$ have opposite signs, sign preservation and nondecreasingness imply $g(1) < g(2) < g(3)$. Hence $ge$ is

injective on the two letters of $p$, whereas $hc$ is constant for every $h : q \to \sigma$. Thus $ge = hc$ is impossible, proving failure of condition I.

For each $\sigma$ define an $R$-linear functional

$$\lambda_\sigma : R[\mathsf{Ty}^{\mathrm{op}}(p, \sigma)] \to R$$

by giving a basis arrow value 1 if it is injective and 0 if it is constant. These are the only possibilities for a map out of $p$. For every $g : q \to \sigma$ we have $\lambda_\sigma(ge - gc) = 1$. Consequently for every finite linear combination $t \in R[\mathsf{Ty}^{\mathrm{op}}(q, \sigma)]$,

$$\lambda_\sigma(t(e - c)) = \mathrm{aug}(t).$$

If $te = tc$, its augmentation is zero. It cannot also be 1 unless $R$ is the zero ring. This is an exact integral dual certificate, valid for all targets and hence for all nonzero coefficient rings.

For the stronger homological claim, work over $\mathbb{Z}$ and put $P_x = \mathbb{Z}[\mathsf{Ty}^{\mathrm{op}}(x, -)]$, $d : P_q \to P_p$, $d(t) = t(e - c)$, $K = \ker d$, and $J = \operatorname{im} d$. The objectwise free-group functor preserves the defining set coequalizer, so $\operatorname{coker} d = \mathbb{Z}[H]$. Representables are projective and their colimits are canonically $\mathbb{Z}$ by augmentation. The functional identity shows that the map $\operatorname{colim} K \to \operatorname{colim} P_q = \mathbb{Z}$ is zero, even when its input is a finite sum of elements represented at different objects. The exact sequence $K \to P_q \to J \to 0$ therefore gives $\operatorname{colim} J \cong \mathbb{Z}$. The map $\operatorname{colim} J \to \operatorname{colim} P_p$ is zero because the augmentation of every $t(e - c)$ is zero. The derived-colimit exact sequence of $0 \to J \to P_p \to \mathbb{Z}[H] \to 0$ now gives

$$L_1 \operatorname{colim} \mathbb{Z}[H] \cong \operatorname{colim} J \cong \mathbb{Z}.$$

The bar complex for the permutation diagram $\mathbb{Z}[H]$ is exactly the integral nerve chain complex of its category of elements, proving the claim. This explicitly witnesses failure of both totality and exactness and does not infer an infinite conclusion from finite tests.

The program `code/finite_ty_verify.py` independently checks the two descriptions on all 961 ordered pairs of words of length at most four, the contravariant composition bijection on 2731 composable pairs of type maps for words of length at most three, and the integral dual certificate on all 7937 arrows from $q$ to the 1023 words of length at most nine. These computations audit the formulas; the preceding proof applies without a length bound. □

## C.12 Constraints on a minimal separator: Lead 9

**Proposition C.41** (Proved: Lead 9)**.** *Suppose $A$ is a small total category that is not exact, and let $a_0, \dots, a_k : p \to q$ be a family without an integral affine equalizer. Assume $k$ is least among all such families in $A$. Then:*

1. *$A(q, p) = \varnothing$.*
2. *The finite cardinalities $|A(p, x)|$ have no uniform finite bound; in symbols, $\sup_x |A(p, x)| = \infty$ in the sense of unboundedness in the natural numbers, allowing infinite values.*
3. *The full cosieve $V_p = \{x \in A : A(p, x) \neq \varnothing\}$ is again a separator, with the same exhibited family. Its object $p$ is weakly initial, and the minimal integer $k$ remains the same after restriction.*
4. *For every arrow $g : q \to x$, the arrows $ga_0, \dots, ga_k : p \to x$ are pairwise distinct.*

*Proof.* Totality gives conditions I and IV. We use precisely the two exclusions proved in the opened source (Isbell and Mitchell, 1976, Corollary 4.3 and Proposition 4.5). Corollary 4.3 first simultaneously equalizes a finite collection of endomorphisms with the identity, by applying

the fixed-point theorem to endomorphisms of a representable act. If $b : q \to p$ existed, apply it to $ba_0, \dots, ba_k, 1_p$. It gives an ordinary arrow $u : p \to r$ with $uba_i = u$ for every $i$. The ordinary arrow $ub : q \to r$ would already be an affine equalizer, proving (1).

For (2), the representable $A$-set $A(p, -)$ is connected because its category of elements has the initial object $(p, 1_p)$. If its values had a uniform finite bound, Proposition 4.5 would imply condition II for this act. That proposition forms the congruence identifying two elements whenever an ordinary arrow equalizes them; condition I makes this a congruence. All arrows act injectively on its quotient. At an object where the finite cardinality is maximal, the maps to subsequent objects are bijections, and the necklace form of IV forces the quotient to have one element there. The conclusion is exactly ordinary equalizability of every pair in the original representable. Apply it successively to the finitely many images of $a_0, \dots, a_k$; composing the resulting ordinary arrows gives an ordinary simultaneous equalizer, a contradiction. This verifies that the cited result applies to the one representable involved here, rather than requiring a bound on every hom-set of $A$.

For (3), $V_p$ is nonempty and is closed under targets. It is connected: condition I applied to a span $p \to x$, $p \to y$ gives a common target inside $V_p$. A covariant set-valued functor on $V_p$ extends to $A$ by the empty set off $V_p$; arrows entering $V_p$ have the unique map from an empty set, and no arrow leaves it. Its category of elements is literally unchanged. Hence totality of $A$ proves totality of $V_p$. Any affine equalizer of the displayed family in $V_p$ would also be one in $A$, so $V_p$ is not exact. Conversely, if a family of arrows in $V_p$ has an affine equalizer in $A$, every target occurring in its finite support is in $V_p$, because it receives an arrow from its source in the cosieve. Thus it is already an affine equalizer in $V_p$. All smaller families therefore remain equalizable. Every object of $V_p$ receives a map from $p$, which is exactly weak initiality. This is the full proof of the cosieve step used from TOTEX D.23.

For (4), suppose $ga_i = ga_j$ for some $i \neq j$. The displayed family at $x$ then has at most $k$ distinct arrows. By the global minimality of $k$ there is an affine $t : x \to y$ equalizing all its distinct members. The composite $tg \in \mathbb{Z}[A(q, y)]$ has augmentation one, because $g$ is an ordinary arrow, and satisfies $(tg)a_i = (tg)a_0$ for every $i$. This contradicts the choice of the original family. This argument uses minimality among all objects and families of $A$, as stipulated in the lead; minimality merely within a fixed target would not suffice. □

# D Class diagram, manuscript readings, and sources

## D.1 The strict class diagram

**Proposition D.1** (Proved: every inclusion in the class diagram is strict)**.** *Among nonempty connected small categories,*

$$\{\textit{filtered}\} \subsetneq \{\textit{exact}\} \subsetneq \{\textit{total}\} \subsetneq \{\textit{quasi-filtered}\}.$$

*The ordinal-injection category witnesses the first strict inclusion. The ordered predimension category $\mathcal{C}_\delta$ of Theorem A.34 witnesses the middle strict inclusion. The selected repaired AKR category of Theorem C.37 witnesses the last strict inclusion.*

*Proof.* Filtered colimits of abelian groups are exact. The complex computing $T_i(\mathbb{Z}[H])$ is the integral nerve-chain complex of $\mathrm{El}_A H$, so exactness implies acyclicity of every component. We check the homotopical step in supplied TOTASPH C.3–C.6. For a component $E \subseteq \mathrm{El}_A H$ and $S : E \to \mathrm{Set}$, the functor

$$a \longmapsto \coprod_{(a,h)\in E} S(a, h)$$

has category of elements isomorphic to $\mathrm{El}_E S$. Hence acyclicity also holds for actions on $E$. For any cyclic subgroup $C \leq \pi_1(BE, e)$, the action $x \mapsto \mathrm{Hom}_{\Pi E}(e, x)/C$ gives the connected

covering with fundamental group $C$, as is checked by edge-path lifting. Its first homology is $C$, forcing $C = 0$. Thus $BE$ is simply connected, and Hurewicz turns its acyclicity into weak contractibility. This does not replace permutation modules by arbitrary diagrams.

For totality implying I, the checked TOTASPH C.11 argument first gives common successors. If two principal cosieves were disjoint, identify the values of the constant two-element action on their union. Its element category is the double of $A$ identified on those cosieves. Its reduced homology sequence has $H_1 = \mathbb{Z}$, since $BA$ is contractible and the two cosieves are nonempty and connected. This contradicts totality. Applying common successors to the inherited total category $p/A = \mathrm{El}_A\, A(p, -)$ completes every span out of $p$ to a commuting square. The case $k = 1$ of Lemma B.2, followed by common successors, gives IV. Conditions I and IV give quasi-filteredness.

We checked the interpolation proof for $\Delta_{\mathrm{inj}}$ in supplied TOTASPH C.8–C.10. Its identity $P_{t\delta^i} = y_i D_i P_t$ encodes gaps of injections as monomials. The integral coordinates $x_i = z_i - z_{i+1}$ reduce extension of compatible homogeneous restrictions to the alternating sum over nonempty coordinate intersections. The prescriptions $D_i C_{n,k} = C_{n-1,2k} \prod_{\ell \neq i} y_\ell^k$ agree on intersections and have value one at the all-ones point. Thus they give an augmentation-one combination equalizing the cofaces; the simplicial identities then equalize all parallel injections. Theorem 1 and Lemma 1 of Isbell and Mitchell (1973), or Theorem 3.1 and Proposition 3.3 of Isbell and Mitchell (1976), give exactness. The two point inclusions $[0] \rightrightarrows [1]$ cannot be post-equalized by an injection. The 1973 Theorem 3 announces the example; the body of Isbell (1974) was not obtained and is not claimed as a proof checked in this run.

Theorem A.34 proves totality of the explicitly specified category $\mathcal{C}_\delta$ for all set-valued actions. Its three point arrows have no integral affine equalizer, or directly its diagram $Q_{\mathbb{Z}}$ has $T_1(Q_{\mathbb{Z}}) = \mathbb{Z}$. Thus exactness is strictly stronger than totality. The same witness works over every nonzero coefficient ring by Corollary A.3.

In repaired AKR, each parallel pair receives a quasi-filter and each object pair a common successor, by the opened Constructions B.9 and C.1 of Adámek et al. (1981). These give I and IV at the union: every finite diagram occurs at a finite stage. Proposition C.4 proves nonexactness for every admissible repaired construction. Theorem C.37 gives a deterministic choice of sufficiently large targets for which the binary category of elements has nonzero second homology. That category is quasi-filtered and is not total, proving the final strict inclusion without assuming the universal choice statement of C.1. □

**Scope of the witnesses.** Part D.1 is settled by these two distinct explicit infinite categories. The middle witness proves totality; the right witness has a specified nonzero second homology class in one category of elements. The stronger universal assertion concerning every admissible repaired AKR choice is treated separately in Part C.1.

## D.2 The corrected manuscript statements

**Proposition D.2** (Refuted conjecture; known corrected implications)**.** *The implication* totalement $W_\infty$-asphérique $\Longrightarrow$ filtrante *at II manuscript page 36ter is false. For nonempty connected small $A$,*

$$A \text{ filtered} \iff A \text{ total and sifted.}$$

*For every fixed integer $n \geq 1$,*

$$A \text{ filtered} \iff A \text{ total and } \operatorname*{hocolim}_A K(M(a), n) \text{ is } n\text{-truncated}$$
$$\text{for every covariant } M : A \to \mathrm{Ab}.$$

*The implication from siftedness alone to totality in Proposition 2.19 at II manuscript page 36 is false. Replacing "filtered" by "exact" in the conjectured implication does not repair it:*

*totality does not imply exact abelian colimits.*

*Proof.* Proposition D.1 refutes the conjecture. We read the wording in the supplied transcription:

> "Cela me fait soupçonner que la conjecture"
>
> $$X \text{ totalement } W_\infty\text{-asphérique} \Longrightarrow X \text{ filtrante}$$
>
> "pourrait être valable."

The proposed replacement by exactness is false by Theorem A.1; $\mathcal{C}_\delta$ supplies the counterexample with the manuscript's total-asphericity hypothesis. Thus the strict diagram above, together with the two displayed additional-hypothesis equivalences, gives the corrected categorical relationships.

For the first corrected equivalence, we checked supplied PRETOT F.3 and TOTEX D.4 and the finite cocone criterion in Gabriel and Ulmer (1971, Sections 15.2(c), 15.9). Siftedness means $C_{p,q} = \mathrm{El}_A(A(p,-) \times A(q,-))$ is nonempty and connected. Inherited totality gives common successors in $C_{p,q}$. For $u, v : p \to q$, a successor of $(q; u, \mathrm{id}_q)$ and $(q; v, \mathrm{id}_q)$ gives $f, g : q \to r$ with $fu = gv$ and $f = g$. Thus $f$ post-equalizes $u, v$. Siftedness also gives common successors in $A$, so $A$ is filtered. The converse is the filtered-colimit criterion.

We checked the spectral-sequence steps of supplied TOTEX D.5, D.7 and D.9; the earlier abelian one-type argument is PRETOT F.6. For $n \geq 2$, $BA$ is contractible and $X_M = \mathrm{hocolim}_A K(M, n)$ is $(n-1)$-connected. Its degree-$n$ homology is $\mathrm{colim}\, M$, so the assumed truncation makes the canonical map $X_M \to K(\mathrm{colim}\, M, n)$ a weak equivalence. The reduced homology bar spectral sequence has first nonzero row $M$ in degree $n$ and zero row in degree $n+1$, since $H_{n+1}K(M,n) = 0$. Hence $T_1(M) = H_{n+1}(X_M) = 0$, first proving exactness. Collapse in all degrees, applied to $M \oplus N$, identifies the natural Künneth cross summand in degree $2n$:

$$\operatorname*{colim}_A (M \otimes N) \cong (\operatorname*{colim}_A M) \otimes (\operatorname*{colim}_A N).$$

There is no Tor term in that degree, since two positive indices are at least $n$. On $P_p, P_q$, this is the free abelian map from the component set of $C_{p,q}$ to a singleton. It is an isomorphism only if that set is a singleton. Thus $A$ is sifted and hence filtered.

For $n = 1$, common successors make the group colimit of an abelian diagram abelian: any two representatives commute after transport to a common object. Since $BA$ is contractible, the fundamental-group presentation of $\mathrm{hocolim}_A BM$ gives $G = \mathrm{colim}_A M$. The one-type assumption identifies it with $BG$. In degree two the homology edge image is that of $\mathrm{colim}_A \Lambda^2 M \to \Lambda^2 G$, with remaining filtration quotient $T_1(M)$. The edge map is surjective by transporting the two factors of an exterior product to a common successor, so $T_1(M) = 0$. After exactness is obtained, the cross summand of $\Lambda^2(M \oplus N)$ gives the same tensor comparison and siftedness. Conversely, filtered homotopy colimits preserve these truncations because spheres, nullhomotopies and horns involve finitely many simplices.

For Proposition 2.19 take $A = \Delta^{\mathrm{op}}$ and $H = \Delta[1]/\partial\Delta[1]$. In our covariant convention, $\mathrm{El}_A H$ is opposite to the usual simplex category, so its classifying space is weakly equivalent to $|H| = S^1$. Thus $A$ is not total. It is sifted: the cospan category in $\Delta^{\mathrm{op}}$ on $[m], [n]$ is opposite to the simplex category of $\Delta[m] \times \Delta[n]$, which is nonempty and connected. This checks the variance of the counterexample. □

**Proposition D.3** (Known, with the comparison maps audited: positive truncations)**.** *For every small category $A$, including empty and disconnected categories, let $\ell : A \to \Pi A$ be groupoid completion. For every fixed finite $n \geq 1$, homotopy colimits over $A$ preserve $n$-types if and only if*

$$C_a = a \downarrow \ell = \mathrm{El}_A(\mathrm{Hom}_{\Pi A}(a, \ell(-)))$$

*is filtered for every $a$. This is supplied TOTEX D.10.*

*Audit of the relied-on reduction.* An object of $C_a$ is $(x, \gamma : a \to \ell(x))$; an arrow to $(y, \delta)$ is $f : x \to y$ with $\ell(f)\gamma = \delta$. All transition maps are bijective, and edge-path lifting identifies $BC_a \to BA$ as the connected universal cover of its component. For $Y : \mathrm{El}_A S \to \mathrm{Spaces}$, the functor $b \mapsto \coprod_{s \in S(b)} Y(b, s)$ has the same simplicial replacement as $\mathrm{hocolim}_{\mathrm{El}_A S} Y$. Preservation of $n$-types therefore descends to $C_a$.

The simply connected case in TOTEX D.9 uses the reduced homology bar complex. The constant point diagram first makes $BC_a$ an $n$-type. For $n \geq 2$, the section of $\mathrm{hocolim}\, K(M, n) \to BC_a$ and the discreteness of the based self-equivalence space of $K(L, n)$ trivialize its Eilenberg–MacLane fibres over this simply connected base. Relative homology in degree $n + 1$ is zero and equals $T_1(M)$. For $n = 1$, the double-cosieve action excludes absence of common successors by a surviving $E^2_{1,1} = \mathbb{Z}$ term; the abelian one-type argument then applies. Thus $C_a$ is exact, total and, by the fixed-$n$ test, filtered. These are the specific steps checked in the supplied proof; we do not attribute the classification to a differently stated theorem on filtered colimits.

Conversely, filteredness makes each $BC_a$ contractible. The pullback of $BC_a \to BA$ along $\mathrm{hocolim}_A X \to BA$ is modeled by $\mathrm{hocolim}_{C_a} X\pi$: on the bar construction, a lift of a chain is determined by a path class from $a$ to its first vertex. Filtered homotopy colimits preserve $n$-types; covering maps induce isomorphisms on homotopy groups in degrees at least two. This proves the converse componentwise. For empty $A$ both assertions hold vacuously. □

**Remark D.4** (Proved: typing corrections and scope)**.** At II manuscript page 36, for $f : X \to Y$, finality must use $y\backslash X$ rather than the printed $y\backslash Y$. The latter always has initial object $\mathrm{id}_y$ and would make the condition vacuous. At XII manuscript page 88, $H' \in (U')^\wedge$, so the correctly typed isomorphism is

$$U'/t' \simeq U'/H',$$

not $U'/t' \simeq U/H'$. A presheaf on $U$ is a covariant set-valued functor on $U^{\mathrm{op}}$; its usual element category is opposite to the corresponding covariant element category. Opposition preserves the classifying-space homotopy type.

Definition 1 at XII manuscript pages 88–89 quantifies over all left Kan extensions and equivalently over ordinary final images. It is the notion used here, distinct from the test-category diagonal condition in *Pursuing Stacks*. We inspected the supplied transcription, not an independent copy of the authoritative edition. These are mathematical typing corrections, not a new editorial collation.

## D.3 Literature and provenance

The access table in `TOTEX2_report.md` records every requested source, exact inspected theorem numbers, failed retrievals and search terms. We obtained Isbell (1968) and Gabriel–Ulmer (1971), which had not been opened in the supplied record. We did not obtain the bodies of Isbell (1974), Oberst (1968) or Adámek–Reiterman (1975). No claim about their unopened contents is used as a proof. Each bibliography entry records body-access status.

The coefficient-ring criterion is the remark on page 296 of Isbell and Mitchell (1976); the weak-terminal and bounded-hom-set exclusions are its Corollaries 4.3–4.4, Proposition 4.5, and Corollary 4.6. The equivalence printed in Adámek and Reiterman (1976, II.3) is false. Adámek et al. (1981, Remark A.4) explicitly corrects the earlier 1975 remark; their counterexample also refutes the assertion repeated in the 1976 paper, which A.4 does not explicitly cite. The fixed-point/exactness question in Husainov (2002, Problem 2.5) has a weaker hypothesis than totality; our separator in particular has the fixed-point property supplied by I and IV, while its affinization is not filtered.

For monoid acts, preservation of monomorphisms and of finite limits are distinct meanings of flatness; Hemelaer and Rogers (2021, Definition 1.6, Proposition 1.8, Theorems 2.32 and 2.49) separates them. We also inspected Laan (nd, Proposition 2.4 and the paragraph after Lemma 2.6). Neither notion may be silently identified with flatness of the constant right $\mathbb{Z}A$-module.

We make no priority claim. Searches included exact colimits, fixed-point properties, homotopy-flat actions, totally aspherical categories, Ramsey classes, and exact titles and DOIs of the requested sources. We also searched strong amalgamation, predimension closure, homogeneous structures, and open-subgroup supports; the precise opened passages are recorded in the report. No opened source in these searches supplied the stated all-action separator theorem. Its proof is included here. This reports the search performed and makes no assertion of priority.

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